

MODULE -1

INTRODUCTION TO NON-TRADITIONAL MACHINING

SYLLABUS

Introduction to Non-traditional machining, Need for Non-traditional machining process, Comparison between traditional and non-traditional machining, general classification, Non-traditional machining processes, classification based on nature of energy employed in machining, selection of non-traditional machining processes, Specific advantages, limitations and applications of non-traditional machining processes.

1.1 Introduction to Non-traditional machining

Non-traditional manufacturing processes is defined as a group of processes that remove excess material by various techniques involving mechanical, thermal, electrical or chemical energy or combinations of these energies but do not use a sharp cutting tool as it needs to be used for traditional manufacturing processes.

Extremely hard and brittle materials are difficult to machine by traditional machining processes such as turning, drilling, shaping and milling. Non-traditional machining processes, also called advanced manufacturing processes, are employed where traditional machining processes are not feasible, satisfactory or economical due to special reasons as outlined below.

- Very hard fragile materials difficult to clamp for traditional machining
- When the work piece is too flexible or slender
- When the shape of the part is too complex.

Several types of non-traditional machining processes have been developed to meet extra required machining conditions. When these processes are employed properly, they offer many advantages over non-traditional machining processes. The common non-traditional machining processes are described in this section.

Definition: A machining process is called Non-traditional machining if its material removal mechanism is basically different than those in the traditional processes, i.e. a different form of energy

(other than the excessive forces exercised by a tool, which is in physical contact with the work piece) is applied to remove the excess material from the work surface, or to separate the workpiece into smaller parts.

1.2 Need for development of Non-Conventional Processes

The strength of steel alloys has increased five folds due to continuous R&D effort. In aero-space requirement of high strength at elevated temperature with light weight led to development and use of hard titanium alloys, nimonic (Nickel based) alloys, and other HSTR alloys. The ultimate tensile strength has been improved by as much as 20 times. Development of cutting tools which has hardness of 80 to 85 HRC which cannot be machined economically in conventional methods led to development of non –traditional machining methods.

1. Technologically advanced industries like aerospace, nuclear power, wafer fabrication, automobiles has ever increasing use of high-strength temperature resistant (HSTR) alloys (having high strength to weight ratio) and other difficult to machine materials like titanium, SST, nimonics, ceramics and semiconductors. It is no longer possible to use conventional process to machine these alloys.
2. Production and processing parts of complicated shapes (in HSTR and other hard to machine alloys) is difficult, time consuming and uneconomical by conventional methods of machining.
3. Innovative geometric design of products and components made of new exotic materials with desired tolerance, surface finish cannot be produced economically by conventional machining.
4. The following examples are provided where NTM processes are preferred over the conventional machining process:
 - ◆ Intricate shaped blind hole – e.g. square hole of 15 mmx15 mm with a depth of 30 mm with a tolerance of 100 microns
 - ◆ Difficult to machine material – e.g. Inconel, Ti-alloys or carbides, Ceramics, composites, HSTR alloys, satellites etc.,
 - ◆ Low Stress Grinding – Electrochemical Grinding is preferred as compared to conventional grinding
 - ◆ Deep hole with small hole diameter – e.g. ϕ 1.5 mm hole with $l/d = 20$
 - ◆ Machining of composites.

Non-Traditional Machining (NTM) Processes are characterized as follows:

- Material removal may occur with chip formation or even no chip formation may take place. For example in AJM, chips are of microscopic size and in case of Electrochemical machining material removal occurs due to electrochemical dissolution at atomic level

- In NTM, there may not be a physical tool present. For example in laser jet machining, machining is carried out by laser beam. However in Electrochemical Machining there is a physical tool that is very much required for machining
- In NTM, the tool need not be harder than the work piece material. For example, in EDM, copper is used as the tool material to machine hardened steels.
- Mostly NTM processes do not necessarily use mechanical energy to provide material removal. They use different energy domains to provide machining. For example, in USM, AJM, WJM mechanical energy is used to machine material.

1.3 Differences between Conventional and Non-Conventional machining processes

Sl No.	Conventional Process	Non-Conventional Process
1.	The cutting tool and work piece are always in physical contact with relative motion with each other, which results in friction and tool wear	There is no physical contact between the tool and work piece, In some non-traditional process tool wear exists
2.	Material removal rate is limited by mechanical properties of work material	NTM can machine difficult to cut and hard to cut materials like titanium, ceramics, nimonics, SST, composites, semiconducting materials
3.	Relative motion between the tool and work is typically rotary or reciprocating. Thus the shape of work is limited to circular or flat shapes. In spite of CNC systems, production of 3D surfaces is still a difficult task	Many NTM are capable of producing complex 3D shapes and cavities
4.	Machining of small cavities, slits, blind holes or through holes are difficult	Machining of small cavities, slits and Production of non-circular, micro sized, large aspect ratio, shall entry angle holes are easy using NTM
5.	Use relative simple and inexpensive machinery and readily available cutting	Non-traditional processes requires expensive tools and equipment as well as skilled

	tools	labour, which increase the production cost significantly
6.	Capital cost and maintenance cost is low	Capital cost and maintenance cost is high
7.	Traditional processes are well established and physics of process is well understood	Mechanics of Material removal of some of NTM process are still under research
8.	Conventional process mostly uses mechanical energy	Most NTM uses energy in direct form. For example: laser, Electron beam in its direct forms are used in LBM and EBM respectively
9.	Surface finish and tolerances are limited by machining inaccuracies	High surface finish (up to 0.1 μm) and tolerances (25 μm) can be achieved
10.	High metal removal rate	Low material removal rate

1.4 Classification of NTM process

- **Mechanical Processes** - Erosion of work material by a high velocity stream of abrasives or fluid (or both) is the typical form of mechanical action. It is characterized by the fact that the material removal is due to the application of mechanical energy in the form of high frequency vibrations or kinetic energy of an abrasive jet.
 - Abrasive Jet Machining (AJM)
 - Ultrasonic Machining (USM)
 - Water Jet Machining (WJM)
- **Electrochemical Processes** – Electro chemical energy to remove material (reverse of electroplating). Electrical energy used in combination with chemical reactions to remove material.
 - Electro Chemical Machining (ECM)
 - Electro Chemical Grinding (ECG)
 - Electro Jet Drilling (EJD)
- **Electro-Thermal Processes** - Thermal energy usually applied to small portion of work surface, causing that portion to be removed by fusion and/or vaporization
 - Electro-discharge machining (EDM)

- Laser Jet Machining (LJM)
- Electron Beam Machining (EBM)
- **Chemical Processes** - Chemical etchants selectively remove material from portions of work part, while other portions are protected by a mask.
 - Chemical Milling (CHM)
 - Photochemical Milling (PCM)

Classification based on energy employed in machining is shown in Fig.1

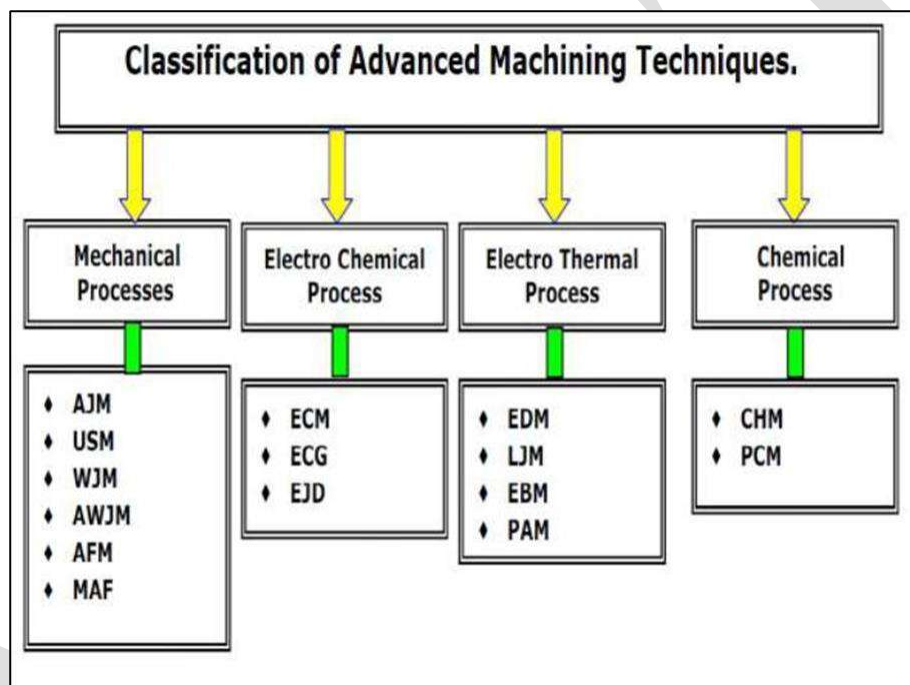


Fig.1 Classification non-traditional machining processes

1.5 Selection of Process

The correct selection of the non-traditional machining methods must be based on the following aspects.

- i) Physical parameters of the process
- ii) Shape to be machined
- iii) Process capability
- iv) Economics of the processes

i) Physical parameter of the process

Parameters	USM	AJM	ECM	CHM	EDM	EBM	LBM	PAM
Potential (V)	220	220	10	—	45	150,000	4500	100
Current (Amp)	12 (A.C.)	1.0	10,000 (D.C.)	—	50 (pulsed D.C.)	0.001 (pulsed D.C.)	2 (average 200 peak)	500 (D.C.)
Power (W)	2400	220	100,000	—	2700	150 (average 200 peak)	—	50,000
Gap (mm)	0.25	0.75	0.20	—	0.025	100	150	7.5
Medium	Abrasive in water	Abrasive in gas	Electrolyte	Liquid chemical	Liquid dielectric	Vacuum	Air	Argon or hydrogen

The physical parameters of the different NTM are given in the Table (refer ppt.) which indicates that PAM and ECM require high power for fast machining. EBM and LBM require high voltages and require careful handling of equipment. EDM and USM require medium power. EBM can be used in vacuum and PAM uses oxygen and hydrogen gas.

ii) Shapes cutting capability

Process	Holes				Through cavities		Surfacing		Through cutting	
	Precision small holes		Standard		Precision	Standard	Double contouring	Surfaces of revolution	Shallow	Deep
	Dia	Dia	Length	Length						
	<.025 mm	>.025 mm	<20 Dia	>20 Dia						
USM	—	—	A	C	A	A	C	—	C	—
AJM	—	—	B	C	C	B	—	—	A	—
ECM	—	—	A	A	B	A	A	B	A	A
CHM	B	B	—	—	C	B	—	—	A	—
EDM	—	—	A	B	A	A	B	—	C	—
LBM	A	A	B	C	C	C	—	—	A	B
PAM	—	—	B	—	C	C	—	C	A	A

NOTE
A Good B Fair C Poor

Process	Materials Applications							
	Material							
	Aluminium	Steel	Super alloys	Titanium	Refractories	Plastics	Ceramics	Glass
USM	C	B	C	B	A	B	A	A
AJM	B	B	A	B	A	B	A	A
ECM	B	A	A	B	B	D	D	D
CHM	A	A	B	B	C	C	C	B
EDM	B	A	A	A	A	D	D	D
EBM	B	B	B	B	A	B	A	B
LBM	B	B	B	B	C	B	A	B
PAM	A	A	A	B	C	C	D	D
NOTE								
A Good Application B Fair C Poor D Not Applicable								

The different shapes can be machined by NTM. EBM and LBM are used for micro drilling and cutting. USM and EDM are useful for cavity sinking and standard hole drilling. ECM is useful for fine hole drilling and contour machining. PAM can be used for cutting and AJM is useful for shallow pocketing.

iii) Process capability

Process	Process Capability				
	Metal removal rate	Tolerance	Surface finish	Surface damage	Corner
	mm ³ /min	μ	CLA μ	depth μ	radii mm
USM	300	7.5	0.2- 0.5	25	0.025
AJM	0.8	50	0.5- 1.2	2.5	0.100
ECM	01500	50	0.1- 2.5	5.0	0.025
CHM	15.0	50	0.4- 2.5	50	0.125
EDM	800	15	0.2-12.5	125	0.025
EBM	1.6	25	0.4- 2.5	250	2.50
LBM	0.1	25	0.4-1.25	125	2.50
PAM	75000	125	Rough	500	—
Conventional milling of steel	50000	50	0.4- 5.0	25	0.050

The process capability of NTM is given in Table 2.0 EDM which achieves higher accuracy has the lowest specific power requirement. ECM can machine faster and has a low thermal surface damage depth. USM and AJM have very material removal rates combined with high tool wear and are used

non metal cutting. LBM and EBM are, due to their high penetration depth can be used for micro drilling, sheet cutting and welding. CHM is used for manufacture of PCM and other shallow components.

iv) Economic consideration

Process	Process Economy				
	Capital investment	Tooling and fixtures	Power requirement	Efficiency	Tool consumption
USM	B	B	B	D	C
AJM	A	B	B	D	B
ECM	E	C	C	B	A
CHM	C	B	D	C	A
EDM	C	D	B	D	D
EBM	D	B	B	E	A
LBM	C	B	A	E	A
PAM	A	B	A	A	A
Conventional machining	B	B	B	A	B

NOTE
A Very Low Cost B Low C Medium D High E Very High

1.6 Applications of NTM

- NTM are used to machine hard and difficult-to-cut materials like superalloys, nimonics, carbides, heat resisting steel, etc, which are widely used in various engineering applications.
- NTM is used for machining very thin and fragile parts, complex geometrical profiles with superior finish, accuracy and tolerance, which are mostly preferred in aerospace, automotive, nuclear, mining and other sectors.
- NTM are used for shaping hard cutting tools having complex geometrical profiles with much precise accuracy.
- Apart from engineering applications, NTM processes are also used for preparation of jewels for watch and timer movements; removing glue and paint from leather objects and paintings; engraving of products with alphanumeric or bar codes, logos, symbols, graphics and other applications.
- In addition, NTM are also used in machining of cavities in forging dies, drilling deeper holes and irregular shaped holes which cannot be obtained by conventional machining methods.

1.7 Advantages

- Difficult to machine materials can be machined with non-conventional process
- Machining materials in the complex shapes is possible with non-conventional process
- Economical for mass production for long duration
- High strength to weight ratio, hardness and heat resisting materials can be machined with non-conventional process
- High accuracy and surface finish
- Burr-free machining(ECM,EDM,CHM)
- Material removed without mechanical contact with the workpiece (ECM,EDM,LBM,CHM).
- Material removal rate is independent of workpiece hardnesss (ECM,LBM,EDM)
- Cutting forces are independent of workpiece hardness (ECM,LBM,EDM,CHM)
- Tool material need not be harder than the workpiece material (ECM,LBM,EDM,CHM,USM)
- Tool wear is not a problem (ECM,LBM,CHM)
- Ability to machine any material.(LBM)
- Stress- free machining (ECM,ECG,CHM)
- Uniform material removal over the entire area simultaneously (ECM,CHM)
- Superior surface integrity possible (ECM,CHM,ECG)
- Intricately shaped, very hard and fragile materials can be machined.(USM)
- Finely focused micro machining
- Micro- hole drilling at shallow entrance angles possible (EDM,ECM,LBM,EBM)
- Easy compatibility with numerical control and mini- computer controls (ECM,EDM,LBM,EBM)

1.8 Limitations

- Work piece and tool must be electrically conductive (EDM,ECM)
- Depth of cut is limited (LBM)
- Recast or heat affected zones (HAZ) of surfaces produced may be troublesome (EDM,LBM,EBM)
- For this purpose the compatibility of the process with the metallurgical state of the work piece materials can be studied before using a particular non-traditional machining process for production work.
- There may be taper in the sidewalls of holes or cavities (EDM,LBM)
- Most of these limitations can be overcome and controlled, so that the advantages can be obtained with good product quality assurance.

QUESTION BANK

1. What is non-traditional machining or unconventional machining or modern machining technique. How are they classified.
2. Differentiate between conventional (traditional) and unconventional (non-traditional) or modern machining processes. (5 to 6 Marks) **(VTU June/July 2016) (VTU June/July 2014) (VTU June/July 2013) (VTU Dec 2011)**
3. What are the applications of non-traditional machining methods.
4. What are the advantages and disadvantages of non-traditional machining.
5. Classify non-traditional machining processes based on nature of energy employed in machining.
6. Write a note on the source of energy harnessed and mechanism of material removal in non-traditional machining. (6 Marks) **(VTU June/July 2011).**
7. Explain the need for development of non-traditional machining. (4 Marks) **(VTU June/July 2011)**
8. Classify the modern machining processes. (7 Marks) **(VTU Dec 2011)**
9. Explain the principle of modern machining process. What are the factors to be considered while selecting a process? (7 Marks) **(VTU Dec 2011)**
10. Explain the need to develop modern machining processes. (4 Marks) **(VTU Dec 2012)**
11. Explain parameter to select to employ the new machining methods. (6 Marks) **(VTU Dec 2012)**
12. Give the broad classification of non-traditional machining processes. (6 Marks) **(VTU Dec 2012)**
13. Justify the need of unconventional manufacturing process in today's industries. (6 Marks) **(VTU June/July 2013)**
14. What are the basic factors upon which the unconventional manufacturing processes are classified? Explain. (10 Marks) **(VTU June/July 2013)**
15. Explain how the non-traditional machining processes are classified. (6 Marks) **(VTU June/July 2014)**
16. List the unconventional machining processes under mechanical energy thermal and chemical energy category. (6 Marks) **(VTU Dec 2014/Jan 2015)**
17. Differentiate between conventional (traditional) and non-traditional machining processes with examples. (6 Marks) **(VTU Dec 2014/Jan 2015)**
18. Make a comparison between traditional and non-traditional machining process in terms of cost, application, scope, machine time and limitations. (8 Marks) **(VTU Dec 2014/Jan 2015)**
19. List and explain the various factors to be considered for selection of machining processes. (7 Marks) **(VTU June/July 2015)**
20. Classify various non-traditional machining process based on energy source used with giving suitable examples. (6 Marks) **(VTU June/July 2015)**
21. Based on the various parameters of machining, compare the conventional and non-conventional

machining processes. (7 Marks) **(VTU June/July 2015)**

22. How modern machining processes are classified? (6 Marks) **(VTU June/July 2016)**

23. What are the essential physical process parameters for an efficient use of modern machining processes ? (5 Marks) **(VTU June/July 2016)**

24. Why Non-traditional machining (NTM) processes are selected for manufacturing? (4 Marks) **(VTU June/July 2016)**

MODULE-2

ULTRASONIC MACHINING (USM), ABRASIVE JET MACHINING (AJM)

SYLLABUS

Ultrasonic Machining (USM): Introduction, Equipment and material process, Effect of process parameters: Effect of amplitude and frequency, Effect of abrasive grain diameter, effect of slurry, tool & work material. Process characteristics: Material removal rate, tool wear, accuracy, surface finish, applications, advantages & limitations of USM.

Abrasive Jet Machining (AJM): Introduction, Equipment and process of material removal, process variables: carrier gas, type of abrasive, work material, stand-off distance (SOD), Process characteristics-Material removal rate, Nozzle wear, accuracy & surface finish. Applications, advantages & limitations of AJM.

Ultrasonic Machining (USM)

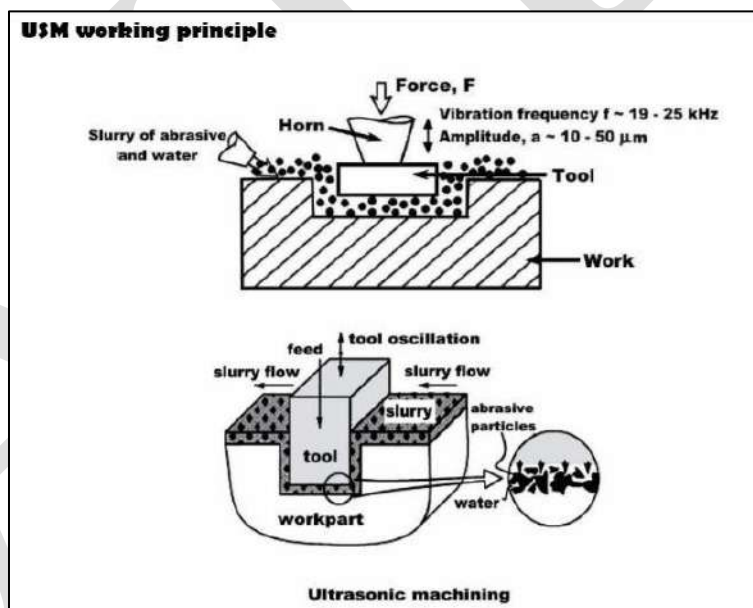
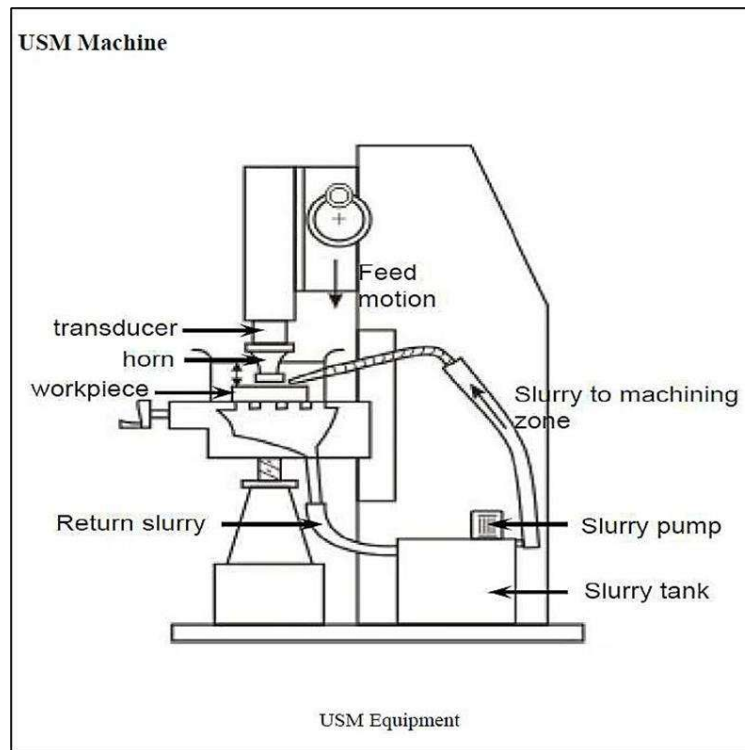
2.1 Introduction to USM

USM is mechanical material removal process or an abrasive process used to erode holes or cavities on hard or brittle work piece by using shaped tools, high frequency mechanical motion and an abrasive slurry. USM offers a solution to the expanding need for machining brittle materials such as single crystals, glasses and polycrystalline ceramics and increasing complex operations to provide intricate shapes and workpiece profiles. It is therefore used extensively in machining hard and brittle materials that are difficult to machine by traditional manufacturing processes. Ultrasonic Machining is a non-traditional process, in which abrasives contained in a slurry are driven against the work by a tool oscillating at low amplitude (25-100 μm) and high frequency (15-30 KHz): The process was first developed in 1950s and was originally used for finishing EDM surfaces. The basic process is that a ductile and tough tool is pushed against the work with a constant force. A constant stream of abrasive slurry passes between the tool and the work (gap is 25-40 μm) to provide abrasives and carry away chips. The majority of the cutting action comes from an ultrasonic (cyclic) force applied. The basic

components to the cutting action are believed to be,

- Brittle fracture caused by impact of abrasive grains due to the tool vibration
- Cavitation induced erosion
- Chemical erosion caused by slurry
- Material removal primarily occurs due to the indentation of the hard abrasive grits on the brittle work material.
- Other than this brittle failure of the work material due to indentation some material removal may occur due to free flowing impact of the abrasives against the work material and related solid-solid impact erosion
- Tool's vibration – indentation by the abrasive grits.
- During indentation, due to Hertzian contact stresses, cracks would develop just below the contact site, then as indentation progresses the cracks would propagate due to increase in stress and ultimately lead to brittle fracture of the work material under each individual interaction site between the abrasive grits and the workpiece. The tool material should be such that indentation by the abrasive grits does not lead to brittle failure.
- Thus, the tools are made of tough, strong and ductile materials like steel, stainless steel and other ductile metallic alloys.

2.2 Equipment and material process



The basic mechanical structure of an USM is very similar to a drill press. However, it has additional features to carry out USM of brittle work material. The workpiece is mounted on a vice, which can be located at the desired position under the tool using a 2 axis table. The table can further be lowered or raised to accommodate work of different thickness.

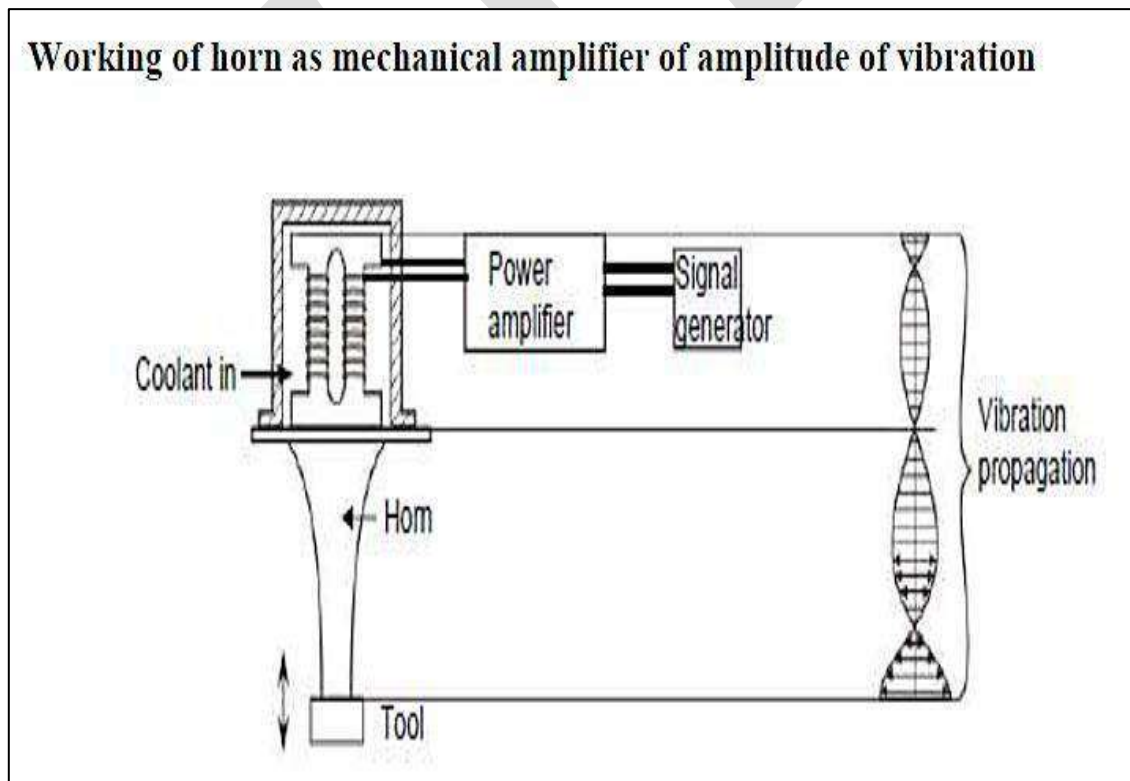
The typical elements of an USM are:

- Slurry delivery and return system
- Feed mechanism to provide a downward feed force on the tool during machining
- The transducer, which generates the ultrasonic vibration
- The horn or concentrator, which mechanically amplifies the vibration to the required amplitude of 15–50 μm and accommodates the tool at its tip.

Working of horn as mechanical amplifier of amplitude of vibration

The ultrasonic vibrations are produced by the transducer. The transducer is driven by suitable signal generator followed by power amplifier. The transducer for USM works on the following principle

- Piezoelectric effect
- Magnetostrictive effect
- Electrostrictive effect

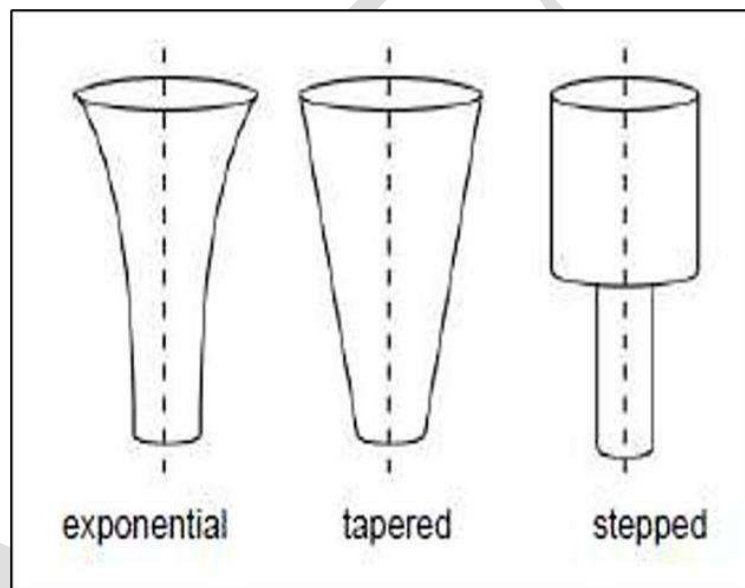


Magnetostrictive transducers are most popular and robust amongst all. Figure shows atypical magnetostrictive transducer along with horn. The horn or concentrator is a waveguide, which amplifies and concentrates the vibration to the tool from the transducer. The horn or concentrator can be of different shape like

- Tapered or conical
- Exponential
- Stepped

Machining of tapered or stepped horn is much easier as compared to the exponential one.

Figure shows different horns used in USM



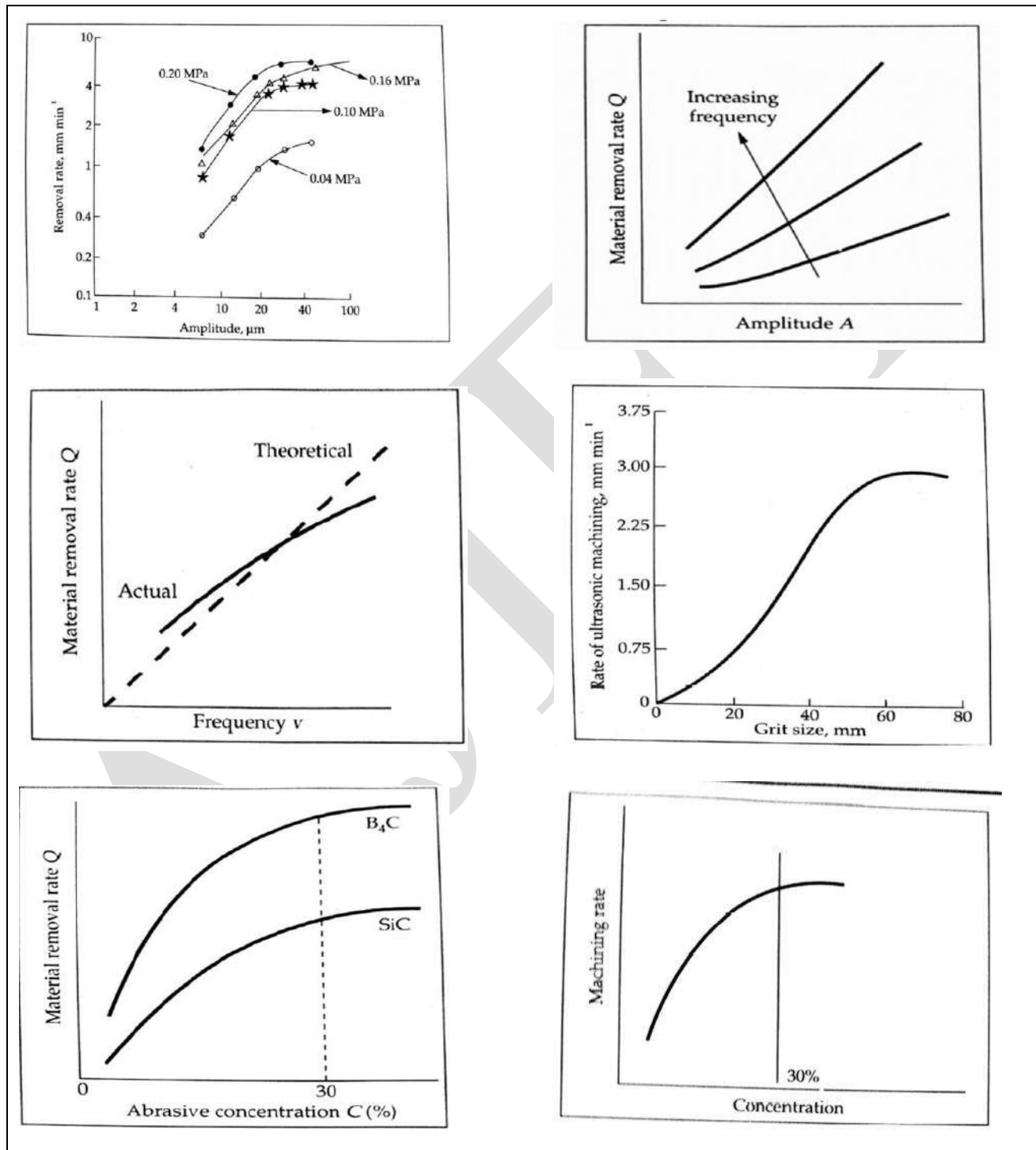
Process variables

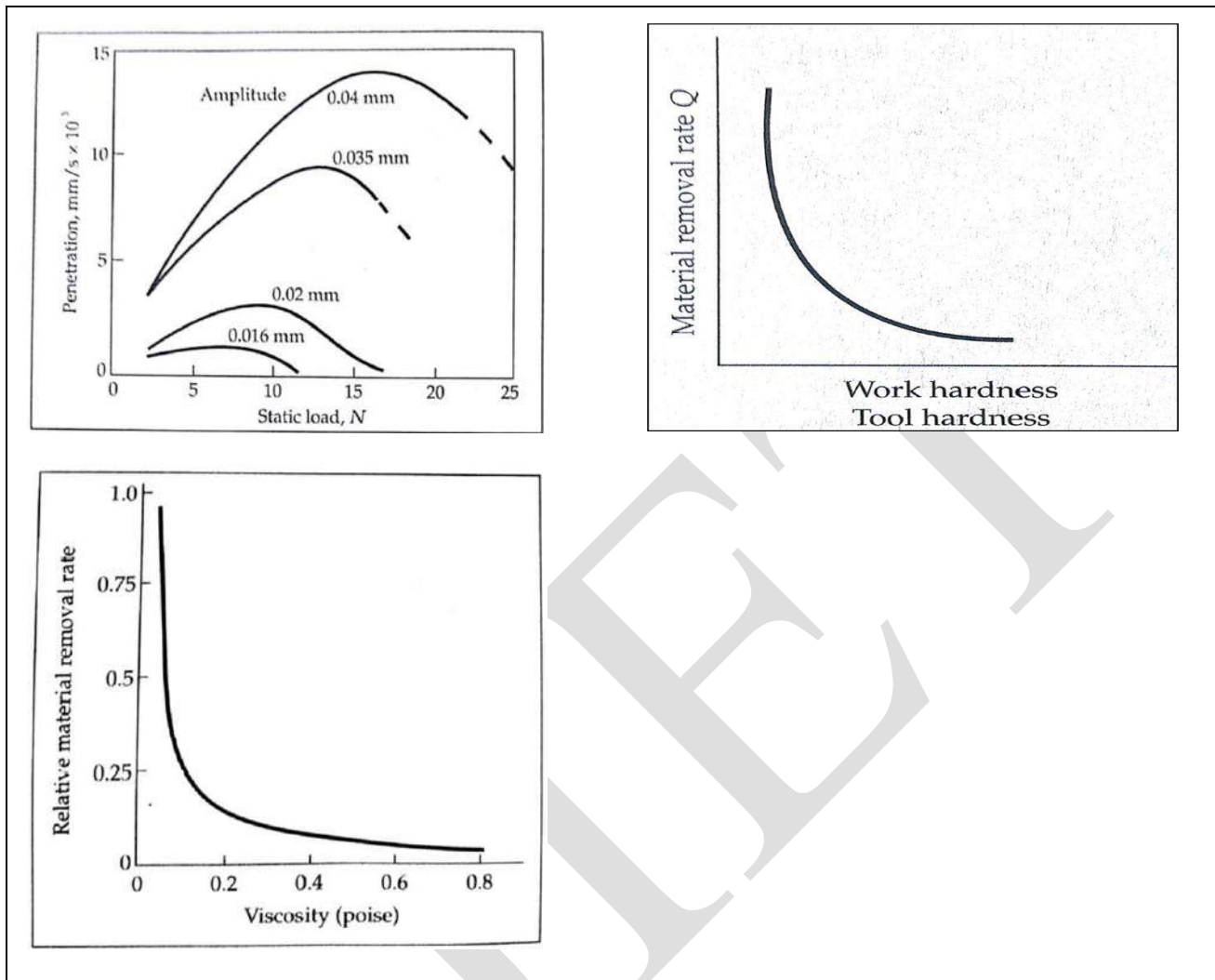
- Amplitude of vibration (a_0) – 15 – 50 μm
- Frequency of vibration (f) – 19 – 25 kHz
- Feed force (F) – related to tool dimensions
- Feed pressure (p)
- Abrasive size – 15 μm – 150 μm
- Abrasive material - Al_2O_3
 - SiC
 - B_4C
 - Boron silicarbide
 - Diamond
- Flow strength of work material
- Flow strength of the tool material
- Contact area of the tool – A
- Volume concentration of abrasive in water slurry – C

2.3 Effect of process parameters

The theoretical as well as experimental investigations on the process have shown that the MRR in the ultrasonic machining is a function of a large number of parameters such as frequency, amplitude, static force, hardness ratio (q) of the work to tool, grain size and concentration of abrasive slurry. A pictorial summary of these effects is presented in Figure. It is understood that the MRR increases with an increase in the frequency and the amplitude of ultrasonic vibrations. The MRR increases in proportion to the increase in the grain size until the grain size becomes more than the amplitude of vibrations, that is, where the particles tend to fracture. The MRR increases proportional to the one-fourth power of slurry concentration, i.e., $C^{1/4}$. However, the increasing trend continues till concentration reaches 30%, beyond which an increase in the concentration does not help. The MRR increases with an increase in the static force, but after a certain increase in the static force, the abrasive particles tend to break thereby reducing the MRR. A decrease in the hardness of workpiece

in the ratio (q) causes a rapid reduction in MRR. This indicates that the brittle work materials are machined rapidly by USM. The MRR decreases significantly with a decrease in the viscosity of the carrier liquid.





Effect of process parameters

2.4 Process Characteristics

1. Material removal rate

There are four main parameters on which the material removal rate of an Ultrasonic machine depends on:

- Workpiece:** Workpiece material properties like fracture toughness and hardness play a crucial role in determining the material removal rate in ultrasonic machining. As the hardness of the work material increases, the material removal rate decreases. Additionally, materials with higher fracture toughness tend to take longer for machining, while brittle and hard materials demonstrate better machining efficiency. Work materials with higher hardness and toughness result in more tool wear compared to softer and brittle materials, which lead to lesser tool wear.

- (b) **Tool:** The material removal rate increases with the hardness of the tool material. Different tool materials exhibit varying levels of performance, with high carbon steel tools achieving higher material removal rates due to their hardness compared to other tools like diamond tips. Additionally, the shape factor and form of the tool play a crucial role in determining the material removal rate. Tools with specific geometries, such as rectangular cross-sections or hollow tools, have been found to yield better material removal rates than others with different shapes. The design of the tool, including factors like tool form and geometry, impacts the resistance to slurry circulation and ultimately affects the material removal rate in ultrasonic machining
- (c) **Slurry:** The material removal rate increases with an increase in slurry concentration. As the concentration of the slurry rises, the material removal rate also increases, leading to a rougher surface finish. This phenomenon is attributed to the clogging of abrasive particles in the slurry, which results in a rougher surface texture. Therefore, higher slurry concentrations tend to produce rougher surfaces due to the increased material removal rate.
- (d) **Machine related factors:** The material removal rate in ultrasonic machining is influenced by various machine-related factors such as the amplitude and frequency of vibrations, grain diameter, and static load. Proper adjustment of parameters like vibration amplitude, frequency, and static load can significantly impact the material removal rate, surface finish, tool wear, and accuracy of the machining process.

2. Tool wear

The tool wear characteristics in ultrasonic machining (USM) are crucial factors that significantly impact the material removal process and overall machining performance.

- (a) **Tool Material Flexibility:** Tool materials with high flexibility have been observed to slow down the wear of abrasives, leading to improved machining efficiency.
- (b) **Machining Time:** Tool wear in ultrasonic drilling shows a linear increase with the number of holes drilled, indicating that longitudinal tool wear rises with an increase in machining time under similar conditions.
- (c) **Effect of Static Load and Depth:** Tool wear is reported to be maximum at specific static loads, and an increase in the depth of the hole drilled results in observable tool wear, subsequently leading to a decrease in the material removal rate.
- (d) **Proportional Relationship with Machining Time:** Tool wear has been found to be directly proportional to the machining time, with wear also being proportional to the previous wear of

the tool.

- (e) **Influence of Abrasive Grain Hardness:** Tool wear increases with the hardness of abrasive grains, and an increase in tool wear is observed with higher hardness and toughness of the workpiece material.
- (f) **Tool Material Properties:** Tool materials with higher hardness, fracture toughness, and abrasion resistance exhibit lower tool wear rates, highlighting the importance of selecting appropriate tool materials for efficient machining processes.

3. Accuracy

The accuracy characteristics in Ultrasonic Machining (USM) are crucial for achieving precise and high-quality machined components. Several factors influence the accuracy of USM processes are:

- (a) **Dimensional Accuracy:** The dimensional accuracy in USM refers to the closeness of the measured dimension to the actual dimension of the machined part. Factors affecting dimensional accuracy include the precision of the machine tool, accuracy of fixtures, quality of assembly parts, abrasive grit size, tool wear, transverse vibration effects, and depth of cut. The aspect ratio, which is the diameter/length ratio, influences lateral vibrations and can impact dimensional accuracy by causing oversize in machined parts.
- (b) **Form Accuracy:** Form accuracy in USM pertains to the conformity of the machined part's shape to the intended design. Factors affecting form accuracy include the precision of the machine tool, accuracy of fixtures, abrasive grit size, quality of assembly parts, tool wear, transverse vibration effects, and the aspect ratio of the machined part.
- (c) **Hole Overcut Size (HOS):** Hole overcut size is a critical parameter affecting accuracy in USM. It refers to the difference between the hole diameter at the entrance and the actual tool diameter before drilling. HOS is influenced by factors such as the diameter of the abrasive particle, grit size, and the mechanical behavior of the workpiece material. The grit size of the abrasive material plays a significant role in controlling the accuracy of holes in USM.
- (d) **Out-of-Roundness:** Out-of-roundness is a measure of deviation from a perfect circular shape in machined holes. Materials with higher hardness to modulus of elasticity ratios tend to exhibit more out-of-roundness. The rotary mode of USM has been reported to offer better performance in reducing out-of-roundness compared to conventional USM.
- (e) **Conicity:** Conicity refers to the tapering effect in machined holes, where the oversize at the entry of the hole is greater than at the exit, leading to unavoidable conicity due to tool wear. Conicity can be reduced by using specific tool materials like tungsten carbide and stainless

steel, employing internal slurry delivery systems, utilizing tools with negative tapering walls, or using fine abrasives.

- (f) **Surface Finish:** Surface finish is a critical aspect of accuracy in USM, defined by characteristics like lay, surface roughness, and waviness. Abrasive grain size plays a significant role in determining surface finish, with smaller grain sizes resulting in smoother surfaces. The surface roughness decreases as the mesh number of the grain size increases, indicating an improvement in surface finish with finer abrasive particles.

4. Surface finish

- (a) **Abrasive Grain Size:** The surface finish of the workpiece in ultrasonic machining is significantly governed by the abrasive grain size. Research findings indicate that surface roughness decreases, leading to improved surface finish, with a decrease in abrasive grain size. Finer abrasive grains result in smoother surfaces by chipping off smaller material chunks, enhancing the overall surface finish.
- (b) **Rotary vs. Conventional Ultrasonic Machining:** Experiments have shown that under the same conditions, the material removal rate in conventional ultrasonic machining is greater than in rotary ultrasonic machining. Consequently, the surface roughness of the machined surface in conventional ultrasonic machining is greater than in rotary ultrasonic machining. Rotary ultrasonic machining has been found to offer superior surface finish compared to conventional ultrasonic machining, attributed to the creation of additional stresses in the rotary mode.
- (c) **Effect of Material Properties:** Materials with high material removal rates tend to exhibit high values of surface roughness. The relationship between the volume of material removed, fracture toughness, hardness, and static load applied influences the surface finish. Materials with low values of the denominator in the relationship tend to have higher volumes of material removed, impacting the surface finish accordingly. Ultrasonic machining produces smooth surfaces in harder materials and materials with high fracture toughness, resulting in low surface roughness and a smooth surface finish.
- (d) **Tool Material and Finish:** The finish of the tool used in ultrasonic machining significantly impacts the surface finish of the workpiece. Any irregularities on the tool surface are reproduced on the workpiece, emphasizing the importance of a tool with better surface finish than required on the workpiece. Additionally, the hardness of the tool material plays a crucial role in determining the surface roughness, with high hardness tool materials producing work

surfaces with less surface roughness.

2.5 Applications of USM

- Used for machining hard and brittle metallic alloys, semiconductors, glass, ceramics, carbides etc.
- Used for machining round, square, irregular shaped holes and surface impressions.
- Machining, wire drawing, punching or small blanking dies.
- Drilling and machining cavities or holes in conductive and non-conductive materials like glass, ceramics etc.
- Used to machine hard materials like tool steel, tungsten and hard carbides.
- Threading of various glass and ceramic materials.
- Used for making tools and dies.
- Soft materials like non-ferrous metals and alloys and brittle materials can be machined.
- Removing flash and parting lines from injection moulded parts.
- Deburring and polishing plastic, nylon and Teflon components.
- To produce high quality surface.
- Hard materials and precious stones such as synthetic ruby for the preparation of jewels for watch and timer movements are successfully machined by this method.

2.6 Advantages of USM

- Any materials can be machined regardless of their electrical conductivity.
- Especially suitable for machining of brittle materials.
- Machined parts by USM possess better surface finish and higher structural integrity. USM does not produce thermal, electrical and chemical abnormal surface.
- No burrs and no distortion of workpiece.
- Hard and brittle metals can be machined.
- Stresses are minimum.

- Thin sheets can be machined.
- Investment cost is low.
- Power consumption is low.

2.7 Limitations of USM

- USM has higher power consumption and lower material-removal rates than traditional fabrication processes.
- Tool wears fast in USM.
- Machining area and depth is restraint in USM.
- Depth of holes and cavities produced are small. Usually the depth of hole is limited to 2.5 times the diameter of the tool. There is a tendency for holes to break out at the bottom owing to high amplitude of vibrations of the tool.
- Not suitable for soft workpiece material.

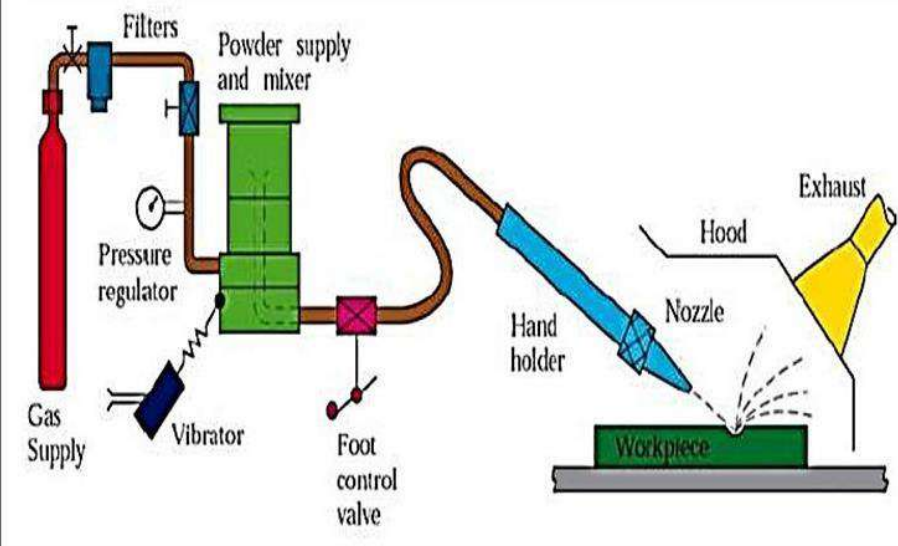
ABRASIVE JET MACHINING (AJM)

2.7 Introduction to AJM

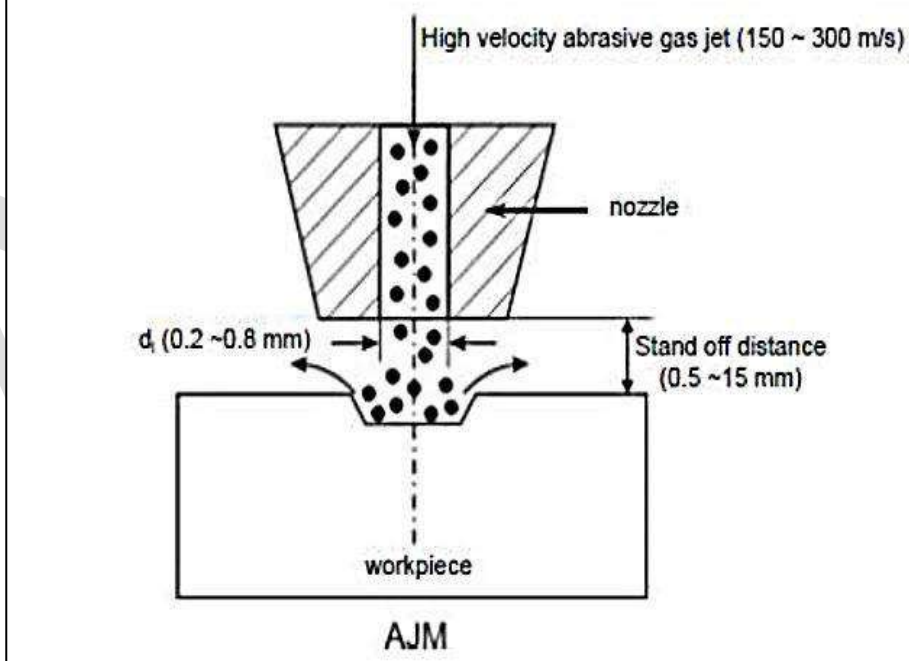
Abrasive water jet cutting is an extended version of water jet cutting; in which the water jet contains abrasive particles such as silicon carbide or aluminum oxide in order to increase the material removal rate above that of water jet machining. Almost any type of material ranging from hard brittle materials such as ceramics, metals and glass to extremely soft materials such as foam and rubbers can be cut by abrasive water jet cutting. The narrow cutting stream and computer controlled movement enables this process to produce parts accurately and efficiently. This machining process is especially ideal for cutting materials that cannot be cut by laser or thermal cut. Metallic, nonmetallic and advanced composite materials of various thicknesses can be cut by this process. This process is particularly suitable for heat sensitive materials that cannot be machined by processes that produce heat while machining.

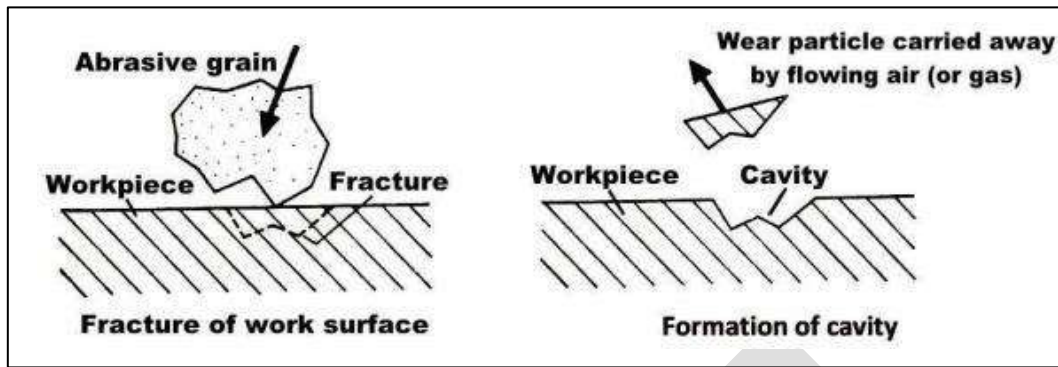
2.8 Equipment and process of material removal

AJM Equipment



Working principle





In Abrasive Jet Machining (AJM), abrasive particles are made to impinge on the work material at a high velocity. The jet of abrasive particles is carried by carrier gas or air. The high velocity stream of abrasive is generated by converting the pressure energy of the carrier gas or air to its kinetic energy and hence high velocity jet. The nozzle directs the abrasive jet in a controlled manner onto the work material, so that the distance between the nozzle and the work piece and the impingement angle can be set desirably. The high velocity abrasive particles remove the material by micro-cutting action as well as brittle of the work material. In AJM, air is compressed in an air compressor and compressed air at a pressure of Around 5 bars is used as the carrier gas. Figure also shows the other major parts of the AJM system. Gases like CO can also be used as carrier gas which may directly be issued from a gas cylinder. Generally oxygen is not used as a carrier gas. The carrier gas is first passed through a pressure regulator to obtain the desired working pressure. To remove any oil vapour or particulate contaminant the same is passed through a series of filters. Then the carrier gas enters a closed chamber known as the mixing chamber. The abrasive particles enter the chamber from a hopper through a metallic sieve. The sieve is constantly vibrated by an electromagnetic shaker. The mass flow rate of abrasive (15 gm/min) entering the chamber depends on the amplitude of vibration of the sieve and its frequency. The abrasive particles are then carried by the carrier gas to the machining chamber via an electro-magnetic on-off valve. The machining enclosure is essential to contain the abrasive and machined particles in a safe and eco-friendly manner. The machining is carried out as high velocity (200 m/s) abrasive particles are issued from the nozzle onto a work piece traversing under the jet.

Abrasive jet machining consists of :

1. Gas propulsion system:

- Supplies clean and dry air.
- Air, nitrogen and carbon dioxide to propel the abrasive particles.

- Gas may be supplied either from a compressor or a cylinder.
- In case of a compressor, air filter cum drier should be used to avoid water or oil contamination of abrasive powder.
- Gas should be non-toxic, cheap, easily available.
- It should not excessively spread when discharged from nozzle into atmosphere.
- The propellant consumption is of order of 0.008 m³/min at a nozzle pressure of 5 bar and abrasive flow rate varies from 2 to 4 g/min for fine machining and 10 to 20 g/min for cutting operation.

2. Abrasive feeder

- Required quantity of abrasive particles is supplied by abrasive feeder.
- The filleted propellant is fed into the mixing chamber where in abrasive particles are fed through a sieve.
- The sieve is made to vibrate at 50-60 Hz and mixing ratio is controlled by the amplitude of vibration of sieve.
- The particles are propelled by carrier gas to a mixing chamber. Air abrasive mixture moves further to nozzle. The nozzle imparts high velocity to mixture which is directed at work piece surface.

3. Machining chamber

- It is well closed so that concentration of abrasive particles around the working chamber does not reach to the harmful limits.
- Machining chamber is equipped with vacuum dust collector. Special consideration should be given to dust collection system if the toxic material are being machined.

4. AJM nozzle

- AJM nozzle is usually made of tungsten carbide or sapphire (usual life - 300 h for sapphire, 20 to 30 h for WC) which has resistance to wear.
- The nozzle is made of either circular or rectangular cross section and head can be straight, or at a right angle.
- It is so designed that loss of pressure due to the bends, friction etc is minimum possible.
- With increase in wear of a nozzle, the divergence of jet stream increases resulting in more stray cutting and high inaccuracy.

5. Abrasives

- Aluminum oxide (Al_2O_3) Silicon carbide (SiC) Glass beads, crushed glass and sodium bicarbonate are some of abrasives used in AJM.
- Selection of abrasives depends on MRR , type of work material , machining accuracy.

2.9 Process parameters and Machining characteristics.

The process parameters are listed below:

• Type of Abrasive

- Material – Al_2O_3 / SiC / glass beads
- Shape – irregular / spherical
- Size – $10 \sim 50 \mu\text{m}$
- Mass flow rate – $2 \sim 20 \text{ gm/min}$
- The choice of abrasive depends on the type of machining operation, for example, roughing, finishing, etc., Work material and cost. The abrasive should have a sharp and irregular shape and be fine enough to remain suspended in the carrier gas and should also have excellent flow characteristics. The abrasives used for cutting are aluminum oxide and silicon carbide, whereas sodium bicarbonate, dolomite, glass beads. Etc. are used for cleaning, etching, deburring and polishing. Reuse of abrasives is not recommended because not only does its cutting ability decrease, but contamination also clogs the orifice of the nozzle.

• Carrier gas

- Composition – Air, CO_2 , N_2
- Density – Air $\sim 1.3 \text{ kg/m}^3$
- Velocity – $500 \sim 700 \text{ m/s}$
- Pressure – $2 \sim 10 \text{ bar}$
- Flow rate – $5 \sim 30 \text{ lpm}$
- Carrier gas, to be used in AJM, must not flare excessively when discharged from the nozzle into the atmosphere. Further, the gas should be nontoxic, cheap, easily available and capable of being dried and cleaned without difficulty. The gases that can be used are air, carbon dioxide or nitrogen. Air is most widely used owing to easy availability and little cost. All abrasive powders supplied by the manufacturers can be run with clean shop air, provided air filters have been installed in the air lines.

• Abrasive Jet

- Velocity – 100 ~ 300 m/s
- Mixing ratio – mass flow ratio of abrasive to gas
- Stand-off distance – 0.5 ~ 5 mm
- Impingement Angle – 60° - 90°

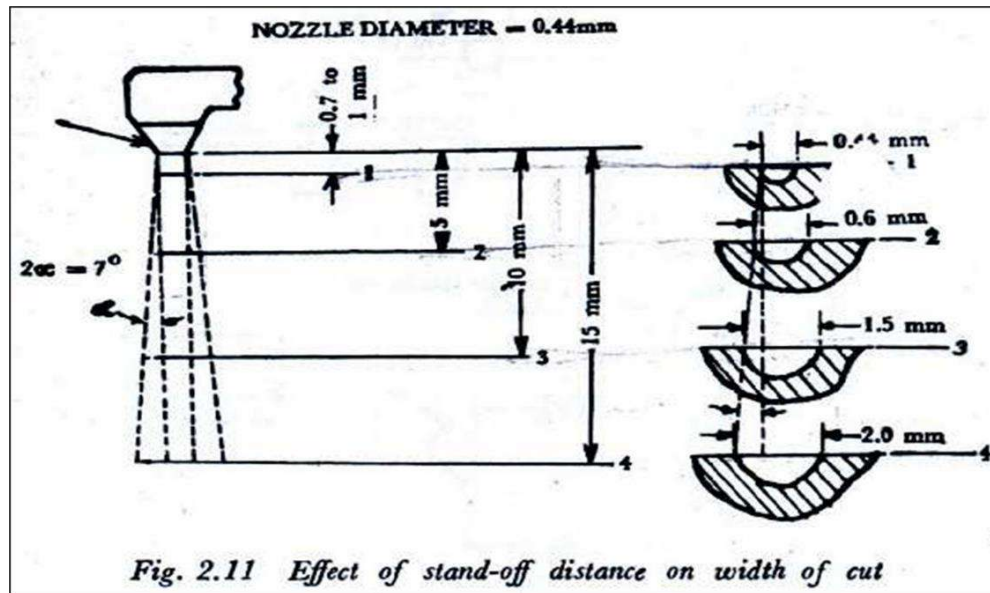
• Abrasive Grain size

- Finer grains are less irregular in shape and hence possess lesser cutting ability. Moreover, finer grains tend to stick together and choke the nozzle. The most favourable grain sizes range from 10 to 50 μm .
- Coarse are recommended for cutting, whereas finer grains are useful in polishing, deburring, etc. For the erosion of glass by silicon carbide (grain size 25 μm), the minimum jet velocity has been found to be around 150 m/s.

Abrasives	Grain Sizes	Application
Aluminum oxide(Al_2O_3)	12, 20, 50 microns	Good for cleaning, cutting and deburring
Silicon carbide (SiC)	25,40 micron	Used for similar application but for hard material
Glass beads	0.635 to 1.27mm	Gives matte finish
Dolomite	200 mesh	Etching and polishing
Sodium bi carbonate	27 micros	Cleaning, deburring and cutting of soft material Light finishing below 50°C

• Stand-off Distance (SOD)

- Stand off distance is defined as the distance between the face of the nozzle and the surface of the work. A large SOD results in flaring of jet which leads to poor accuracy.
- It is clear from figure that MRR increase with nozzle tip distance or Stand off distance up to certain distance and then decreases. Penetration rate also increases with SOD and then decreases.
- Decrease in SOD improves accuracy, decreases kerf width, and reduces taper in machined groove. However light operation like cleaning, frosting etc are conducted with large SOD.

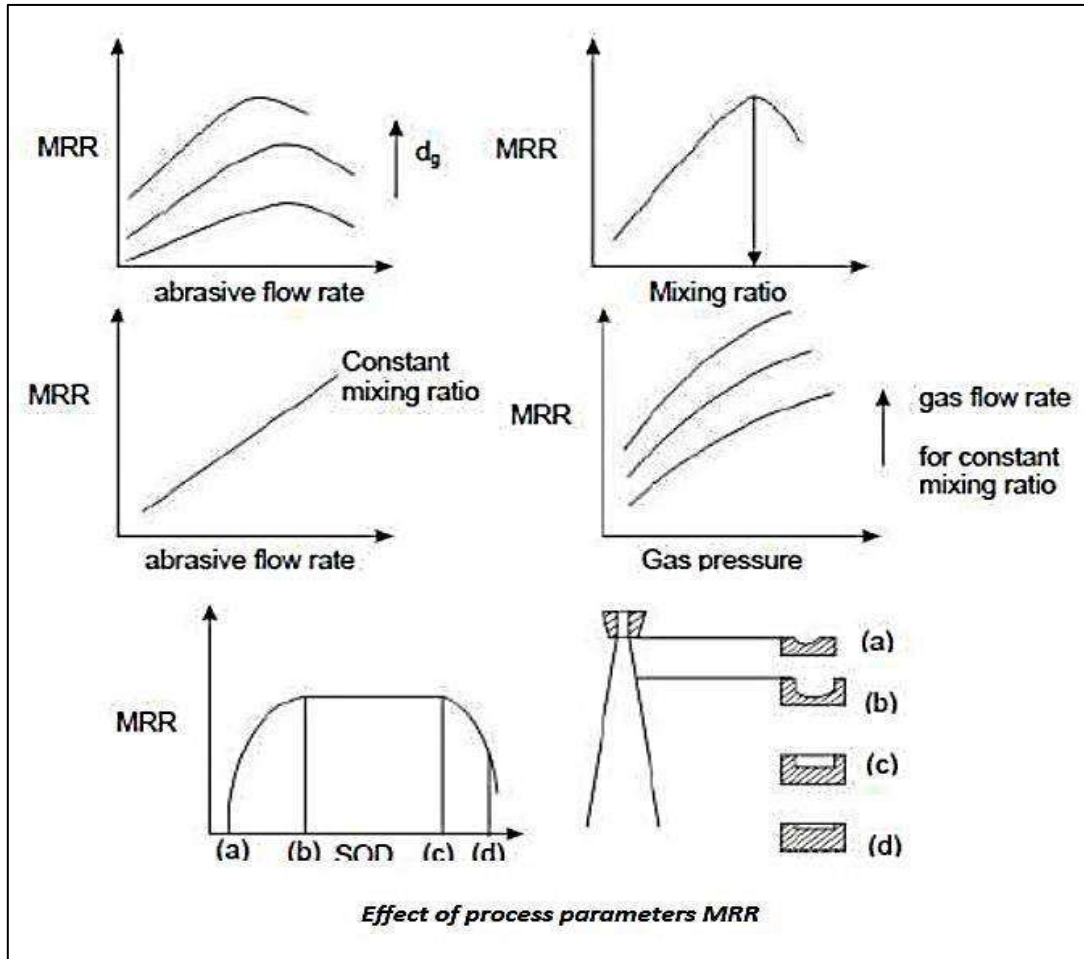


• Nozzle

- Material – WC / sapphire
- Diameter – (Internal) 0.2 ~ 0.8 mm
- Life – 10 ~ 300 hours

The important machining characteristics in AJM are:

- The material removal rate (MRR) mm³/min or gm/min
- The machining accuracy
- The life of the nozzle



Parameters of Abrasive Jet machining (AJM) are factors that influence its Metal Removal Rate (MRR). In a machining process, Metal Removal Rate (MRR) is the volume of metal removed from a given work piece in unit time. The following are some of the important process parameters of abrasive jet machining:

1. Abrasive mass flow rate
2. Nozzle tip distance
3. Gas Pressure
4. Velocity of abrasive particles
5. Mixing ratio
6. Abrasive grain size

Abrasive mass flow rate:

Mass flow rate of the abrasive particles is a major process parameter that influences the metal removal

rate in abrasive jet machining. In AJM, mass flow rate of the gas (or air) in abrasive jet is inversely proportional to the mass flow rate of the abrasive particles. Due to this fact, when continuously increasing the abrasive mass flow rate, Metal Removal Rate (MRR) first increases to an optimum value (because of increase in number of abrasive particles hitting the work piece) and then decreases. However, if the mixing ratio is kept constant, Metal Removal Rate (MRR) uniformly increases with increase in abrasive mass flow rate.

Nozzle tip distance:

Nozzle Tip Distance (NTD) is the gap provided between the nozzle tip and the work piece. Up to a certain limit, Metal Removal Rate (MRR) increases with increase in nozzle tip distance. After that limit, MRR remains constant to some extent and then decreases. In addition to metal removal rate, nozzle tip distance influences the shape and diameter of cut. For optimal performance, a nozzle tip distance of 0.25 to 0.75 mm is provided.

Gas pressure:

Air or gas pressure has a direct impact on metal removal rate. In abrasive jet machining, metal removal rate is directly proportional to air or gas pressure.

Velocity of abrasive particles:

Whenever the velocity of abrasive particles is increased, the speed at which the abrasive particles hit the work piece is increased. Because of this reason, in abrasive jet machining, metal removal rate increases with increase in velocity of abrasive particles.

Mixing ratio:

Mixing ratio is a ratio that determines the quality of the air-abrasive mixture in abrasive Jet Machining (AJM). It is the ratio between the mass flow rate of abrasive particles and the mass flow rate of air (or gas). When mixing ratio is increased continuously, metal removal rate first increases to some extent and then decreases.

Abrasive grain size:

Size of the abrasive particle determines the speed at which metal is removed. If smooth and fine surface finish is to be obtained, abrasive particle with small grain size is used. If metal has to be removed rapidly, abrasive particle with large grain size is used.

2.10 Process characteristics

1. Material removal rate

The factors that affect the Material Removal Rate (MRR) in Abrasive Jet Machining (AJM) include:

- (a) **Type of Abrasive:** The type of abrasive used in AJM influences the MRR, with harder abrasives typically resulting in higher MRR rates compared to softer abrasives. Harder abrasives can achieve higher material removal rates due to their increased cutting efficiency.
- (b) **Abrasive Grain Size:** The size of the abrasive particles impacts the MRR in AJM. Larger abrasive particles create bigger indentations and lead to higher MRR rates, while smaller particles provide smoother surface finishes but lower MRR rates. The grain size of the abrasive directly affects the material removal efficiency in the machining process..
- (c) **Abrasive Flow Rate:** The mass flow rate of the abrasive material is another crucial factor affecting the MRR in AJM. The MRR is directly proportional to the abrasive flow rate, meaning that higher flow rates result in increased material removal rates during the machining process..
- (d) **Gas Pressure:** The pressure of the carrier gas used in AJM affects the MRR by converting into kinetic energy. Higher gas pressure leads to higher jet velocities, resulting in increased material removal rates. Optimal gas pressure is essential for achieving high MRR in abrasive jet machining operations.
- (e) **Nozzle Design and Material:** The design and material of the nozzle play a significant role in determining the MRR in AJM. Factors such as the inner diameter of the nozzle, wear of the nozzle, and the material used for the nozzle impact the jet velocity and efficiency of material removal. Proper selection and maintenance of the nozzle are crucial for achieving desired MRR levels.

2. Nozzle wear

Factors that affect the nozzle wear in Abrasive Jet Machining (AJM) include:

- (a) **Mixing Tube Material:** The material of the mixing tube, where the abrasive particles are mixed, accelerated, and focused with high-pressure waterjet, plays a crucial role in nozzle wear. The wear of the mixing tube is a significant contributor to nozzle wear in AJM processes.
- (b) **Nozzle Geometry:** The geometry of the nozzle, including parameters like length, inlet angle, diameter, and orifice diameter, influences the wear characteristics of the nozzle in AJM.

Variations in these geometrical factors can impact the wear patterns and longevity of the nozzle during the machining process.

- (c) **Abrasive Particle Size:** The size of the abrasive particles used in AJM affects the wear of the nozzle. Larger abrasive particles can lead to more significant wear on the nozzle, impacting its performance and longevity. Proper selection of abrasive particle size is essential to minimize wear on the nozzle.
- (d) **Water Jet Characteristics:** The characteristics of the water jet, such as pressure and flow rate, can affect the wear of the nozzle in AJM. High-pressure water jets can accelerate wear on the nozzle, emphasizing the importance of controlling water jet parameters to mitigate wear effects.
- (e) **Nozzle Material Properties:** The material properties of the nozzle, including hardness and toughness, are crucial factors influencing nozzle wear in AJM. Nozzle materials with high hardness and toughness are essential for effective performance and extended nozzle lifetime. Proper material selection is key to reducing wear and improving nozzle durability.

3. Accuracy

The factors that affect the accuracy of Abrasive Jet Machining (AJM) include:

- (a) **Stand-Off Distance (SOD):** The gap between the nozzle tip and the work surface, known as the Stand-Off Distance, significantly influences the accuracy of machined profiles in AJM. Deviations in SOD can lead to dimensional errors in the machined features, affecting the depth and width of the channels created.
- (b) **Abrasive Grain Size:** The size of the abrasive particles used in AJM plays a crucial role in determining the surface finish and accuracy of the machining process. Larger abrasive particles can create larger cavities or indentations, while finer abrasives can result in better surface finishes. Inaccurate selection of abrasive size can lead to variations in surface roughness, tolerance, and corner radius, impacting accuracy.
- (c) **Gas Pressure and Flow Rate:** The carrier gas pressure and flow rate in AJM determine the velocity of the abrasive jet. Variations in gas pressure or flow rate can affect the cutting accuracy by influencing the jet velocity and material removal rate, leading to inaccuracies in the machining process.
- (d) **Nozzle Diameter and Wear:** The diameter of the nozzle and its wear over time are critical factors affecting the accuracy of AJM. As the nozzle wears down, its diameter increases, impacting the kerf width of the machined features. Increased wear can lead to wider kerfs

and inaccurate machining results.

- (e) **Impingement Angle:** The impingement angle of the abrasive jet on the work surface plays a role in determining the depth and width of the machined slots or holes. Deviations from the optimal 90-degree impingement angle can result in improper cutting, affecting the accuracy of the machining process.
- (f) **Material Properties:** The homogeneity of the material being machined can also impact the accuracy of AJM. Variations in material properties such as surface treatment, residual stress, and surface asperities can lead to inconsistencies in material removal rates and inaccurate machining features.

4. Surface finish

Factors that affect the surface finish of Abrasive Jet Machining (AJM) include:

- (a) **Type of Abrasive:** The type of abrasive used in AJM impacts the surface finish. Different abrasive particles have varying hardness levels, where harder abrasives can achieve higher MRR but may produce rougher surfaces, while softer abrasives can crumble when striking hard surfaces, affecting the surface finish.
- (b) **Abrasive Grain Size:** The size of the abrasive particles influences the surface finish in AJM. Larger abrasive particles create bigger indentations, leading to rougher surfaces and potentially higher MRR rates. In contrast, smaller abrasive particles provide smoother surface finishes but may result in lower MRR rates.
- (c) **Variation in Grain Size:** The consistency of abrasive grain size within the overall mass affects the surface quality and MRR in AJM. Lesser variation in grain size leads to better predictions for surface quality and machining efficiency.
- (d) **Impingement Angle:** The impingement angle, which is the operating angle between the work surface and the jet diameter, impacts the surface finish in AJM. Maintaining an optimal impingement angle between $60^\circ - 90^\circ$ can influence the depth of cut and the resulting surface finish.
- (e) **Stand-Off Distance (SOD):** The Stand-Off Distance in AJM affects the surface finish by influencing the jet diameter. Higher SOD results in a wider jet but lower velocity, potentially affecting the surface quality. Conversely, a small SOD may not provide sufficient passage for used grits to leave the machining zone, impacting the surface finish.
- (f) **Work Material Properties:** The properties of the work material, such as surface roughness, homogeneity, and stress concentration, can affect the surface finish in AJM. High surface

roughness can lead to inaccuracies in the cut, while stress concentration may reduce the MRR and surface quality.

2.11 Applications of AJM

- Abrasive water jet cutting is highly used in aerospace, automotive and electronics industries.
- In aerospace industries, parts such as titanium bodies for military aircrafts, engine components (aluminium, titanium, heat resistant alloys), aluminium body parts and interior cabin parts are made using abrasive water jet cutting.
- In automotive industries, parts like interior trim (head liners, trunk liners, door panels) and fibre glass body components and bumpers are made by this process. Similarly, in electronics industries, circuit boards and cable stripping are made by abrasive water jet cutting.



Figure: Steel gear and rack cut with an abrasive water jet

2.12 Advantages of AJM

- In most of the cases, no secondary finishing required
- No cutter induced distortion
- Low cutting forces on workpieces
- Limited tooling requirements

- Little to no cutting burr
- Typical finish 125-250 microns
- Smaller kerf size reduces material wastages
- No heat affected zone
- Localises structural changes
- No cutter induced metal contamination
- Eliminates thermal distortion
- No slag or cutting dross
- Precise, multi plane cutting of contours, shapes, and bevels of any angle

2.13 Limitations of AJM

- Cannot drill flat bottom
- Cannot cut materials that degrades quickly with moisture
- Surface finish degrades at higher cut speeds which are frequently used for rough cutting
- The major disadvantages of abrasive water jet cutting are high capital cost and high
- Noise levels during operation

QUESTION BANK

Ultrasonic machining (USM)

1. With a neat sketch explain Ultrasonic machining, equipment and mechanism of metal removal.
2. What are the parameters and characteristics of Ultrasonic machining.
3. What are the applications of Ultrasonic machining?
4. Write a note on abrasive and liquid media in USM. (10 Marks) **(VTU June/July 2011)**
5. Discuss the influence of the following parameters on USM process: i) Amplitude and frequency of vibration. ii) Grain size. iii) Slurry. (10 Marks) **(VTU June/July 2011)**
6. Write a note on process capability of USM. (5 Marks) **(VTU June/July 2011)**
7. With a neat sketch, explain the working principle of ultrasonic machining process. (8 Marks) **(VTU June 2012)**
8. Explain the effect of different process parameters on machining performance in USM process. (10 Marks) **(VTU June 2012)**

9. What are the advantages, disadvantages and application of USM process? (10 Marks) **(VTU June 2012)**
10. Explain USM process with the required figure of the set up and a magnified view at tool tip/workpiece. (10 Marks) **(VTU Dec 2012)**
11. Explain with help of neat sketch the working principle ultrasonic machining process, and also mention its advantages. (10 Marks) **(VTU June/July 2013)**
12. Explain how various process parameters influence the material removal rate in ultrasonic machining process. (10 Marks) **(VTU June/July 2013)**
13. Explain with sketch the working principle ultrasonic machining process. (8 Marks) **(VTU June/July 2014)**
14. Explain with sketch, different tool feeding mechanisms in ultrasonic machining. (8 Marks) **(VTU June/July 2014)**
15. Explain different parameters which affect ultrasonic machining. (6 Marks) **(VTU June/July 2014)**
16. What are the advantages and disadvantages of Ultrasonic machining ? (6 Marks) **(VTU Dec 2014/Jan 2015)**
17. Explain with graph the effect of various parameters on material removal rate (MRR) in USM process. (10 Marks) **(VTU June/July 2015)**
18. With neat sketch, explain the main elements of ultrasonic machining process. (10 Marks) **(VTU June/July 2015)**
19. Explain with graph the effect of various parameters on material removal rate (MRR) in USM process. (10 Marks) **(VTU June/July 2015)**
20. With neat sketch, explain the main elements of ultrasonic machining process. (10 Marks) **(VTU June/July 2015)**
21. Explain with graph the effect of various parameters on material removal rate (MRR) in USM process. (10 Marks) **(VTU June/July 2015)**
22. With neat sketch, explain the main elements of ultrasonic machining process. (10 Marks) **(VTU June/July 2015)**
23. Explain with neat diagram construction and working of USM processes. (10 Marks) **(VTU June/July 2016)**

24. Explain the following parameters with respect to USM: i) Effect of amplitude and frequency of vibration. ii) Effect of grain diameter. iii) Effect of applied static load. iv) Effect of slurry. (10 Marks) **(VTU June/July 2016)**

Abrasive jet machining (AJM)

1. What are the advantages and disadvantages of Abrasive jet machining?
2. With a neat sketch explain Abrasive jet machining, equipment and mechanism of metal removal.
3. What are the parameters and characteristics of Abrasive jet machining.
4. What are the applications of Abrasive jet machining?
5. How does the following parameters affect MRR in abrasive jet machining? i) Nozzle gap distance. ii) Abrasive size. iii) Nozzle pressure. (12 Marks) **(VTU June/July 2011)**
6. Write a note on abrasives used in AJM with examples. (8 Marks) **(VTU June/July 2011)**
7. List the variables that influence the rate of metal removal and accuracy of machining in abrasive jet machining. Explain any three. (10 Marks) **(VTU Dec 2011)**
8. Sketch a explain AJM process. (10 Marks) **(VTU June 2012)**
9. During AJM process the mixing ratio is 0.2. calculate the mass ratio if the ratio of density of abrasive and density of carrier gas is 20. (4 Marks) **(VTU June 2012)**
10. Write a note on an abrasive slurry used in AJM indicating types of abrasive, and their properties, sizes used and liquid media with functions and characteristics. (10 Marks) **(VTU Dec 2012)**
11. List the variables in AJM. Explain any four variables. (10 Marks) **(VTU Dec 2012)**
12. Explain how the following parameters influence the metal removal rate in abrasive jet machining process: i) Nozzle tip distance. ii) Velocity of abrasive. iii) Abrasive flow rate. iv) Gas pressure. (10 Marks) **(VTU June/July 2013)**
13. Explain the desired properties of abrasive materials used in abrasive jet machining. (5 Marks) **(VTU June/July 2013)**
14. Explain with block diagram the principle of operation of abrasive jet machining. (6 Marks) **(VTU June/July 2014)**
15. Explain the different variables that influence the rate of material removal and accuracy in abrasive jet machining. (8 Marks) **(VTU June/July 2014)**

16. How does ASM differ from conventional sand blasting process. (4 Marks) **(VTU Dec 2014/Jan 2015)**
17. What are the different types of abrasives used in ASM? (4 Marks) **(VTU Dec 2014/Jan 2015)**
18. State and explain the working and principle of abrasive jet machining. (12 Marks) **(VTU Dec 2014/Jan 2015)**
19. Draw the schematic diagram of abrasive jet machining and explain working principle. (8 Marks) **(VTU June/July 2015)**
20. List the applications of Abrasive jet machining. (4 Marks) **(VTU June/July 2015)**
21. Draw the schematic diagram of Abrasive Jet Machining (AJM). Explain its construction and working. (6 Marks) **(VTU June/July 2016)**
22. List and explain the variables used in AJM. (12 Marks) **(VTU June/July 2016)**

MODULE-3

ELECTROCHEMICAL MACHINING (ECM)

CHEMICAL MACHINING (CHM)

SYLLABUS

ELECTROCHEMICAL MACHINING (ECM)

Introduction, Principle of electro chemical machining: ECM equipment, elements of ECM operation, Chemistry of ECM. ECM Process characteristics: Material removal rate, accuracy, surface finish. Process parameters: Current density, Tool feed rate, Gap between tool & work piece, velocity of electrolyte flow, type of electrolyte, its concentration temperature and choice of electrolytes. ECM Tooling: ECM tooling technique & example, Tool & insulation materials. Applications ECM: Electrochemical grinding and electrochemical honing process. Advantages, disadvantages and application of ECG, ECH.

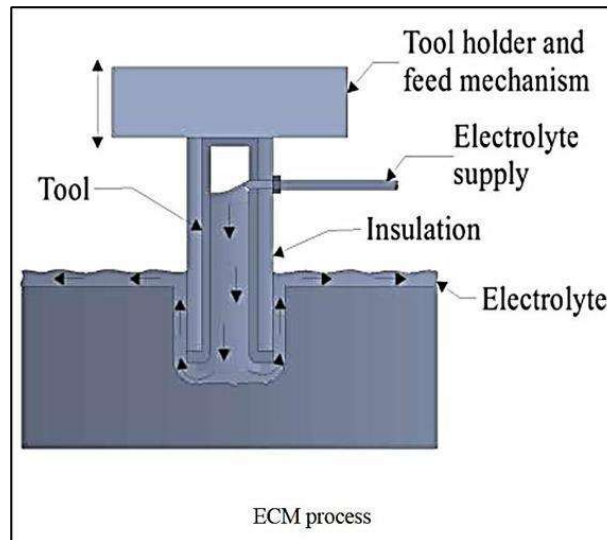
CHEMICAL MACHINING (CHM)

Elements of the process: Resists (maskants), Etchants. Types of chemical machining process, chemical blanking process, chemical milling process. Process characteristics of CHM: material removal rate, accuracy, surface finish, advantages, limitations and applications of chemical machining process.

ELECTROCHEMICAL MACHINING (ECM)

3.1 Introduction to ECM

Electrochemical machining (ECM) is a metal-removal process based on the principle of reverse electroplating. In this process, particles travel from the anodic material (workpiece) toward the cathodic material (machining tool). A current of electrolyte fluid carries away the depleted material before it has a chance to reach the machining tool. The cavity produced is the female mating image of the tool shape.



Similar to EDM, the workpiece hardness is not a factor, making ECM suitable for machining difficult-to-machine materials. Difficult shapes can be made by this process on materials regardless of their hardness. A schematic representation of ECM process is shown in Figure. The ECM tool is positioned very close to the workpiece and a low voltage, high amperage DC current is passed between the workpiece and electrode.

3.2 ECM Equipment

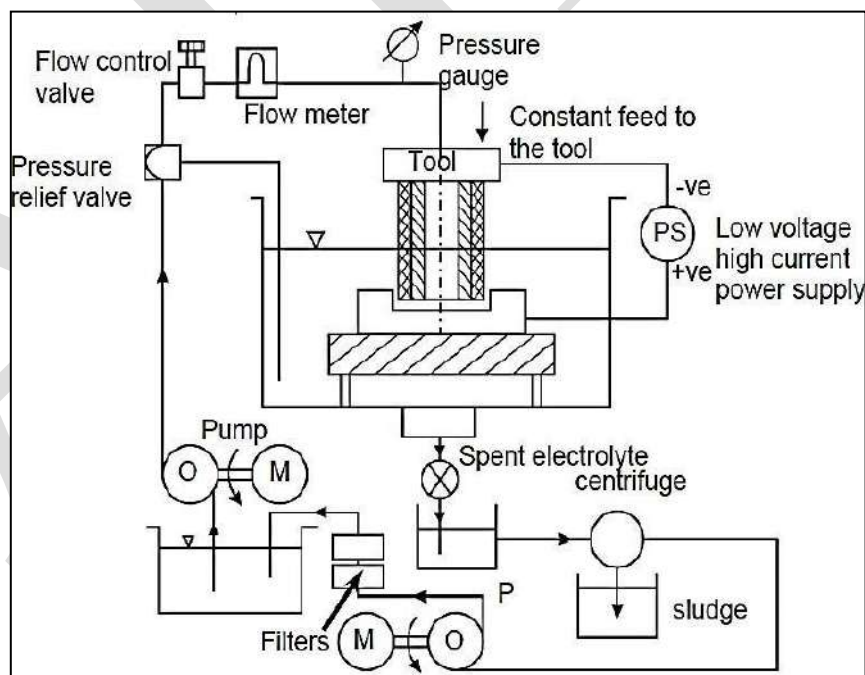


Fig. Electrochemical Drilling Unit

The electrochemical machining system has the following modules:

1. Power supply

During ECM, a high value of direct current (may be as high as 40000 A) and a low value of electric potential (in range of 5-25 V) across IEG (Interelectrode gap) is desirable. The highest current density achieved so far is around 20,000 A/cm². Hence, with the help of a rectifier and a transformer, three phase AC is converted to a low voltage, high current DC. Silicon controlled rectifier (SCRs) are used both for rectification as well as for voltage regulation because of their rapid response to the changes in the process load and their compactness. Voltage regulation of $\pm 1\%$ is adequate for most of the precision ECM works. However, lack of process control, equipment failure, operator's error, and similar other reasons may result in sparking between tool and work. The electrical circuitry detects these events and power is cut off (using the device like SCRs) within 10 microseconds to prevent the severe damage to the tool and work. In case of precision works even a small damage to an electrode is not acceptable. It may be minimized by using a bank of SCRs placed across the DC input to ECM machine.

2. Electrolyte supply and cleaning system

The electrolyte supply and cleaning system consisting of a pump, filter, piping's, control valves, heating or cooling coils, pressure gauges, and a storage tank (or reservoir). Electrolyte supply ports may be made in the tool, work or fixture, depending upon the requirement of the mode of electrolyte flow. Small inter electrode gap, usually smaller than 1mm, should be maintained for achieving High MRR and high accuracy. For this purpose, smooth flow of electrolyte should be maintained and any blockade of such a small gap by particles carried by electrolyte, should be avoided. Hence, electrolyte cleanliness is imperative. It is normally done with the help of filters made of SS steel, Monel or any other anticorrosive material. It should be ensured that the piping system does not introduce any foreign material like corroded particles, scale or pieces of broken seal material. Piping system is therefore made of SS steel, Glass fibre reinforced plastic (GFRP), plastic lined MS or similar other anti-corrosive material. The required minimum capacity of electrolyte tank is 500 gallons for each 10000 A of current. ECM is supposed to machine different metals and alloys at optimum machining conditions and with varying requirements of accuracy, surface texture, etc. Under such situations, a single tank system is not recommended because of loss of time and wastage of electrolyte during drilling cleaning, mixing or filling of new electrolyte in the tank. It results in

higher cost and poor accuracy of electro chemically machined surface and also poor control of operating conditions. More than one tank therefore, can be used and their number would depend upon the range of electrolytes needed to meet the work load.

3. Tool and Tool feed system

- Use of anti-corrosive material for tools and fixtures is important because they are required for a long period of time to operate in the corrosive environment of electrolyte. Easy machining of tool material is equally important because dimensional accuracy and surface finish of the tool directly affect the work piece accuracy and surface finish. Aluminium, Brass, Bronze, copper, carbon, stainless steel and monel are a few of the material used for this purpose. Further, those areas on the tool where ECM action is not required, should be insulated. For example, lack of insulation on the sides of die sinking tool causes unwanted machining of work and results in a loss of accuracy of the machined work piece. Use of non – corrosive and electrically non conducting material for making fixtures is recommended. Also, the fixtures and tools should be rigid enough to avoid vibration or deflection under the high hydraulic forces to which they are subjected. The tools (electrodes) are also part of the machine & should be:
 - A good conductor of electricity.
 - Rigid enough to take up the load due to fluid pressure.
 - Chemically inert to the electrolyte.
 - Easily machinable to make it in the desired shape.

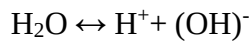
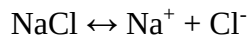
4. Work piece and Work holding system

Only electrically conductive material can be machined by this process. The chemical properties of anode (work) material largely govern the material removal rate (MRR). Work holding devices are made of electrically nonconductive materials having good thermal stability, and low moisture absorption properties, For Example, graphite fibres reinforced plastics, plastics, Perspex, etc., are the materials used for fabricating the work holding device.

3.3 Chemistry of ECM

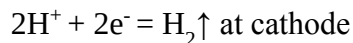
During ECM, there will be reactions occurring at the electrodes i.e. at the anode or workpiece and at the cathode or the tool along with within the electrolyte Let us take an example of machining of low carbon steel which is primarily a ferrous alloy mainly containing iron. For electrochemical machining

of steel, generally a neutral salt solution of sodium chloride (NaCl) is taken as the electrolyte. The electrolyte and water undergoes ionic dissociation as shown below as potential difference is applied

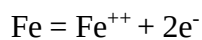


As the potential difference is applied between the work piece (anode) and the tool (cathode), the positive ions move towards the tool and negative ions move towards the workpiece.

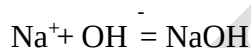
Thus the hydrogen ions will take away electrons from the cathode (tool) and from hydrogen gas as:



Similarly, the iron atoms will come out of the anode (work piece) as:



Within the electrolyte iron ions would combine with chloride ions to form iron chloride and similarly sodium ions would combine with hydroxyl ions to form sodium hydroxide



In practice FeCl_2 and $\text{Fe}(\text{OH})_2$ would form and get precipitated in the form of sludge. In this manner it can be noted that the work piece gets gradually machined and gets precipitated as the sludge. Moreover there is not coating on the tool, only hydrogen gas evolves at the tool or cathode. Figure below depicts the electro-chemical reactions schematically. As the material removal takes place due to atomic level dissociation, the machined surface is of excellent surface finish and stress free.

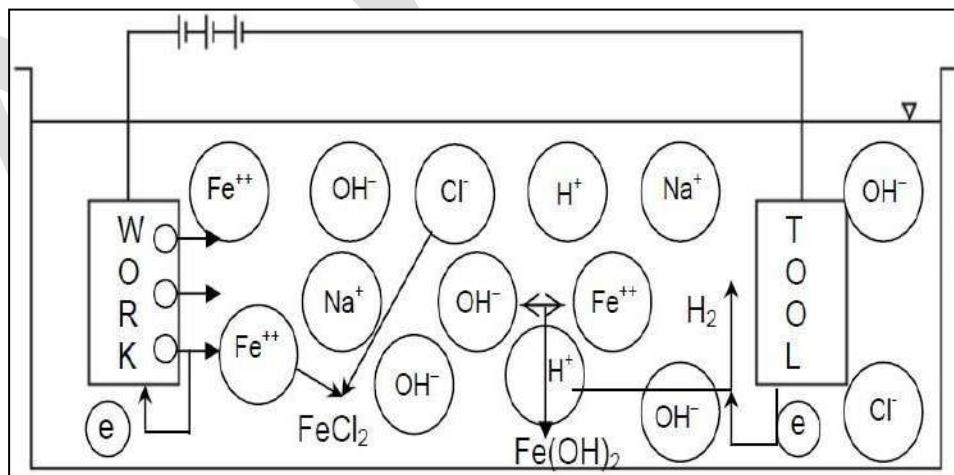
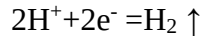
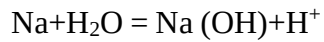
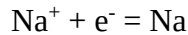
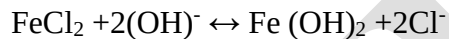
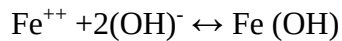
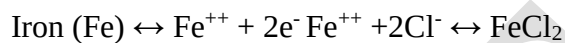


Fig. Schematic representation of electrochemical process

Summary of cathode and anode reaction is given below:

Cathode Reaction

It shows that there is no deposition on tool but only gas is formed, whereas, in cathode in machining an iron.

Anode Reaction

It shows that metal (work piece) i.e., Fe goes into solution and hence machined to produce reaction products as iron chloride and iron-hydroxide as a precipitate. Interesting part is that the removal is an atom by atom, resulting in higher surface finish with stress and crack free surface, and independent of the hardness of work material.

Smaller the interelectrode gap (IEG) the gap, greater will be the current flow because resistance decreases and higher will be rate of metal removal from the anode. Higher current density, in small spacing (usually about 0.5mm or less), promotes rapid generation of reaction products.

The voltage is required to be applied for the electrochemical reaction to proceed at a steady state. That voltage or potential difference is around 2 to 30 V. The applied potential difference, however, also overcomes the following resistances or potential drops. They are:

- The electrode potential
- The activation over potential
- Ohmic potential drop
- Concentration over potential
- Ohmic resistance of electrolyte

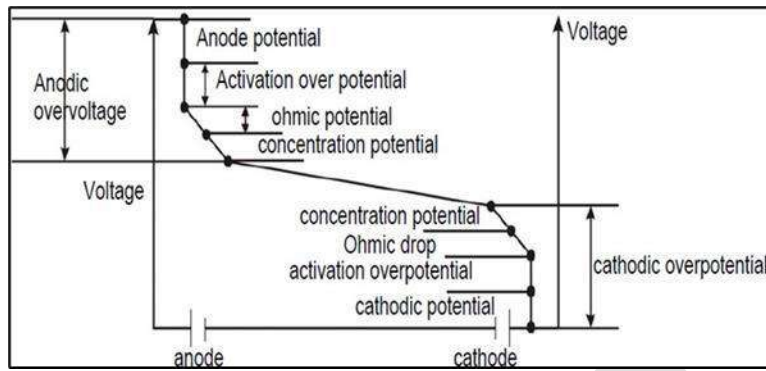


Fig. Potential drop during process

3.4 Process Parameters

1. **Current density** - Current density is simply the current that can be passed into a square inch of work area. At low current densities, metal removal rate is small. The relationship between current density and metal removal rate as shown. The electrochemical machine used for a particular application must have sufficient current available to maintain a current density of 50-1500 A/in².
2. **Tool feed rate** - Tool feed rate is directly proportional to current density. If the feed rate is increased the electrical resistance of the tool-work gap reduces to allow current to flow resulting in high metal removal rates. Also, it produces good surface finish and accuracy is improved.
3. **Gap between workpiece and tool** - The tool and the workpiece are positioned as close together to encourage efficient electrical transmission. Small gap results in high current densities and hence, more metal removal. It is important to maintain a uniform gap between the tool and the workpiece. Any physical contact between the tool and the workpiece results in arcing and serious damage to both the members. The gap size may vary from 0.25-0.76 mm. A gap size of 0.25 mm is often used.
4. **Velocity of electrolyte flow** - Electrolyte flow may be between 15-60 m/sec. If the electrolyte flow is too low, the heat and by-products of electrolytic reaction (hydrogen gas bubbles, sludge etc) build in gap of tool and workpiece causing non-uniform metal removal. Too high velocity will cause cavitations, also promoting non-uniform metal removal.
5. **Type of electrolyte, its concentration and temperature** - The type of electrolyte selected depends on the tool and the workpiece material. For instance, sodium chloride is cheap and provides good conductivity. However, it is corrosive and hence, cannot be used on tungsten carbide or molybdenum materials. Sodium nitrate is also popular due to its less corrosive nature. But it does not produce a good surface finish as that of sodium chloride. Also, it is more expensive than sodium chloride. It is preferred for machining aluminum and copper. The electrolyte in water at various concentrations

affect the surface finish produced. Low concentrations decrease the equilibrium machining gap resulting in better surface finish and tolerance control. Electrolyte temperature seriously affects the overcut. The power loss in electrolyte reaction gives rise to an increase in the temperature of the electrolyte. The heat must be carried away from the cutting area so as to maintain stable and steady conditions. Low temperature of electrolyte is conducive to better surfaces finish and tolerance.

$$\text{MRR in ECM} = C \cdot I \cdot h \text{ (cm}^3\text{/min)}$$

C: specific (material) removal rate (e.g., 0.2052 cm³/amp-min for nickel);

I: current (amp); h: current efficiency (90–100%).

The rates at which metal can electrochemically remove are in proportion to the current passed through the electrolyte and the elapsed time for that operation. Many factors other than current influence the rate of machining. These involve electrolyte type, rate of electrolyte flow, and some other process conditions. The rates at which metal can electrochemically remove are in proportion to the current passed through the electrolyte and the elapsed time for that operation. Many factors other than current influence the rate of machining. These involve electrolyte type, rate of electrolyte flow, and some other process conditions.



Fig. Parts made by ECM

1. Power supply Type:

- DC Voltage – 30V
- Current - 40000A
- Current Density – 500 A/Cm²

2. Electrolyte:

- Type – NaCl, NaNO₃, Proprietary mixtures.

- Temperature – 26 to 50 deg.
 - Flow rate – 16 to 20 LPM
 - Velocity – 1500 to 3000 m/min
 - Inlet pressure – 2200 kPa.
 - Outlet Pressure- 300 kPa
3. Working Gap: 0.075 to 0.75 mm
 4. Side over cut: 0.125 to 1 mm
 5. Feed rate: 0.500 to 13 mm/min
 6. Electrode material: Copper, Brass, Bronze
 7. Tolerance: 0.025 mm (2D) and 0.050 mm(3D)
 8. Roughness: 1.5 microns

3.5 ECM Tooling

Electrochemical Machining (ECM) tooling involves the use of a cathode (tool) that acts under DC current in the presence of an electrolyte fluid to remove material from the workpiece through controlled electrically-assisted chemical reactions. The tooling technique in ECM is critical for maintaining a gap between the cathode and the workpiece surface, ensuring efficient material removal without direct contact between the tool and the workpiece. The design of the cathode is essential in determining the final shape of the workpiece or the impression placed upon it, emphasizing the importance of precise tooling for achieving desired machining outcomes.

Example of ECM Tooling:

An example of ECM tooling is the fixture including the cathode, which is a crucial element in the ECM process. The cathode is designed to maintain a gap between itself and the workpiece surface, allowing the electrolyte flowing between the cathode and the workpiece to remove material efficiently when flushing. In ECM, the tooling plays a significant role in ensuring the accuracy, quality, and consistency of the machining process by controlling the gap, designing appropriate fixtures, and selecting materials that are corrosion-resistant and suitable for the specific application.

Tool and insulation material

1. Tool in ECM:

- The tool in ECM, known as the cathode, is made of electrically conductive materials like copper, brass, graphite, or copper-tungsten to facilitate the electrochemical machining process.
- The cathode is designed to maintain a small gap between itself and the workpiece surface,

allowing the electrolyte to flow between them for material removal without direct contact.

- The shape and design of the cathode determine the final shape of the workpiece, making it essential for achieving precise machining outcomes in ECM.
- Unlike traditional machining processes, the cathode in ECM does not experience wear during operation, contributing to its efficiency and longevity.

2. **Insulation Material in ECM:**

- Insulation materials are used in ECM to protect specific areas of the workpiece from the electrochemical machining process, ensuring selective material removal and precise machining.
- Side-insulation of the tool electrode in ECM can significantly enhance machining accuracy, especially for intricate contours or cavities in hard and exotic metals.
- Introducing an insulation layer, such as aluminum oxide, on the tool electrode can improve ECM performance, as demonstrated by drilling experiments on stainless steel workpieces.
- Insulation materials are crucial for controlling the gap between the tool and the workpiece, maintaining process stability, and achieving high precision in ECM operations.

Common insulation materials used in Electro Chemical Machining (ECM) include:

1. **Epoxy Resins:** Sprayed or dipped coatings of epoxy resins are among the most effective insulation materials used in ECM processes. These resins provide effective insulation to protect specific areas of the workpiece during machining operations.
2. **Aluminum Oxide:** Insulation of electrodes for ECM can be achieved through hot-dip aluminizing and microarc oxidation, where aluminum oxide is used as an insulation layer. This material helps in protecting the electrode and controlling the electrochemical machining process.
3. **Chemical Vapor Deposition (CVD) Materials:** Sidewall-insulation layers on micro-hollow electrodes in ECM are often fabricated using chemical vapor deposition (CVD) with various insulation materials. These materials, such as those used in CVD processes, contribute to effective insulation and precision in ECM operations.

3.6 Applications of ECM

- Machining of cavities in forging dies, drilling deeper holes and irregular shaped holes which cannot be obtained by conventional machining methods.
- Machining of complex profiles like turbine wheels, turbine and jet blades.

- Die sinking: Electro Chemical Machining is often used as an alternative to the cavity type electric discharge machining (EDM).
- Fabrication of thin walled parts: Electrochemical machining does not produce surface stress in the work piece therefore even very brittle and easily deformed materials may be machined in thin walled shapes.
- Grinding of a work piece by a rotating wheel, which performs grinding operation through an electrolyte. The wheel is conductive and cathodically connected. Non-conductive hard particles are set on the wheel surface. The particles provide a constant gap through which an electrolyte is continuously fed. Hard and brittle materials are ground by the method.
- Rough corners or edges can be turned into very smooth parts and the process is known as deburring.

3.7 Advantages of ECM

- The rate of machining does not depend on the hardness of the work piece material
- No physical contact between tool and workpiece. Hence no tool wears
- As the tool does not wear. Soft materials (e.g., copper) may be used for tool fabrication
- No stresses are produced on the work piece surface and have high surface finish
- No burrs form in the machining operation
- High surface quality may be achieved
- Accurate shape with good surface finish can be obtained
- Suitable for mass production applications
- Several tools could be connected to a cassette to make many cavities simultaneously
- ECM process can be easily automated
- Capable of machining metals and alloys irrespective of their strength and hardness.
- Metal removal is due to anodic (workpiece) dissolution. Hence, no thermal effects on the workpiece.

3.8 Limitations of ECM

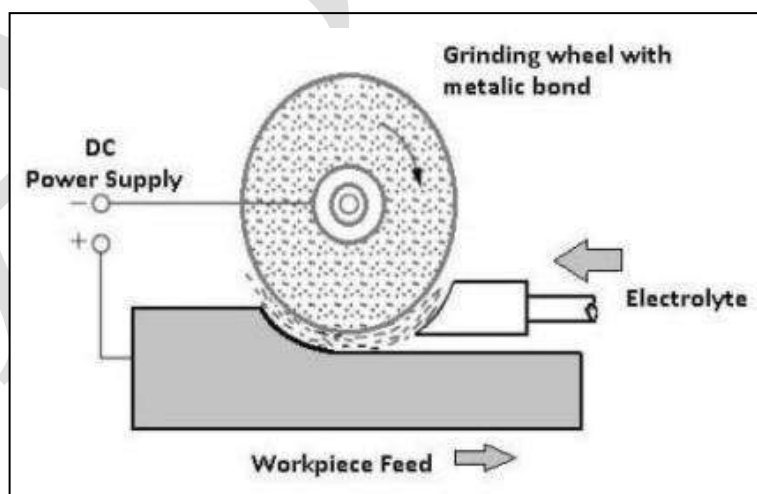
- Suitable only for conductive materials.
- Inability to machine sharp interior edges and corners because of very high current densities at

these points.

- Space and floor requirements are higher than conventional machines.
- High electrical power is consumed during the process.
- Higher cost and frequent maintenance of equipment.
- Post machining and cleaning is a must to reduce the corrosion of workpiece.
- Electrolyte may cause corrosion of the equipment.
- Only electrically conductive materials may be machined.
- Not environmentally friendly process. The sludge produced in large amounts need to be disposed off.

3.9 Electrochemical grinding (ECG)

Electrochemical Grinding is a variety of ECM that consolidates electrolytic movement with the physical evacuation of material by use of charged grinding wheels. It can deliver burr and stretch free parts without any metallurgical harm brought on by grinding, wiping out the need for secondary machining operations. Similar to ECM, ECG creates no heat to contort fragile segments. It can handle any electrochemically reactive material. The most well-known reason clients pick ECG is for the burr free nature of cut.



Electrochemical grinding

Principle:

Electrochemical grinding is also called electrolytic grinding is a variation process of the basic electrochemical machining, wherein material removal of the electrically conductive work material takes place through the combined effect of electrochemical process and the mechanical action of abrasive particles (grinding) on the work material. The process makes use of a metallic grinding wheel which is a circular metal plate and is embedded with insulating abrasive particle such as aluminum oxide or diamond, set in a conducting binding material. In this technique, the workpiece plays the role of the anode while the grinding wheel works as a cathode.

Applications of ECG:

- Any material which is conductive may be ground by the electrolytic process. But its most useful application is concerned with hardened steel, cemented carbides, and similar materials.
- This is mainly applied to re-sharpening and reconditioning of carbide tools and other materials that are difficult to grind.
- The grinding pressure is low. It is possible to grind and cut thin sections and thin-wall tubing of difficult materials without distortion or burr.

Advantages of ECG:

- Ability to grind any electrically conductive material regardless of hardness.
- MRR are 5 to 10 times greater than Broaching, milling or conventional machining.
- Frequent grinding dressing is unnecessary.

Limitations of ECG:

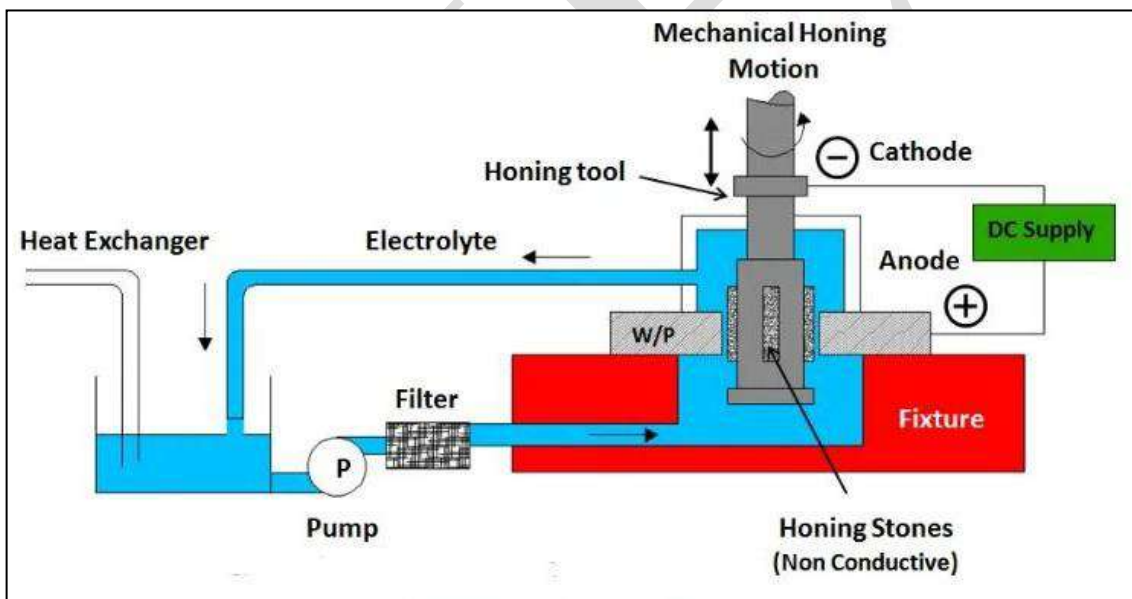
- Workpiece must be electrically conductive.
- High equipment cost.
- Not suitable for grinding low hardness metals.

3.10 Electrochemical Honing (ECH)

Electrochemical honing is a modification of conventional honing. ECH is a process in which the metal removal capabilities of ECM are combined with the accuracy capabilities of honing. The

process consists of a rotating and reciprocating tool inside a cylindrical component. Material is removed through anodic dissolution and mechanical abrasion – 8% or more, of the material removal occurs through electrolytic action. As with conventional ECM, the workpiece is the anode and a stainless-steel tool is the cathode.

Tool consists of a hollow stainless-steel body that has expandable, nonconductive honing stones protruding from at least three locations around the circumference. The tool is rotated and reciprocated on a rigid spindle for precise metal removal from internal cylindrical hole of the workpiece. The honing stones are identical with those used in conventional honing operations, except that they must resist the corrosiveness of the electrolyte. The honing stones are mounted on the tool body with a spring-loaded mechanism so that each of the stones exerts equal pressure against the workpiece. The length of the stones is selected to be approximately one-half the length of the bore being processed.



Electrochemical Honing

At the beginning of the ECH cycle, the stones protrude only 0.075-0.127 mm from the stainless-steel body, establishing the gap through which the electrolyte flows. The electrolyte enters the tool body via a sliding inlet sleeve from which it exits into the tool-workpiece gap through small holes in the tool body. After passing through the gap, the electrolyte flows from the workpiece through the gap at the top and bottom of the bore. The mechanical action of the tool is the same as with conventional honing; the tool is rotated and reciprocated so that the stones abrade the entire length

of the bore. Electrolytes used in ECH are essentially the same as those used in ECM, although the control of pH, composition and sludge is less critical because the abrasive action of the stones tends to correct any resulting surface irregularities. As in ECM, the electrolytes are re-circulated and reused after passing through appropriate filtration, and the most commonly used electrolytes are sodium chloride and sodium nitrate.

Applications of ECH

- Primarily used in the grinding of Tungsten carbide tool bits.
- Grinding of cutting tools, chilled iron castings, magnet alloys, contour milling of honey- comb structures.
- Used for machining of cemented carbides, satellites, refractory materials, stainless steel and high alloys steels without any burr.
- Chromium plated materials, flame hardened materials and temperature sensitive alloys can be machined without forming thermal cracks and distortion.
- Grinding of super alloy turbine blades.
- Burr free sharpening of hypodermic needles.

Advantages of ECH

- The most important advantage of this process is that we get both the surface finishing and shaping as a result of this process. That means no need to spend your time on the surface finishing and shaping separately.
- In case hard materials, the tradition machining techniques are not applicable. But this is the perfect process for the hard or tough materials.
- No heat is produced during this whole technique.
- It gives you one of the best surface finishing.
- This method gives you the desired low tolerance.

Limitations of ECH

- More number of equipments are required during this process.

- The cost of machines is too much high to carry out this process.
- Skilled labour is needed to implement this metal removing process.
- It is only applicable to the hard materials mostly.

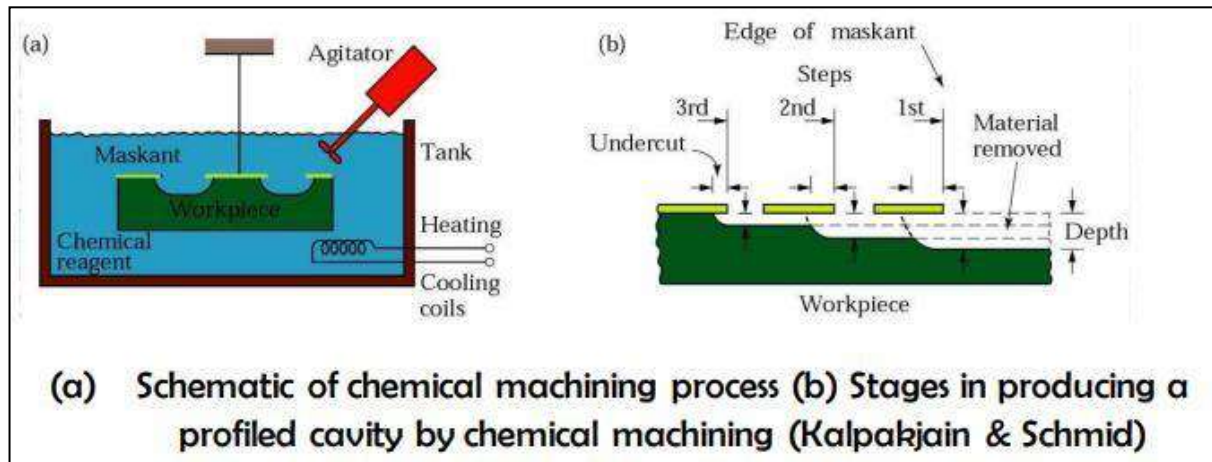
CHEMICAL MACHINING (CHM)

3.11 Introduction to CHM

Chemical machining is well known as nontraditional machining process and is controlled by chemical dissolution of the machines work piece material by contact with a strong chemical reagent. It is also known as a chemical etching and almost all the material from metal to ceramics can be chemically machined. Chemical machining process is employed where metal removal is difficult. Chemical machining is a controlled dissolution of the work material by means of contact with a strong acidic or alkaline chemical reagent. The major characteristics of chemical machining are material removal rate, accuracy and surface finish.

3.12 Elements of process in CHM

The working principle of chemical machining is based on chemical etching. The part of the work piece metal where material is to be removed is brought into contact with a strong corrosive chemical called etchant. The etchant react with the workpiece in the material to be cut and causes the solid material to be removed. Thus the metal is removed by the chemical attack of the etchant. The portion of the work material where material is not to be removed is protected from chemical attack by means special coating called as maskants. So by this chemical machining process from all metal to ceramic substance can be machined with chemical machining process. This process of chemical reaction between the material of the work piece and some chemical reagent use to produce a reaction between then so that it can be removed easily .thus the surface of the workpiece is etched away, exposing the lower layers, and the process is continued until the desired amount of material is removed.



Procedure for Chemical Machining

- 1. Workpiece pre cleaning process:** The surface of workpiece metal is cleaned thoroughly, degreased and pickled by acid or alkalis. Pre cleaning is the most important method to remove oil, grease, dirt, rust, or any foreign substance from the work surface to produce a good adhesion of masking material.
- 2. Masking and scribing mask:** Masking involves covering the portion of the workpiece metal where material is not to be removed by the chemical action. Masking with adhesive types or paints is a common practice although rubber (elastomers) and plastics are also used. Since it is difficult to apply maskant on small surface, the maskant is initially applied on a large surface.
- 3. Etching:** The unmasked surface of the workpiece is machined chemically with selected etchant. Etching is carried out by immersing the work material in a tank of agitated etchant. The process is carried out at high temperature depending on the etched material. Temperature control and agitation during chemical machining. Erosion of the work material takes place from the exposed surface. The work piece is converted into metallic salt, which is then dissolved and carried away in the etchant solution.
- 4. Demasking:** When etching is completed the mask is removed either through mechanical or chemical means. Any etchant on the work material is also removed by cold water to clean. A deoxidizing bath may also be required in order to remove the oxide coating or films left on the surface of the work material.

3.13 Maskants (Resists)

Masking material is known as Maskant which is used to protect workpiece surface from chemical etchant. In another words maskants protect the portion of workpiece metal where material is not to be removed by chemical action of the etchant. Polymer or rubber based materials are generally used as a maskant material. Various maskant application methods can be used such as dip, brush, spray, roller, and electro coating as well as adhesive tapes.

The type of maskant to be selected for machining is based on the following factors:

- Be inert to the chemical reagent used
- Chemical resistance required
- Be tough enough to withstand handling
- Adhere well to the work piece
- Allow itself to be scribed easily
- Be removed easily after etching
- Be inexpensive after etching
- Be able to withstand the heat generation by etching
- Availability and low cost

Masking material for various work material	
Work material	Maskant
Aluminium & Alloys	Polymer, butyl rubber & neoprene
Copper & alloys	Polymer
Iron based alloys	Polymer, poly vinyl chloride, polyetilien butyl rubber
Nickel	Neoprene
Magnesium	Polymer
Titanium	Polymer

3.14 Etchants

Etchants are acid or alkaline solutions maintained within a controlled range of chemical composition and temperature. The workpiece material to be removed is sprayed or immersed in a suitable etchant. Various etchant are available for machining different material as listed in table below.

The type of etchant to be selected for machining is based on following factor:

- Type of workpiece metal that is being etched
- Rate of metal removal
- Surface finish required
- Type of maskant used
- Depth of etch required
- Ability to regenerate the etchant solutions
- Un harmful Or non toxic to human operator
- Availability at low cost

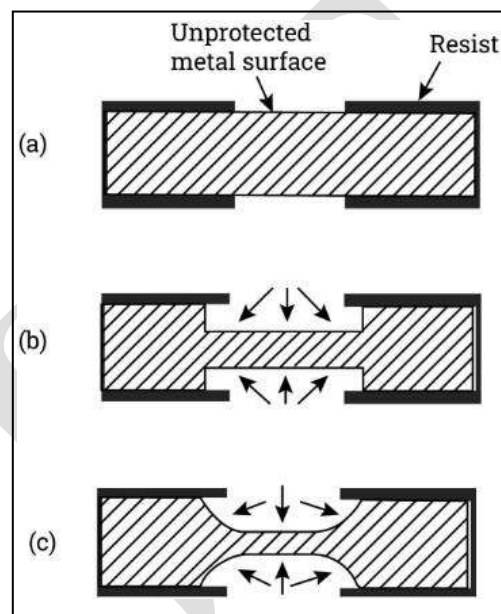
Etchant characteristics		
Work material	Etchant	Etchant temperature
Aluminum & alloys	Ferric Chloride	49
Copper & alloys	Cupric Chloride	49
Steel	Ferric Chloride	54
Nickel	Ferric Chloride	49
Magnesium	Nitric acid	49
Titanium	Hydrogen fluoride	-
Glass	Hydrogen fluoride	-

3.15 Types of CHM process

There are two type of chemical machining:

1. Chemical blanking method

Chemical blanking is a process of producing a part from thin sheet metal by chemically etching the periphery of the desired shape. The material is removed by chemical dissolution. Chemical blanking is used for parts that are otherwise typically produced by mechanical blanking presses from thin plates and foil material. With mechanical presses, vibrations, backlash, and part distortion will make smaller parts difficult to produce.



Chemical blanking process

Steps of chemical blanking: Chemical blanking process has several steps for producing machine parts. These are given below:

1. Workpiece pre cleaning process: The surface of the workpiece metal is cleaned thoroughly, degreased and pickled by acid and alkalis. Pre cleaning is of utmost importance in order to remove oil, grease, dirt, rust or any foreign substance for the work surface so as to produce a good adhesion of the masking material. The material is allowed to dry.

2. Masking: Masking involves covering the portions of the workpiece metal where material is not to be removed by the chemical action of the etchant. A suitable maskant, say a polymer, rubber, or any other material is selected based on the workpiece material. The maskant is applied on the work surface

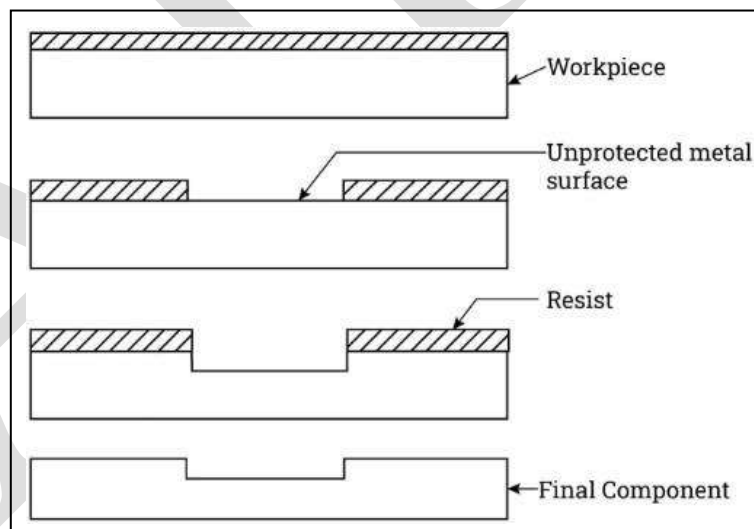
by various methods like dip, brush, spray, roller, electro-coating, and as well as adhesive tapes.

3. Etching: Removal of material from the workpiece take place by etching process. The workpiece metal is either sprayed continuously with a selected etchant like ferric chloride on those portions where the material is to removed, or immersed in a tank of agitated etchant, where the etchant chemically attack those portions not masked. Erosion of the work material take place both inward and laterally from the exposed(unmasked) surface. The work material is converted in to metallic salt, which is the dissolved and carried away in the etchant solution.

4. Demasking: When etching is completed, the mask is removed either through mechanical or chemical means Any etchant on the material is also removed with a wash or clear, cold water. A deoxidizing bath may also be required in order to remove the oxide films left on the surface of the work material.

2. Chemical milling or contour machining method

Chemical milling is a process used to produce shapes by chemical etching selective portion of material from the relatively large surface area of work metal. The main purpose is to produce shallow cavity with complex profiles on plates, sheets forging, generally for the overall reduction of weight. Chemical milling as shown in Figure below



Chemical milling process

Steps of Chemical Milling: Chemical machining process has several steps for producing machine parts. These are given below

1. Workpiece pre cleaning process: The workpiece material has to be cleaned in the beginning of chemical machining process. The cleaning operation is carried out to remove the oil, grease, dust, rust or any substance from the surface of material. A good cleaning process produces a good adhesion of the masking material. There are two cleaning methods; mechanical and chemical methods. The most widely used cleaning process is chemical method due to less damages occurred comparing to mechanical one. Ultrasonic cleaning machine is applied with using special cleaning solution and heating is beneficial during the cleaning process.

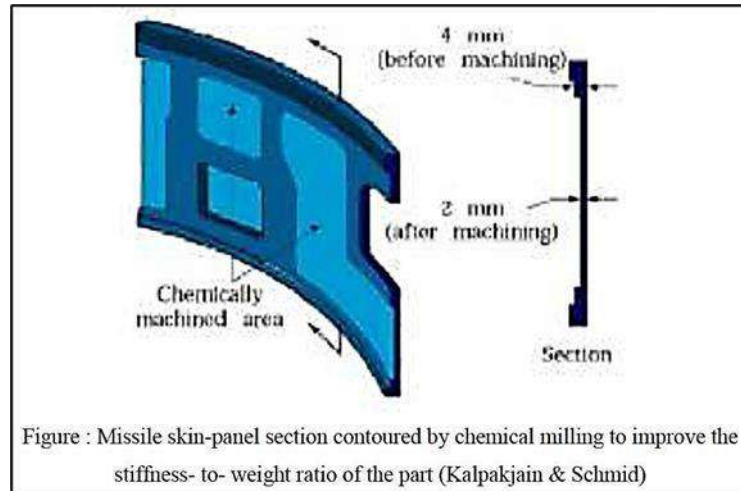
2. Masking and Scribing mask: Masking involves covering the portions of the workpiece metal where material is not to be removed by the chemical action of the etchant. The selected masking material should be readily strippable mask, which is chemically impregnable and adherent enough to stand chemical abrasion during etching. Scribing mask is guided by templates to expose the areas that receive chemical machining process. The selection of mask depends on the size of the workpiece material, the number of parts to be produced, and the desired detail geometry. Silkscreen masks are preferred for shallow cuts requiring close dimensional tolerances.

3. Etching: This step is the most important stage to produce the required component from the sheet material. This stage is carried out by immerse type etching machine. The workpiece material is immersed into selected etchant and the uncovered areas were machined. This process is generally carried out in elevated temperatures which are depended on the etched material. Then the etched workpiece is rinsed to clean etchant from machined surface.

4. Cleaning masking material: Final step is to remove masking material from etched part. The inspections of the dimensions and surface quality are completed before packaging the finished part.

3.16 Applications of CHM

- Chemical milling is used in the aerospace industry to remove shallow layers of material from large aircraft components missile skin panels (Figure), extruded parts for airframes.



- This process is best suited for production of printed circuits , where the basic connections of the circuit consist of thin metal strips attached to a insulating board faced with a thin layer of copper .
- Besides, it removes metal from a portion of entire surface of formed or irregularly shaped parts such as forging casting, extrusions or formed wrought stock. The surface finish obtained in this process is in the range of 0.5 to 2 μm .
- One of the major applications of chemical milling and blanking is in the manufacture.

3.17 Advantages of CHM

- Low tooling cost.
- Multiple machining can be done on a workpiece simultaneously.
- No application of force so on risk of damage to delicate or low strength workpiece.
- Complicated shapes/patterns can be machined.
- Machining of hard and brittle material is possible.
- Easy weight reduction
- No stress introduction to the workpiece

3.18 Limitations of CHM

- Slower process, very low MRR so high cost of operation.
- Small thickness of metal can be removed.
- Sharp corners cannot be prepared.
- Difficult to chemically machine thick material (limit is depended on workpiece material, but the thickness should be around maximum 10 mm)

QUESTION BANK

ELECTROCHEMICAL MACHINING (ECM)

1. Mention advantages and disadvantages of Electrochemical machining (ECM).
2. Mention the applications of Electrochemical machining (ECM).
3. With a neat sketch explain Electrochemical grinding process (ECG).
4. With a neat sketch explain Electrochemical honing process (ECH).
5. Mention the advantages, disadvantages and applications of Electrochemical grinding process (ECG).
6. Mention the advantages, disadvantages and applications of Electrochemical honing process (ECH).
7. Describe various process parameters affecting ECM. (10 Marks) **(VTU June/July 2011)**
8. Differentiate between “fludging” and “non-fludging” electrolyte. (3 Marks) **(VTU June/July 2011)**
9. Explain with a neat diagram, electrochemical machining process. (6 Marks) **(VTU Dec 2011)**
10. With a neat schematic diagram, explain the electrochemical honing process. (8 Marks) **(VTU Dec 2011)**
11. Explain the advantages and limitations of ECM. (6 Marks) **(VTU Dec 2011)**
12. With a neat sketch, explain the working principle of ECM process. (8 Marks) **(VTU June 2012)**
13. Sketch and explain different types of tools used in ECM process. (6 Marks) **(VTU June 2012)**
14. List the advantages, limitations and applications of ECM process. (6 Marks) **(VTU June 2012)**
15. Explain ECM with the schematic diagram. (5 Marks) **(VTU Dec 2012)**
16. Explain functions of electrolytes used in ECM and name 3 electrolytes with their specialities. (5 Marks) **(VTU Dec 2012)**
17. Explain chemistry of ECM with the circuit. (5 Marks) **(VTU Dec 2012)**
18. Theoretically estimate metal removal rate and electrode feed rate of ECM. (5 Marks) **(VTU Dec 2012)**
19. With suitable sketches. Explain the metal removal mechanism in electro chemical grinding. (8 Marks) **(VTU June/July 2013)**
20. Why are chemical machining and electro chemical machining considered as chipless machining? Explain the mechanisms of metal removal on both cases and compare it with conventional grinding process. (12 Marks) **(VTU June/July 2013)**
21. Explain different elements of electrochemical machining process. (6 Marks) **(VTU June/July 2014)**
22. Explain with sketch the electrochemical grinding operation. (6 Marks) **(VTU June/July 2014)**
23. What are the advantages and disadvantages of ECM? (6 Marks) **(VTU Dec 2014/Jan 2015)**

24. What are the factors that influences oxidation of ECM? (4 Marks) **(VTU Dec 2014/Jan 2015)**
25. Explain the principle of electro chemical grinding, with a neat sketch. (10 Marks) **(VTU Dec 2014/Jan 2015)**
26. Explain the chemistry of ECM process with diagram. (8 Marks) **(VTU June/July 2015)**
27. List the functions of electrolyte in ECM process. (8 Marks) **(VTU June/July 2015)**
28. Explain with diagram, working of electro chemical grinding (ECG). (8 Marks) **(VTU June/July 2015)**
29. Draw schematic sketch of electro chemical machining and explain briefly the elements of ECM process. (10 Marks) **(VTU June/July 2016)**
30. Explain with neat schematic diagram of electro chemical grinding and their advantages and application. (10 Marks) **(VTU June/July 2016)**

CHEMICAL MACHINING (CHM)

1. Mention advantages and disadvantages of Chemical machining (CHM).
2. Mention the applications of Chemical machining (CHM).
3. Discuss the factors to be considered for selection of 'Maskants' and the types that are used in 'chemical machining'. (10 Marks) **(VTU June/July 2011)**
4. Differentiate between 'chemical milling' and chemical blanking'. (4 Marks) **(VTU June/July 2011)**
5. Write a note on 'etchants' in 'chemical milling'. (6 Marks) **(VTU June/July 2011)**
6. What is chemical machining? Explain briefly the elements of the process. (7 Marks) **(VTU Dec 2011)**
7. Explain briefly, the chemical milling process with the help of neat flow chart. (7 Marks) **(VTU Dec 2011)**
8. Discuss the advantages and applications of CHM. (6 Marks) **(VTU Dec 2011)**
9. What are Maskants used in chemical machining? Explain the different types of it. (10 Marks) **(VTU June 2012)**
10. What are the factors to be considered in the selection of etchant? (4 Marks) **(VTU June 2012)**
11. Explain the chemical blanking process stepwise with the flow chart. (10 Marks) **(VTU Dec 2012)**
12. Write a note on etchants indicating factors to select. (6 Marks) **(VTU Dec 2012)**
13. Write advantages of chemical machining. (4 Marks) **(VTU Dec 2012)**
14. Explain in brief the following in chemical machining process: i) Maskants. ii) Etchants. (8 Marks) **(VTU June/July 2013)**
15. With the help of neat sketches. Explain the different steps involved in chemical blanking. (12 Marks)

(VTU June/July 2013)

16. Explain with flow chart the chemical blanking process. Mention its applications. (10 Marks) **(VTU June/July 2014)**
17. Explain with flow chart the chemical milling process. Mention its advantages and applications. (10 Marks) **(VTU June/July 2014)**
18. Explain the principle steps involved in chemical milling to produce pockets and contours. (6 Marks) **(VTU Dec 2014 / Jan 2015)**
19. List out the advantages and disadvantages of chemical machining. (8 Marks) **(VTU Dec 2014 / Jan 2015)**
20. List out the major applications of CHM. Further process application related to improving the characteristics. (6 Marks) **(VTU Dec 2014 / Jan 2015)**
21. Explain the elements of chemical machining process. (6 Marks) **(VTU June/July 2015)**
22. Explain with flow chart the chemical blanking process. (10 Marks) **(VTU June/July 2015)**
23. Mention the limitations of chemical machining process. (4 Marks) **(VTU June/July 2015)**
24. What are the factors on which the selection of a resist for all in chemical machining depend? (3 Marks) **(VTU June/July 2016)**
25. Explain the elements of process (i) Maskants or resist. (ii) Etchants in CHM. (8 Marks) **(VTU June/July 2016)**
26. Explain with sketch progressive stages of metal removal in chemical blanking. (6 Marks) **(VTU June/July 2016)**
27. List the applications of chemical machining. (3 Marks) **(VTU June/July 2015)**

Module-4

ELECTRICAL DISCHARGE MACHINING (EDM)

PLASMA ARC MACHINING (PAM)

SYLLABUS

ELECTRICAL DISCHARGE MACHINING (EDM)

Introduction, mechanism of metal removal, EDM equipment: spark erosion generator (relaxation type), dielectric medium-its functions & desirable properties, electrode feed control system. Flushing types; pressure flushing, suction flushing, side flushing, pulsed flushing. EDM process parameters: Spark frequency, current & spark gap, surface finish, Heat Affected Zone. Advantages, limitations & applications of EDM, Electrical discharge grinding, Traveling wire EDM.

PLASMA ARC MACHINING (PAM)

Introduction, non-thermal generation of plasma, equipment mechanism of metal removal, Plasma torch, process parameters, process characteristics. Safety precautions. Safety precautions, applications, advantages and limitations.

ELECTRICAL DISCHARGE MACHINING (EDM)

4.1 Introduction to EDM

Electro discharge machining is also known as spark erosion, electro-erosion or spark machining. It is a non-traditional machining process based on the principle of erosion of metals by an interrupted electric spark discharge. These discharges are repeated many thousand times/sec in the selected area of workpiece. In this method, all operations are carried out in a single setup. It can be applied to machine steels, super alloys, refractories, etc. No complicated fixtures are needed for holding the job and even very thin jobs can be machined to the desired dimensions and shape. The main attraction of EDM over traditional machining processes such as metal cutting using different tools and grinding is that this technique utilizes thermoelectric process to erode undesired materials from the workpiece by a series of discrete electrical sparks between the workpiece and the electrode. A picture of EDM machining is shown in Figure.

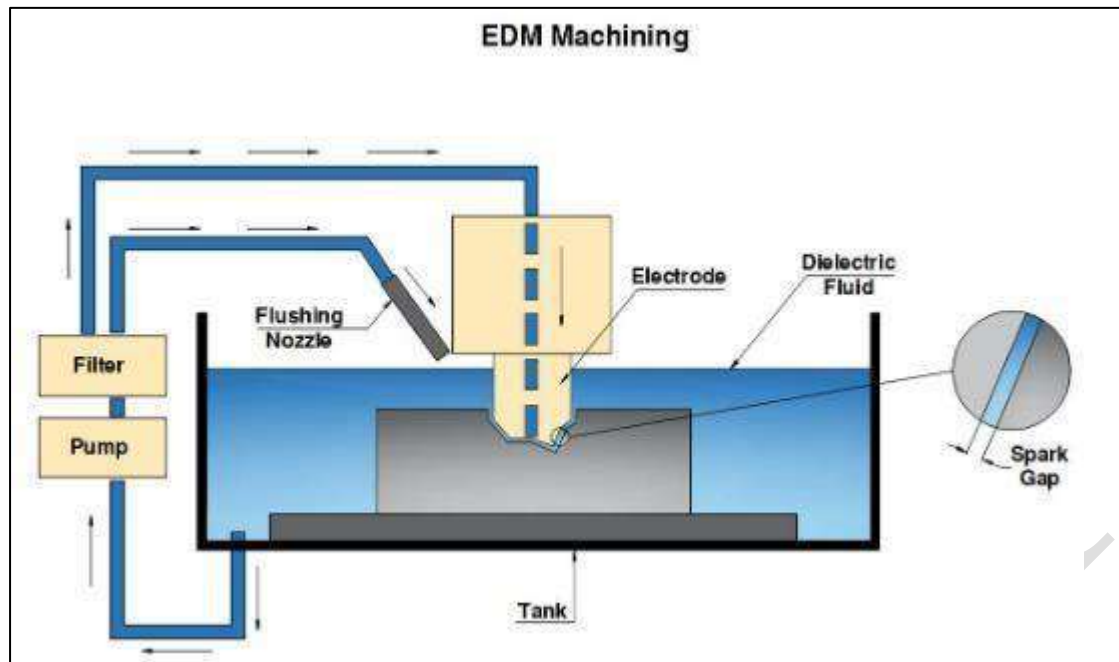


Fig. EDM machining

The traditional machining processes rely on harder tool or abrasive material to remove the softer material whereas non-traditional machining processes such as EDM uses electrical spark or thermal energy to erode unwanted material in order to create desired shape. So, the hardness of the material is no longer a dominating factor for EDM process. A schematic of an EDM process is shown in Figure 2, where the tool and the work piece are immersed in a dielectric fluid.

Definition:

- Electro-discharge machining is an electro-thermal non-traditional machining process, where electrical energy is used to generate electric spark and material removal mainly occurs due to thermal energy of the spark.
- EDM is mainly used to machine difficult to machine materials and high strength temperature resistant alloys, used for making stamping tools, wire drawings and extrusion dies, intricate mould cavities etc.
- It can be used to machine complex geometries in small batches. The work material to be machined should be electrically conductive.

4.2 Mechanism of metal removal

- EDM works on the principle that heat energy generated by a spark is used to remove material from the work piece.
- The tools and work piece are separated by a small gap called as spark gap. The gap varies from 0.01 mm to 0.5 mm. The tool and work piece both immersed in the dielectric fluid.
- When the supply is made 'ON', thousands of sparks are produced per second. The duration of each spark is very short.
- When the spark comes in contact with the dielectric fluid in the spark gap, the fluid gets ionized. It allows current to flow between the tools and workpiece.

As shown in Figure, at the beginning of EDM operation, a high voltage is applied across the narrow gap between the electrode and the workpiece. This high voltage induces an electric field in the insulating dielectric that is present in narrow gap between electrode and workpiece. This cause conducting particles suspended in the dielectric to concentrate at the points of strongest electrical field. When the potential difference between the electrode and the workpiece is sufficiently high, the dielectric breaks down and a transient spark discharges through the dielectric fluid, removing small amount of material from the workpiece surface. The volume of the material removed per spark discharge is typically in the range of 10^{-6} to 10^{-6} mm³.

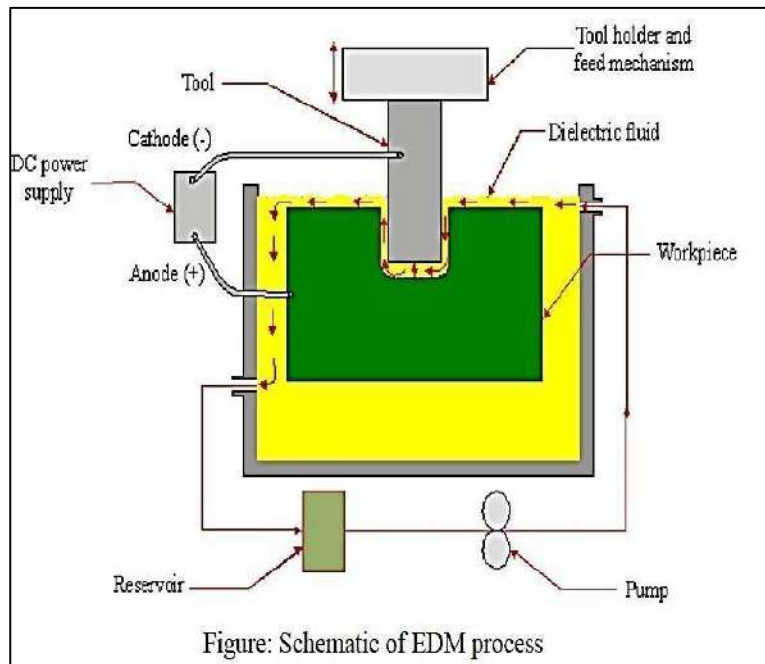
The material removal rate, MRR, in EDM is calculated by the following formula:

$$MRR = 40 I / T_m$$

$$1.23 (cm^3/min)$$

Where, I is the current amp,

T_m is the melting temperature of workpiece in °C



EDM removes material by discharging an electrical current, normally stored in a capacitor bank, across a small gap between the tool (cathode) and the workpiece (anode) typically in the order of 50 V/10 A.

4.3 EDM equipment

1. **Tool Electrode and Workpiece:** The tool and work piece is kept in a reservoir and is connected to a DC power supply. The tool is connected to negative terminal, so that it becomes cathode, while work piece is connected to positive terminal and become anode. Tool electrode has the shape basically same to that of the product desired with allowance for side clearance and over cut.
2. **Dielectric fluid system:** The dielectric fluid is spark conductor, coolant and also flushing medium. The common dielectric fluids used are paraffin oil, transformer oil and kerosene. Dielectric fluid is stored in a tank and circulated through a pump with the help of nozzle at the gap between tool and workpiece. Dielectric is continuously flushed into the spark gap. The used dielectric is filtered and recirculated into the reservoir. The dielectric fluid is pumped at a pressure of 2 kg/cm² or less.

Functions of Dielectric fluid:

1. It should act as an insulator until the required breakdown voltage is attained.

2. It should act as a conductor, once the breakdown voltage is reached.
3. It should clean the spark gap by carrying away the molten metal.
4. It should deionize the spark gap rapidly after the discharge has occurred.
5. It should cool the tool, workpiece and spark region.

Characteristics of Dielectric fluid

1. Stable dielectric strength: The dielectric fluid should have sufficient and stable dielectric strength to serve as an insulation between the tool and work till the breakdown voltage is reached.
2. Rapid deionisation: It should de-ionise rapidly after the spark discharge has taken place.
3. Low velocity and good wetting capacity: It should provide effective cooling and remove the swarf particles from the machining gap.
4. Chemically neutral: It should be chemically neutral so as not to attack the electrode, work piece other components of setup.
5. High flash point: Dielectric should have high flash point
6. Nontoxic: It should not emit any toxic vapours or have an unpleasant odour.
7. Stable properties: It should maintain above properties in varying temperature contaminations of wear debris, etc.
8. Economical and easily available: It should not be too costly. It should be easily available in the nearby market.

Selection parameters of any Dielectric fluid

- a) Workpiece size
 - b) Types of shape
 - c) Tolerance
 - d) Surface finish
 - e) Metal removal rate
3. **Electrical power supply and spark generator:** The D.C. power supply with current density in the range of $10,000 \text{ A/cm}^2$ and the power density of 500 mw/cm^2 is used. The voltage is about 40-450 volts is applied. Spark generator is the circuit consist of RC combination. The capacitor is continued to charge as long as the voltage in the capacitor reaches to the value of breakdown voltage. This circuit helps to produce spark between the gap of tool and workpiece. When the supply is made 'ON' the capacitor voltage starts rising continuously. When the capacitor voltage

equals the breakdown voltage, (of dielectric fluid) a spark discharge will occur in the spark gap. The spark persists until the capacitor voltage falls below that which is required to maintain sparking. After the capacitor discharge, sparking ceases and the dielectric fluid in the spark gap gets deionized. The capacitor is then recharged and the cycle repeats itself. The time taken by the capacitor to recharge upto the breakdown voltage should be sufficient to allow the dielectric to ionize. Resistor R in the circuit prevents the charging of capacitor before the spark gap is ionized.

4. **Tool feed mechanism/ Servo system:** A servo-controlled electrode feeding arrangement is provided which continuously senses the spark gap and moves the tool electrode to maintain the gap. The servo system advances the tool electrode according to the machining required. The tool feed control prevents.

- Too large gaps which may prevent the formation of spark.
- Short circuits which will damage both tool and the workpiece.

The servo system may work electromechanically or hydraulically. Since during operation both the tool and work piece are eroded, it is necessary to feed the tool continuously towards the workpiece so as maintain the spark gap. This can be achieved by a suitable tool feed control mechanism along with servo mechanism system.

4.4 Flushing types

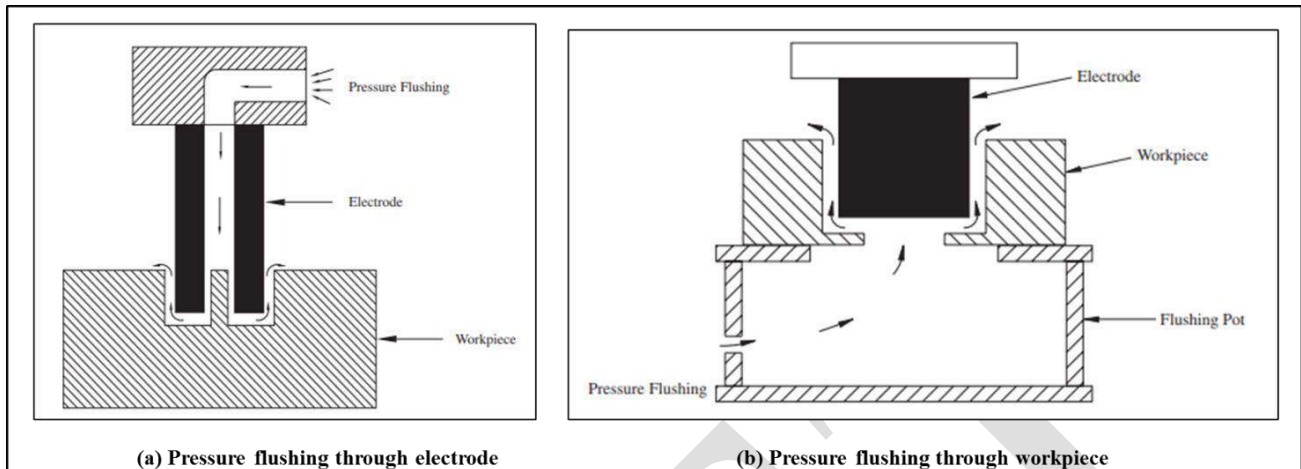
Flushing is defined as providing a fresh dielectric fluid between the tool and workpiece. Due to successive spark discharge, the particles eroded from the tool and workpiece contaminates the dielectric. This reduces its insulation strength and hence spark may occur easily. To prevent this premature occurrence of discharge, due to the eroded particles, flushing is necessary. The various methods of flushing are:

1. **Pressure / Injection flushing:** Pressure flushing, also called injection flushing, is the most common and preferred method for flushing. One great advantage of pressure flushing is that the operator can see the amount of oil that is being used for flushing. With pressure gauges, this method of flushing is simple to learn and use. Pressure flushing may be performed in two ways: (a) through the electrode or (b) through the workpiece.

(a) Pressure flushing through electrode

With pressure flushing, there is the danger of a secondary discharge. Since electricity takes the path of least resistance, secondary discharge machining can occur as the eroded particles pass between the walls of the electrode and the workpiece, as presented in Figure. This secondary discharge can

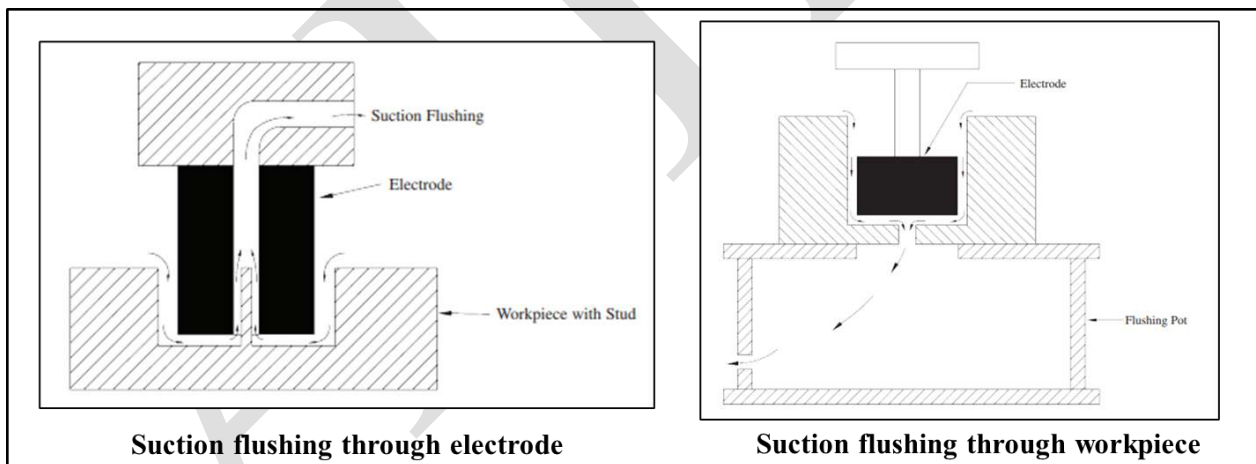
cause side wall tapering. Suction flushing can prevent side wall tapering.



(b) Pressure flushing through workpiece

Pressure flushing can also be done by forcing the dielectric fluid through a workpiece mounted over a flushing pot. This method eliminates the need for holes in the electrode.

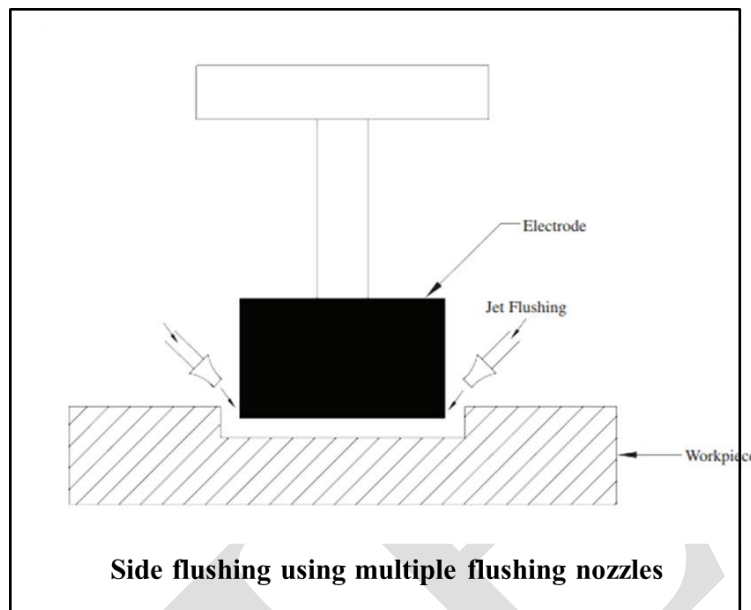
2. **Suction flushing:** Dielectric fluid is sucked either through workpiece or tool. Suction or vacuum flushing can be used to remove eroded gap particles. Suction flushing can be done (a) through the electrode or (b) through the workpiece.



Suction flushing minimizes secondary discharge and wall tapering. Suction flushing sucks oil from the worktank, not from the clean filtered oil as in pressure flushing. For suction cutting, efficient cutting is best accomplished when the work tank oil is clean. A disadvantage of suction flushing is that there is no visible oil stream as with pressure flushing. Also, gauge readings are not always reliable regarding the actual flushing pressure in the gap. A danger of suction flushing is that gases may not be sufficiently removed, this can cause the electrode to explode. In addition, the created vacuum can be so great that the electrode can be pulled from its mount, or the workpiece pulled

from the magnetic chuck.

3. **Side / Jet flushing:** Jet or side flushing is done by tubes or flushing nozzles which direct the dielectric fluid into the gap. Dielectric fluid is flushed from side of the tool.



4. Pulsed flushing:

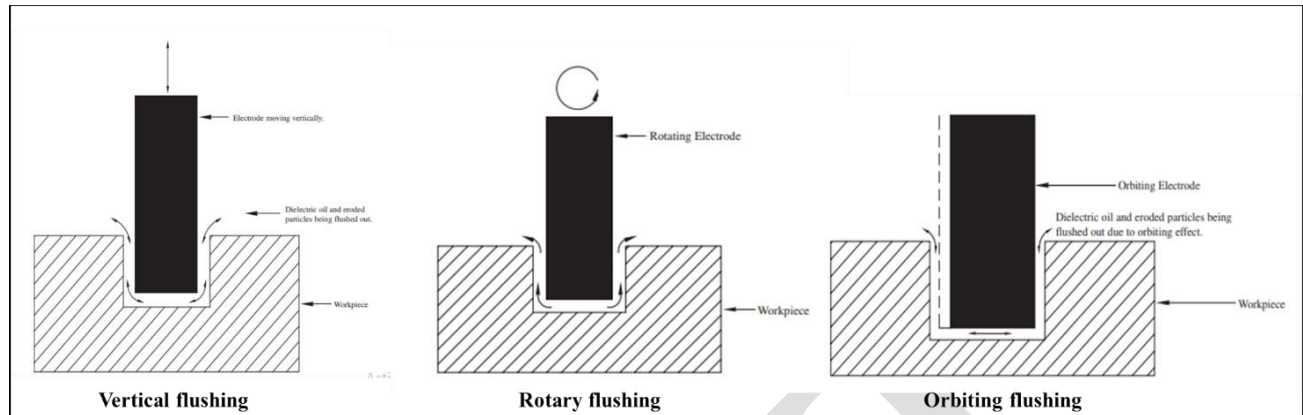
Three types of pulse flushing are:

- a. Vertical flushing: the electrode moves up and down.
- b. Rotary flushing: the electrode rotates.
- c. Orbiting flushing: the electrode orbits.

a. Vertical flushing: In vertical flushing, the electrode moves up and down in the cavity. This up and down motion causes a pumping action which draws in fresh dielectric oil. Many machines are now equipped with jump control which causes the electrode to jump rapidly in and out of the cavity which aids in flushing out the eroded particles.

b. Rotary flushing: In rotary flushing, the electrode rotates within the cavity. Rotating the electrode aids in flushing out the EDM particles from the cavity. For small round electrodes, manufacturers make multiple cavities in these electrodes to aid in flushing. This is a very efficient method of producing holes without a stud.

c. Orbiting flushing: Orbiting an electrode in a cavity allows the electrode to mechanically force the eroded particle from the cavity. Orbiting flushing is the most efficient method for cutting. Furthermore, if the orbiting is larger than the radius of the flushing holes in the electrode, it will produce no studs.



4.5 EDM process parameters

1. **Spark gap**: It is very small gap between tool and workpiece about 0.0125-0.125 mm. Decrease in spark gap results in lower metal removal rate.
2. **Current**: The D.C. power supply with a current of 0.5 – 400 A is used in this process.
3. **Voltage (DC)**: It is the voltage which is provided by the power supply system. The voltage supplied in the range of 50 – 400 volts.
4. **Pulse duration**: It is the duration between successive pulses for producing sparks. It is about 2 – 2000 μ s. Pulse duration has effect on tool wear rate. Increase in pulse duration results in lower metal removal rate.
5. **Dielectric pressure**: It is the pressure under which the dielectric fluid is forced or circulated. It is less than 0.2 MPa.
6. **Surface finish**: It is the finish obtained after machining. It is about 3 – 10 μ m for rough machining and 0.8 – 3 μ m for machining.
7. **Material removal rate**: It is the volume of metal removed per unit time. It depends of the material to be cut and spark frequency.
8. The material removal rate (MRR) varies inversely as the melting point of the metal.
9. During roughing out of steel with graphite, electrode MRR is about 400 mm³ / min with 50 A.

The selection of EDM parameters is important in determining the accuracy and surface finish obtained for a particular application.

Spark frequency, current & surface finish

- As current is increased, each individual spark removes a larger crater of metal from the work piece. Although the net effect is an increase in material removal rates, when holding all other parameters constant it also has the effect of increasing surface roughness.

- The same effect is also observed when spark voltage is raised.
- Electrical discharge machining equipment is available that is able to operate between 0.5 and 400 amp and with voltages ranging from 40 to 400 V DC.
- Increasing spark frequency and holding all other parameters constant, results in a decrease in surface roughness.
- The frequency capability of EDM machines ranges from a low of 180 Hz when performing roughing cuts, to a high of several hundred kilohertz when generating the fine finishes required for finishing cuts.

Spark gap

- The gap between the electrode and workpiece is determined by the spark voltage and current.
- Typical values for the gap range from 0.012 to 0.050 mm (0.0005 to 0.002 in.).
- The smaller the gap, the closer the accuracy with a better finish and slower material removal rate.
- As the gap decreases, efficient flushing becomes difficult to achieve.
- Increasing the pulse duration of the sparks has the effect of increasing the removal rate, increasing the surface roughness, and decreasing the electrode wear.
- The values of pulse duration available with currently available EDM machines range from a few microseconds to several milliseconds.

Heat affected zone

- Heat affected zone is the area of the base material, which has its properties altered by heat intensive machining in EDM.
- Heat from the arc formation and subsequent flushing of molten material leaves behind un-melted material that cools unevenly forming a layer whose properties are altered.

4.6 Process characteristics

1. Metal removal rate:

- It is the volume of metal removed from the workpiece per unit time.
- The metal removal rate depends upon the current density.
- It increases with increases in the current.
- The metal removal rate in EDM is 5 to 80 mm³/sec, which is very low as compared to conventional machining rate.

2. Tool wear:

- As sparking takes place, tool wear is inevitable.
- Increases in pulse duration decreases tool wear.
- It depends upon metal removal rate, current and the spark gap.

3. Surface finish:

- Surface finish depends on the metal removal rate.
- High metal removal rate produces poor surface finish.
- For rough cuts heavy current is used and for finish cuts less current is used.
- Increase in spark frequency results in improved surface finish.

4. Accuracy:

- Tolerance value of ± 0.05 could be easily achieved in EDM.
- By close control of several variables, a tolerance of ± 0.003 mm can be obtained.
- An overcut of minimum of 100 micron is produced.
- The overcut increases with high current

4.7 Advantages of EDM

- Any material can be machined irrespective of its hardness.
- Any complicated shape can be produced on the work piece.
- There is no contact between the tool and workpiece; therefore, no stresses are produced in the workpiece.
- Delicate workpiece (small electronic parts) which cannot withstand cutting forces can be machined.
- Very fine holes can be easily drilled. Micro holes, deep hole, like in fuel injecting nozzles.
- A very good surface finish can be obtained.
- The process is generally automated, hence little operator skill is required.
- Time of machining is less than conventional machining.

4.8 Limitations of EDM

- High tool wear rate: the wear rate of tool electrode is considerably higher, sometimes more than one electrode may be required to finish the job.
- More energy required: the EDM process needs more energy for the machining than

conventional machining. high specific power consumption.

- Workpiece should be electrically conductive to be machined by EDM.
- Reproduction of sharp corners: it causes reproduction sharp corners at the workpiece surface.
- Surface cracking due to sparks: some materials may surface cracking due to higher energy given by the sparks.
- High machining cost: The setup requires high cost and power hence higher machining cost.
- Dielectric fluid: Dielectric fluid is used to fill the gap between tool and workpiece. It is stored in a tank and circulated through high pressure pump. A dielectric fluid is a liquid which acts as an insulator between consecutive spark discharge and as a conductor during the spark discharge. It has dielectric properties, to act as an insulator at one time and to acts as conductor at other time. Most commonly dielectric fluids are hydrocarbon fluids, silicon-based oils, deionised water, kerosene and water with glycol are also used. Kerosene is suitable for machining intricate shaped jobs as it will give high accuracy, lower tool wear and better quality of surface.

4.9 Application of EDM

- Machining dies for forging, blanking, extrusion etc.
- Machining of hydraulic valve spools.
- Internal threads, internal helical gears can be cut in hardened materials.
- Felting of delicate parts in to vacuum tubes.
- Machining of exotic materials used in aerospace industries, refractory metals, hard carbides, hardened steels.
- Some of the shapes made by EDM process are shown in Figure.



Figure: Difficult internal parts made by EDM process

4.10 Electrical Discharge Grinding (EDG)

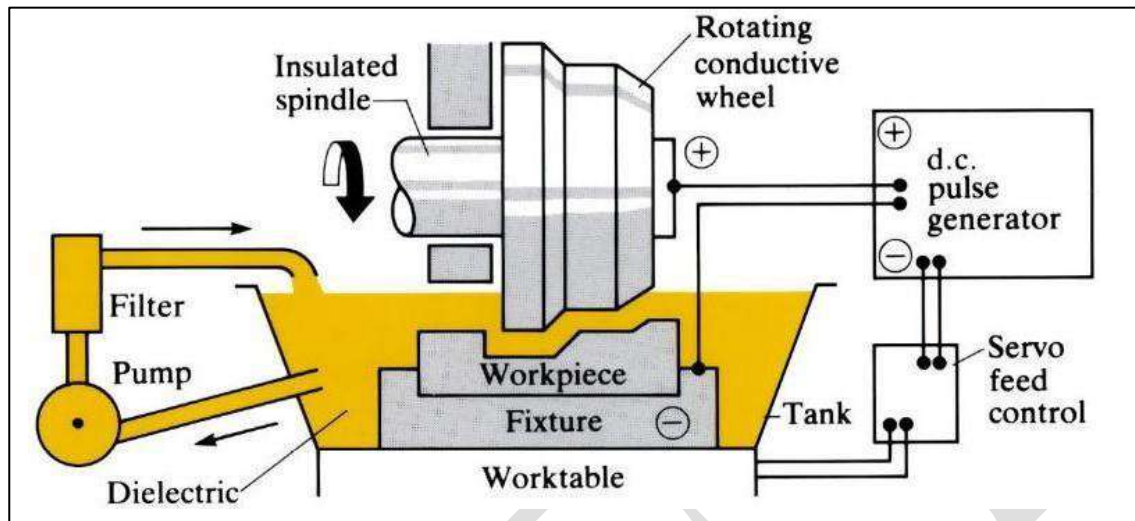


Fig. Electrical discharge grinding

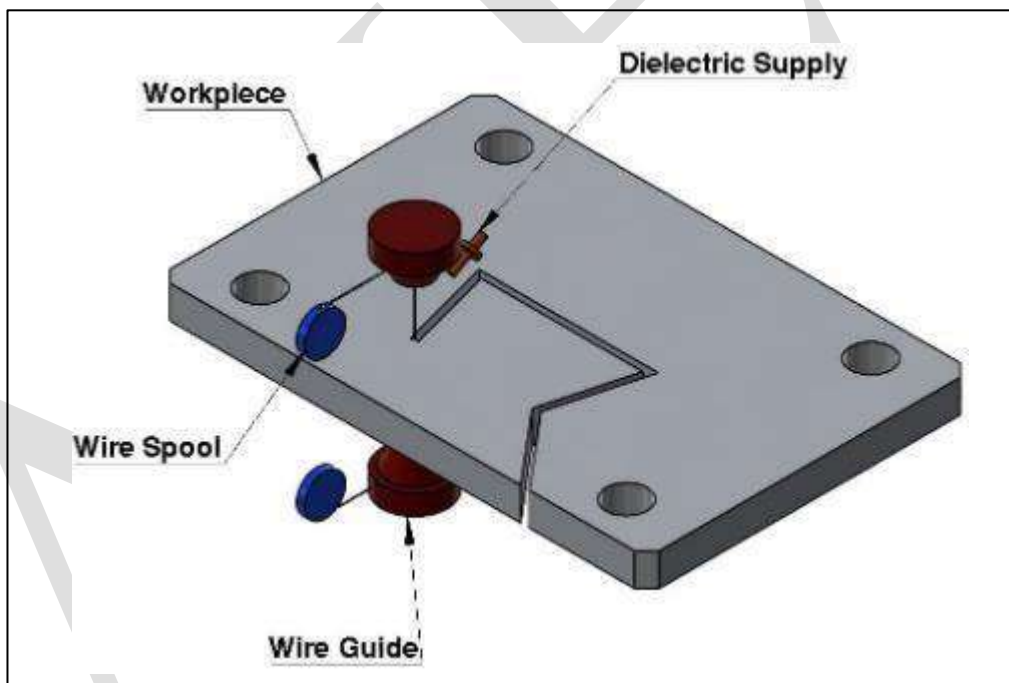
EDG is abrasive machining method where controlled spark discharges from an electrically charged, rotating wheel grinds the workpiece surface. The tool consists of a disc, which rotates around its horizontal axis. Connection via power supply makes the disc act as the cathode and workpiece as the anode. When the disc is brought close to the workpiece, intermittent spark discharges occur, which means that the workpiece is melted and vaporized. The discharge is supported by a dielectric liquid, which in combination with the rotation of the disc helps to cool the process and carry away the removed material while the disc is moved over the workpiece. The spark gap makes the machined surface slightly below the disc's outer diameter. It is important that the electrode, i.e., the rotating wheel is electrically conductive so it's normally made of metal or graphite.

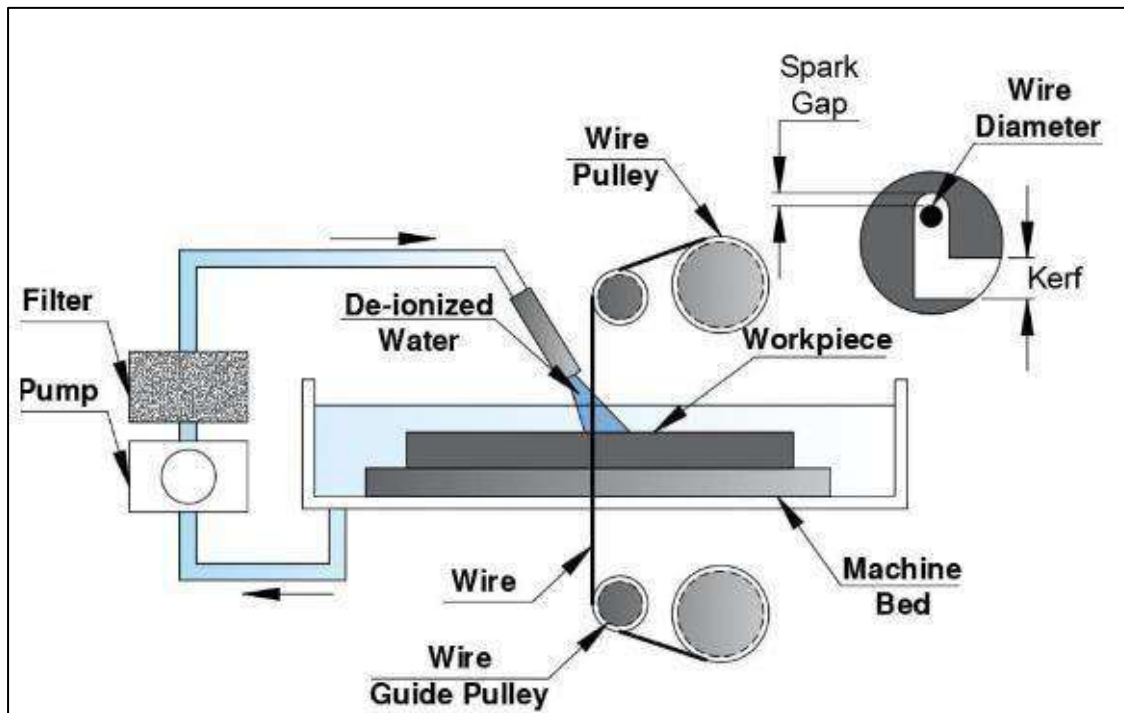
4.11 Wire EDM

A method of precision machining called electrical discharge machining (EDM) removes material from a workpiece using thermal energy rather than mechanical force. A thin, single-strand metal wire and deionized water used to conduct electricity are used in the electrical discharge machining method known as wire electrical discharge machining (EDM) to cut through metal while preventing rust. In wire EDM, the tool electrode is a metallic wire, typically constructed of brass or layered copper. To cut or shape a workpiece, utilize the metallic wire. The thin electrode wire travels along a predetermined route. Although smaller and larger diameters are available, the typical electrode

diameters range from 0.004 inches to 0.012 inches (0.10mm to 0.30mm). The wire is coiled between two spools so that the active portion of the wire is continually changing. This design prevents the wire from eroding to the point where it breaks. The concept of wire EDM is shown in Figure.

In this process, a slowly moving wire travels along a prescribed path and removes material from the work piece. Wire EDM uses electro- thermal mechanisms to cut electrically conductive materials. The material is removed by a series of discrete discharges between the wire electrode and the work piece in the presence of dielectric fluid, which creates a path for each discharge as the fluid becomes ionized in the gap. The area where discharge takes place is heated to extremely high temperature, so that the surface is melted and removed. The removed particles are flushed away by the flowing dielectric fluids. The wire EDM process can cut intricate components for the electric and aerospace industries. This non-traditional machining process is widely used to pattern tool steel for die manufacturing.





The wires for wire EDM is made of brass, copper, tungsten, molybdenum. Zinc or brass coated wires are also used extensively in this process. The wire used in this process should possess high tensile strength and good electrical conductivity. Wire EDM can also employ to cut cylindrical objects with high precision. The sparked eroded extrusion dies are presented in Figure. This process is usually used in conjunction with CNC and will only work when a part is to be cut completely through. The melting temperature of the parts to be machined is an important parameter for this process rather than strength or hardness. The surface quality and MRR of the machined surface by wire EDM will depend on different machining parameters such as applied peak current, and wire materials.

PLASMA ARC MACHINING (PAM)

4.12 Introduction to PAM

A plasma is defined as a superheated, electrically ionized gas. PAC uses a plasma stream operating at temperatures in the range from 10,000 to 14,000°C to cut metal by melting. The cutting action takes place by directing the high velocity plasma stream at the work, thus melting it and blowing the molten metal through the kerf. Plasma is encountered in electrical discharges, such as fluorescent tubes and electric arcs, lightning, high temperature combustion flames and the sun. Most applications of PAC involve cutting of flat metal sheets and plates. Operations include hole piercing and cutting

along a defined path. PAC was initially employed to cut metals that are difficult to machine by conventional methods. However, in recent years, PAC has also been used to cut plain carbon steel, stainless steel and aluminum.

Plasma:

Plasma is a mixture of free electrons positively charged ions and neutral atoms. Plasma is obtained by heating a gas to a very high temperature so that it is partially ionized. Plasma torch confirms the formation of gas in an arc chamber and the arc supplies a large input of electrical energy.

The changes that takes place when gases are heated to a few 1000 degrees are:

1. The number of collisions, elastic or inelastic between atoms increases.
2. The gas ionizes, so that a portion of the atoms are stripped off their outer electrons resulting in the creation of electrons and ions.
3. The electrons thus produced, in turn, collide with atoms, heat them through relaxation processes so that their thermal kinetic energy increases, excite them so that de-excitation light is emitted from the atoms and ionize them so that more electrons and ions are produced.

4.13 Non thermal generation of plasma

- The method of heating the gas by first ionizing is one of the popular methods of generating hot plasma. This can be done either by applying a suitable electric field across the gas column or by imposing the gas to ionizing radiation.
- When gases are heated by an applied electric field ignite supplies the initial electrons, which accelerates in the field before colliding and ionizing the atoms
- The free electrons in turn gets accelerated (increased) and cause further ionization and heating of the gases.
- This continues till steady state is obtained in which the rate of production of free charges is balanced by recombination and loss of the free charges to the walls and electrodes.
- The actual heating of the gas takes place due to the energy liberated when free ions and electrons recombine into atoms or when atoms recombine into molecules.
- The bonding energy thus liberated is manifested in the form of kinetic energy of the atom or molecule formed by the recombination.
- It is thus clear that the heat content of molecular gases is higher than that of mono atomic gases.
- The energy input for such a mechanism is determined by the enthalpy of gas at the required

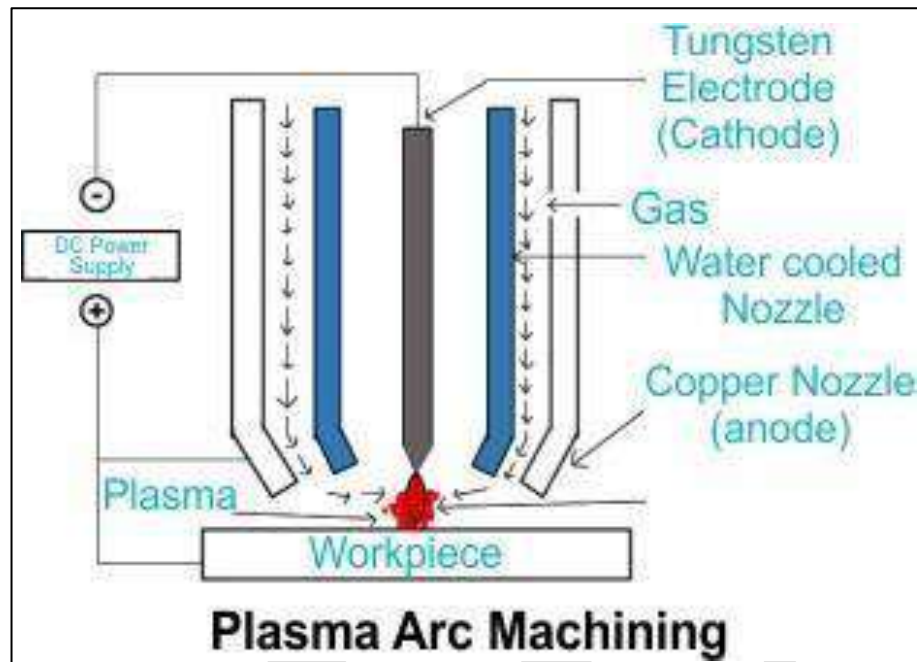
temperature, thermal conductivity of the gas and loss of energy in the form of radiation.

- The electrical conductivity of the gas is a relevant parameter for ascertaining the current density in the plasma for the given field.
- The high velocity electrons of the arc collide with the gas molecules and produce dissociation of diatomic molecules followed by ionization of the beam.
- The plasma forming gas is forced through the nozzle duct in such a manner as to stabilize the arc.
- Heating of the gas takes place in the constricted region of the nozzle duct, resulting in relatively high exit gas velocity and very high core temperatures up to 16000°C.

4.14 Mechanism of Metal Removal

- The metal removal in plasma arc machining is basically due to the high temperature produced.
- The heating of workpiece is as a result of anode heating, due to direct electron bombardment plus convection heating from the high temperature plasma that accompanies the arc.
- The heat produced is sufficient to raise the workpiece temperature above its melting point and the high velocity gas stream effectively blows the molten metal away.
- Under optimum conditions, upto approximately 45% of the electrical power delivered to the torch is used to remove metal from the workpiece.
- Plasma arc cutting was primarily employed to cut metals that form a refractory oxide outer skin.
- The arc heat is concentrated on a localized area of the workpiece and it raises it to its melting point.
- The quality of cut is affected by the heat flow distribution; uniform heat supply throughout the thickness of the material produces a cut of excellent quality.
- The optimum cutting speed (for a given torch) is dependent upon the uniform distribution of heat flow at the plasma to material surface.

4.15 Elements of PAC system



The setup of the process consists of:

- 1) **Plasma cutting torch:** A plasma cutting torch carries a tungsten electrode fitted in a small chamber. At other end of the torch is a small converging orifice called as nozzle. One side of the torch provides a passage for supply of gas into the torch.
- 2) **Tool and workpiece:** The electrode is connected to negative terminal of D.C. power supply and therefore acts as a cathode. The nozzle is made anode by connecting to the positive terminal of the power supply through a suitable resistor. This resistor limits the current through the nozzle to about 50 A. The workpiece to be machined is also connected to the positive terminal of the supply. The anode and cathode are separated by an insulator.
- 3) **Gas supply unit:** It consists of gas cylinder, regulators and gas supply hoses. The commonly used gases are argon or nitrogen or the mixture of two. For certain useful purposes, a percentage of hydrogen may be added. The choice of the gas depends upon the material to be cut, economics and the quality of the cut edge desired. The flow rate of the gas varies directly with the thickness of the workpiece. Typical gas flow rate is 2 to 11 m³/hr.
- 4) **Cooling system:** A provision is made for circulating the water around the torch so that the electrodes and the nozzle both remains water cooled.
- 5) **Power supply:** A D.C. power supply of 400 V, 200 KW and upto 10000 A is supplied to the

nozzle. When supply is made ON, a strong arc is struck between the electrode and the nozzle and then gas is forced into the chamber. When the gas molecules collide with the high velocity electrons of the arc, plasma is formed. This plasma is forced through the nozzle (anode) onto the workpiece. The heat produced from this jet of plasma is sufficient to raise the workpiece temperature above its melting point and high velocity gas stream effectively blows the molten metal away.

Working of Plasma Arc Machining

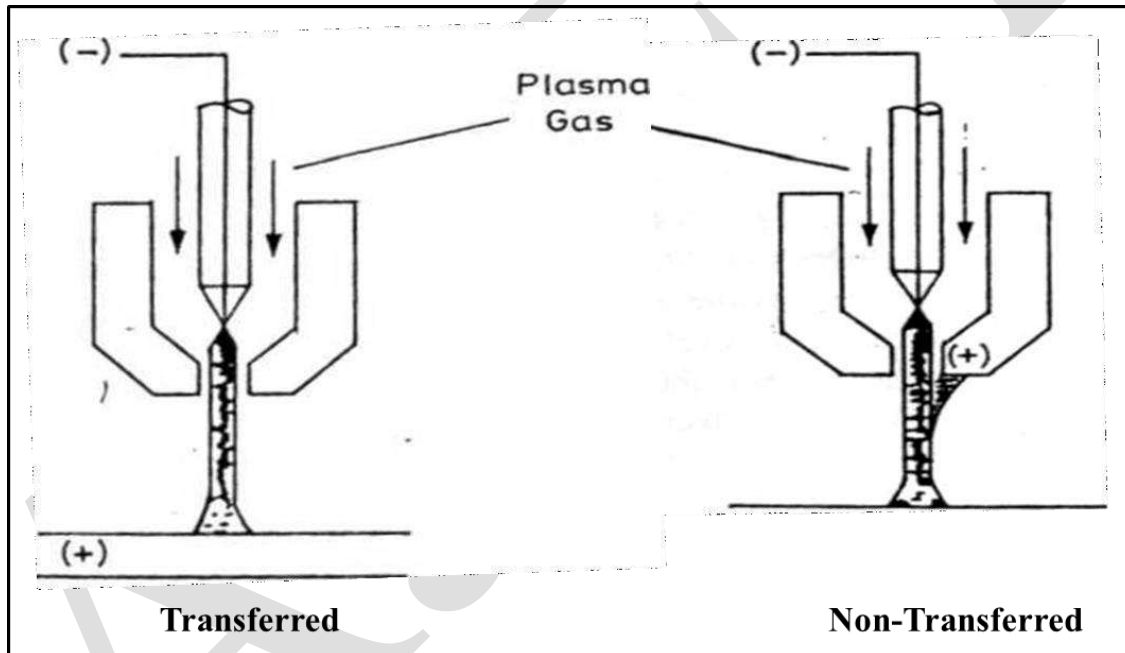
- When DC power is applied to the circuit, a potent arc emerges between the cathode (electrode) and the anode (nozzle). Subsequently, gas is introduced into the chamber, selected from options like hydrogen, nitrogen, argon, or a mixture tailored to the specific metal being worked upon. The gas is then heated to extraordinarily high temperatures, ranging from 11,000°C to 28,000°C, utilising the arc formed between the cathode and anode. As the arc interacts with the gas, electrons collide with gas molecules, causing them to dissociate into individual atoms.
- Due to the arc's intense heat, certain atoms lose their electrons, leading to ionisation, thereby transforming the gas into plasma—an electrically charged state. This ionised gas releases a significant amount of thermal energy. Directed towards the workpiece with high velocity, the plasma jet benefits from the electric arc in several ways. It further elevates the ionised gas's temperature, almost aligns the beam in parallel, and enhances the gas's velocity.
- Upon reaching the workpiece, the plasma jet efficiently melts the material, while the high-velocity gas effectively blows away the molten metal. This process of plasma arc machining effectively removes material from the workpiece, demonstrating its remarkable utility in various industrial applications.

4.16 Plasma torch

- The modified stabilized arcs came to be known as plasma torches or plasma jets because the plasma of the arc column is pushed out of the nozzle in the form of a high velocity jet.
- An additional feature of the plasma torches is that the anode tube is in the form of a constricting nozzle, so that further confinement and acceleration of the arc takes place, in addition to the constriction due to the stabilizing flow and 'Maecker effect'.
- The plasma torch consists of a non consumable cathode rod of 2 percent thoriated tungsten and a converging anode nozzle with a suitable orifice.
- The two electrodes are separated by an insulator of high carbonate resin or some suitable rubber.

Usually for vortex stabilized torches, the gas is fed tangentially through an inlet in the insulator. For sheath stabilization, the gas is fed through small parts around the cathode. Both the electrodes are water cooled.

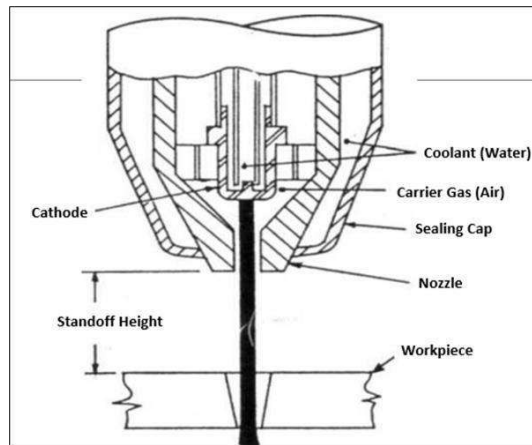
PAC system uses DC power source. PAC systems operate either on non-transferred arc mode or transferred arc mode. In case of a transferred arc mode, the arc is maintained between the electrode (negative polarity) and the electrically conductive work piece (positive polarity). In non-transferred arc mode, the thermal efficiency is low (65-75%) and power is transferred between the electrode and the nozzle. This non-transferred arc ionizes a high velocity gas that is streaming towards the work piece. The work piece may be electrically conductive or non-conductive.



There are many torch design practically used:

1. Air plasma:

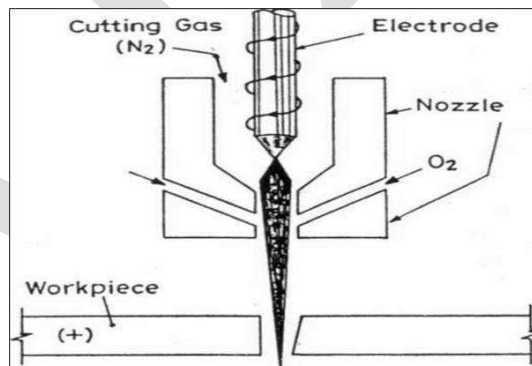
- Air plasma torch uses compressed air as the gas that ionizes and does cutting.
- The air to be used should be uncontaminated.
- The nozzle of this torch may result in premature failure because of double arcing i.e., arcing between the electrode and the nozzle, and between the nozzle and the work piece.
- Zirconium or hafnium are used as electrode material because of their higher resistance to oxidation.



Air plasma torch

2. Oxygen injected:

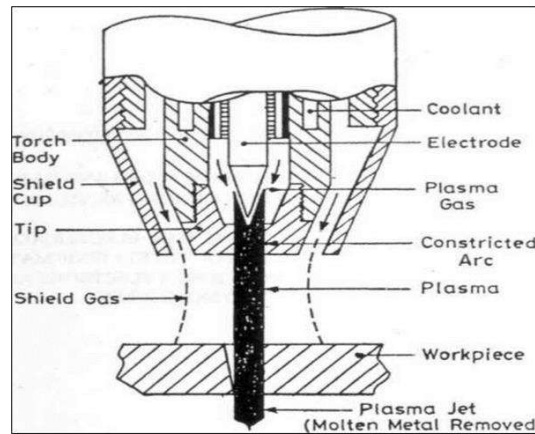
- To avoid oxidation of electrode (or to enhance the life of the electrode), oxygen injected torch uses nitrogen as the plasma gas.
- Oxygen is injected downstream of the electrode. However, it lowers down the nozzle life. This torch gives high MRR and poor squareness of the cut edges. It is commonly used for mild steel plate cutting.
- The presence of oxygen in the air helps in increasing MRR in case of oxidizable materials like steel.



Oxygen injected plasma torch

3. Dual gas system:

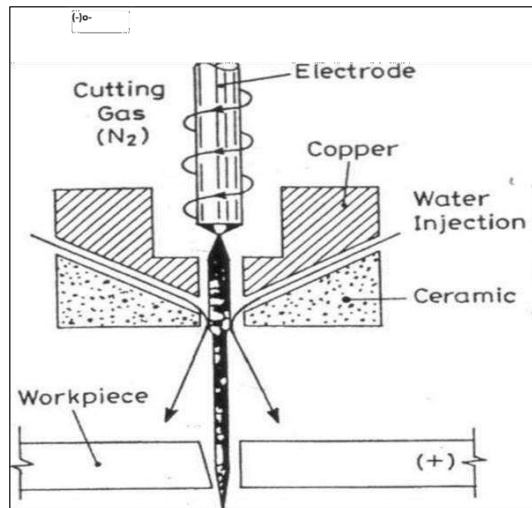
- It uses one gas (nitrogen) as the plasma gas while another gas as the shielding gas (O_2 , CO_2 , argon-hydrogen etc).
- Secondary or shielding gas is chosen according to the material to be cut. Secondary gas system helps in maintaining sharp corners on the top side of the cut edges.



Dual gas plasma torch

4. Water injected torch:

- Water (pressure = 1.2 MPa) is injected (radially or swirling vertically) to constrict the plasma. A small quantity (about 10%) of water vaporizes. This thin layer of steam constricts the plasma and also insulates the nozzle.
- Nitrogen at about 1 MPa is used as the plasma.
- To avoid double arcing, the lower part of the nozzle is made of ceramic.
- Water constriction helps in reducing smoke, enhancing nozzle life, reducing HAZ, and limiting formation of oxides on the cut edges of the workpiece.
- Water muffler (a device that produces a covering of water around the plasma torch and extends down to the work surface) helps in reducing smoke and noise.
- Water mixed with a dye also absorbs part of the ultraviolet rays produced in PAC.
- In some cases, a water table is also used to reduce the level of noise and extent of sparks. Water below the workpiece quenches sparks and damps sound level.
- Underwater PAC systems are also available which effectively reduce the noise and smoke levels.



Water injected plasma torch

4.17 Process parameters

1) Stand-off distance: increase in the stand-off distance reduces the depth of penetration and hence narrows the cut width at the bottom. The stand-off distance depends on the thickness of the metal to cut. The typical value of the stand-off distance varies from 5 mm to 10 mm.

2) Cutting speed: increase in the cutting speed reduces the depth of immersion of the plasma jet, leading to narrowing of the cut in the lower portion. Decrease in the cutting speed will cause the opening of the cut at the bottom of the workpiece. For example, the typical cutting speed for aluminium at 80 KW power and 4 mm orifice diameter is 8 mm/sec.

3) Gas: the gas flow rate is directly proportional to the thickness of the material. The selection of a particular of a particular gas depends on the quality of cut and the economics.

4.18 Process Characteristics

- The cutting rate in PAM are 250 to 1700 mm/min. depending on the thickness and material of workpiece. A 25 mm thick aluminium plate can be cut at 4000 mm/min. A 6 mm carbon steel sheet can be cut at 4000 mm/min. the use of water injection can increase the cutting rate in carbon steel to 6000 mm/min for a 5 mm thick plate. Surface cut by plasma torch are smoother than cut by oxyacetylene flame but edges are round. The corner radius is a minimum of 4 mm on thinner plates. The walls of a cut have a V shape with an angle of 5 to 10°.
- Accuracy on width of the slots and diameter of holes is from ± 0.8 mm on 6-30 mm thick plates and ± 0.3 mm on 100 -150 mm thick plates.
- The depth of heat affected zone depends on the work material, thickness and cutting speed.
- On a work piece of 25 mm thickness the heat affected zone is about 4 mm and is less at high cutting speed.

4.19 Safety Precautions

- Machines are operated after ensuring that all health safety measures are observed.
- The plasma flame emits a highly intense beam, particularly strong in ultraviolet and infrared radiations.
- These radiations, if taken in large doses, might cause permanent damage to the eye in the form of cataract.
- Over exposure results in reddening and a gritty feeling in the eyes due to loss of sleep.
- Over exposure to ultraviolet rays may also cause painful skin burns, and in extreme cases, these lead to cancer.

- Though these radiations are very intense in all the plasma flames, the laminar plasma flame is much more dangerous.
- Hence. It is imperative that proper glasses and proper dresses are worn before going close to the torch.
- The glasses should be good at ultraviolet and infrared cut off and most of the body should be covered.
- It is necessary to operate the torch in an airy room with proper exhaust, as many toxic gases, etc) are synthesized in the atmosphere.
- These are very much above permissible levels when proper exhausting facilities do not exist.
- The noise level while operating the torch is also very high. Hence, ear muffs or plugs should be used.
- Asbestos gloves with an inner layer of leather should be worn for operating hand torches.
- Health physicians should be consulted regarding the number of hours the operator can remain to the plasma torch.
- In the spraying and chemical synthesis of toxic powders, it is preferable to mechanise the torch and operate it from a distance.

4.20 Applications of PAM

- **Cryogenic and High-Temperature Alloys:** PAM is extensively employed in working with cryogenic and high-temperature corrosion-resistant alloys, thanks to its ability to handle challenging materials effectively.
- **Titanium Plate Cutting:** PAM is an ideal choice for cutting titanium plates up to 8 mm in thickness, providing precise and efficient machining capabilities.
- **Aerospace and Defense:** PAM plays a vital role in the aerospace and defence industries, being utilised in nuclear submarine pipe systems and welding steel rocket motor cases, where precision and reliability are paramount.
- **Stainless Tubes and Tube Mills:** PAM is a staple for applications related to stainless tubes and tube mills, enabling accurate cutting and shaping of these materials.
- **Medical Device Manufacturing:** PAM is also employed in manufacturing medical devices, particularly in intricate and precision-focused components, where its versatility and accuracy are crucial.
- **Automotive and Power Generation:** The automotive industry utilises PAM for fabricating

critical components, such as engine parts and exhaust systems. Additionally, power generation equipment benefits from PAM's capability to work with high-temperature materials.

4.21 Advantages of PAM

- **Versatility:** PAM can easily handle hard and brittle metals, making it applicable to a wide range of metal materials.
- **Universal Applicability:** Almost all types of metals can undergo plasma arc machining, showcasing its broad range of applications.
- **Enhanced Cutting Rate:** One of the primary benefits is the ability to achieve higher cutting rates, ensuring increased productivity and efficiency.
- **Superior Dimensional Accuracy:** PAM excels in machining small cavities, delivering superior dimensional accuracy, and allowing for intricate and precise work.
- **Simplicity and Efficiency:** The process of plasma arc machining is straightforward to execute, and its efficiency contributes to streamlined manufacturing operations.
- **Vital Role in Jet Engine Repairs:** PAM plays a significant role in the automatic repair of jet engine blades, showcasing its importance in critical industries like aerospace and aviation.

4.22 Limitations of PAM

- **High Equipment Cost:** PAM requires various specialised equipment, which can be expensive, posing a significant initial investment for implementation.
- **Inert Gas Consumption:** The process consumes substantial amounts of inert gases like nitrogen or argon, contributing to increased operating costs.
- **Narrow Surfaces:** PAM can produce narrow and unnecessary surfaces, which may not be desirable in certain applications.
- **Surface Changes:** One drawback is the occurrence of surface changes on the workpiece, which may require additional finishing or post-processing steps.
- **Safety Precautions:** The operator or personnel involved in the process must take proper safety precautions due to the intense heat and potential hazards associated with plasma arc machining.
- **Eye Protection:** PAM can emit intense light that may harm the human eye. Operators must wear appropriate goggles or helmets with protective filters to shield their eyes during the operation.

QUESTION BANK

Electrical Discharge Machining (EDM)

1. Define Electrical Discharge Machining (EDM).
2. List advantages and disadvantages of Electrical Discharge Machining (EDM).
3. Mention the applications of Electrical Discharge Machining (EDM).
4. Explain the mechanism of Metal removal in EDM process.
5. Briefly explain the rotary pulse generator in EDM process, with a neat sketch. (6 Marks) **(VTU Dec 2011)**
6. Explain the various parameters that govern the metal removal rate, using a R-C circuit. (8 Marks) **(VTU Dec 2011)**
7. Explain with a neat sketch, the traveling wire EDM. (6 Marks) **(VTU Dec 2011)**
8. List the commonly used dielectric fluids in EDM process. What properties should they possess? (6 Marks) **(VTU June 2012)**
9. Define dielectric. Write a note on it indicating its functions and characteristics. (10 Marks) **(VTU Dec 2012)**
10. Explain the mechanism of EDM showing the circuit and movements of ions. (10 Marks) **(VTU Dec 2012)**
11. Explain the advantages of EDM. (5 Marks) **(VTU Dec 2012)**
12. Draw a diagram of electrode feed control of EDM and label it. (5 Marks) **(VTU Dec 2012)**
13. Discuss the factors influencing the choice of electrode material in EDM. (5 Marks) **(VTU June/July 2013)**
14. Explain with help of neat sketches any two types of flushing methods used in EDM. (5 Marks) **(VTU June/July 2013)**
15. Explain with the help of neat sketches, the mechanism of metal removal in EDM process, and also mention its advantages and disadvantages. (10 Marks) **(VTU June/July 2013)**
16. Explain with sketch the electrode feed control in electric discharge machining process. (8 Marks) **(VTU June/July 2014)**
17. Explain the different dielectric flow patterns of electric discharge machining process (6 Marks) **(VTU June/July 2014)**
18. Explain with sketch the traveling wire electric discharge machining process. (6 Marks) **(VTU**

June/July 2014)

19. Name some of the dielectric fluids commonly used in EDM. Name some of the tool material used in EDM. (6 Marks) **(VTU Dec 2014/Jan 2015)**
20. What are the basic requirements of the dielectric fluid used in EDM? (4 Marks) **(VTU Dec 2014/Jan 2015)**
21. With the help of neat sketch, explain wire cut electrical discharge machining. (10 Marks) **(VTU Dec/Jan 2015)**
22. Explain with sketch, the mechanism of metal removal in electric discharge machining. (7 Marks) **(VTU June/July 2015)**
23. Explain the elementary relaxation circuit for EDM. (7 Marks) **(VTU June/July 2015)**
24. Explain the different methods of dielectric flushing in Electrical Discharge Machining. (6 Marks) **(VTU June/July 2015)**
25. Draw neat diagram of EDM (Electrical Discharge Machining). Explain its construction and working. (10 Marks) **(VTU June/July 2016)**
26. Explain briefly EDM process characteristics. (10 Marks) **(VTU June/July 2016)**

Plasma Arc Machining (PAM)

1. Define Plasma Arc Machining (PAM).
2. List advantages and disadvantages of Plasma Arc Machining (PAM).
3. Mention the applications of Plasma Arc Machining (PAM) process.
4. Discuss the various process parameters and characteristics involved in Plasma Arc Machining (PAM).
5. Explain the mechanism of Metal removal in Plasma Arc Machining (PAM) process.
6. Explain with a neat sketch, the non-thermal generation of plasma and mechanism of metal removal. (9 Marks) **(VTU Dec 2011)**
7. What are the different modes of operation of plasma torches? Explain. (4 Marks) **(VTU Dec 2011)**
8. Discuss the plasma arc surfacing and plasma arc spraying. (7 Marks) **(VTU Dec 2011)**
9. Sketch and explain transferred and non-transferred plasma arc system. (10 Marks) **(VTU June 2012)**
10. Write a note on process performance in plasma arc cutting process. (4 Marks) **(VTU June 2012)**
11. Explain the non-thermal generation of plasma with the related diagram. (10 Marks) **(VTU Dec 2012)**

12. With a neat sketch, explain the plasma arc machining (PAM) process and also mention its applications. (10 Marks) **(VTU June/July 2013)**
13. Which are the important considerations are to be made in the design of plasma torch? (6 Marks) **(VTU June/July 2013)**
14. Mention any two advantages and disadvantages of plasma arc machining. (4 Marks) **(VTU June/July 2013)**
15. Explain with sketch the principle of working of plasma arc machining. (8 Marks) **(VTU June/July 2014)**
16. Explain the transferred arc and non transferred arc method of plasma arc machining. (6 Marks) **(VTU June/July 2014)**
17. Mention the advantages, disadvantages and applications of plasma arc machining. (6 Marks) **(VTU June/July 2014)**
18. Explain the basic principle of PAM (4 Marks) **(VTU Dec 2014/Jan 2015)**
19. With a neat sketch, explain the working of PAM. List out the advantages and limitations of PAM process. (16 Marks) **(VTU Dec 2014/Jan 2015)**
20. Explain with diagram the working of plasma arc machining. (10 Marks) **(VTU June/July 2015)**
21. What are the factors that govern the performance of plasma arc machining? Explain any one of them. (6 Marks) **(VTU June/July 2015)**
22. Write the applications of plasma arc machining. (4 Marks) **(VTU June/July 2015)**
23. Explain the construction and working principle of Plasma Arc Machining (PAM) with neat sketch. (8 Marks) **(VTU June/July 2016)**
24. List the general guideline for designing the torch. (6 Marks) **(VTU June/July 2016)**
25. What are the application of PAM and also mention advantages and limitations ? (6 Marks) **(VTU June/July 2016)**

Module-5

LASER-BEAM MACHINING (LBM)

ELECTRON BEAM MACHINING (EBM)

SYLLABUS

LASER BEAM MACHINING (LBM)

Introduction, generation of LASER, Equipment and mechanism of metal removal, LBM parameters and characteristics, Applications, Advantages & limitations.

ELECTRON BEAM MACHINING (EBM)

Introduction, Principle, equipment and mechanism of metal removal, applications, advantages and limitations.

LASER BEAM MACHINING (LBM)

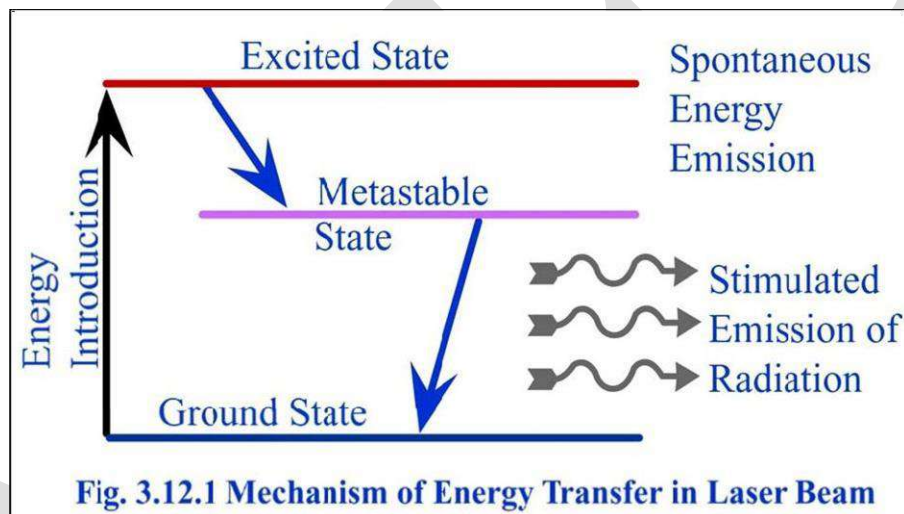
5.1 Introduction to LBM

Laser-beam machining is a thermal material-removal process that utilizes a high-energy, coherent light beam to melt and vaporize particles on the surface of metallic and nonmetallic workpieces. Lasers can be used to cut, drill, weld and mark. LBM is particularly suitable for making accurately placed holes. A schematic of laser beam machining is shown in Figure.

Laser machining is a technology that uses a laser beam (narrow beam of intense monochromatic light) to cut required shapes or profile or pattern in almost all types of materials. Some of the examples include metals, ceramics, food products, leather etc. In this process, the output of a high power laser beam is directed in a programmed manner towards the material required to be cut. The high amount of heat thus generated either melts, burns, or vaporizes away the material at the focused region. The process can be used to make precise holes in thin sheets and materials. The laser beam cutting finds its applications in a variety of fields. The fields where laser beam has been successfully used are cloth and plastic cutting, laser marking, laser welding, laser drilling, cleaning and surface treatments.

5.2 Generation of Laser

The word laser is an acronym for Light Amplification by the Stimulated Emission of Radiation. When an atom absorbs a quantum of energy from a light source, the orbital electron of an atom jumps to a higher energy level. The electron later drops to its original orbit and emits the absorbed energy. If the electron, which is already at high energy level, absorbs the second quantum of energy, it emits two quanta of energy and after emitting the energy it returns back to its original orbit. The energy that is radiated has the same wave length as the stimulating energy. The laser material when placed in an optical cavity and exposed to light energy keeps storing the energy. The energy initially builds up in the laser material and finally gets emitted in the form of a highly amplified light beam. The basic mechanism of energy transfer in laser beam is shown in Fig. 3.12.1 and laser beam generation is schematically shown following figure.



Attributes of Laser Beam

The attributes of laser light are as follows:

- It is coherent i.e. all photons that make up the beam are in phase with each other. This optical property of light that mostly distinguishes the laser from other light source is 'coherence'. The laser is regarded, quite correctly as the first truly coherent light source. Other light sources, such as the sun or a gas discharge lamp, are at best only partially coherent.
- It is highly collimated; i.e. a parallel beam is produced. Light rays are almost perfectly parallel.
- It is monochromatic means the light is of one color, or of one wavelength.

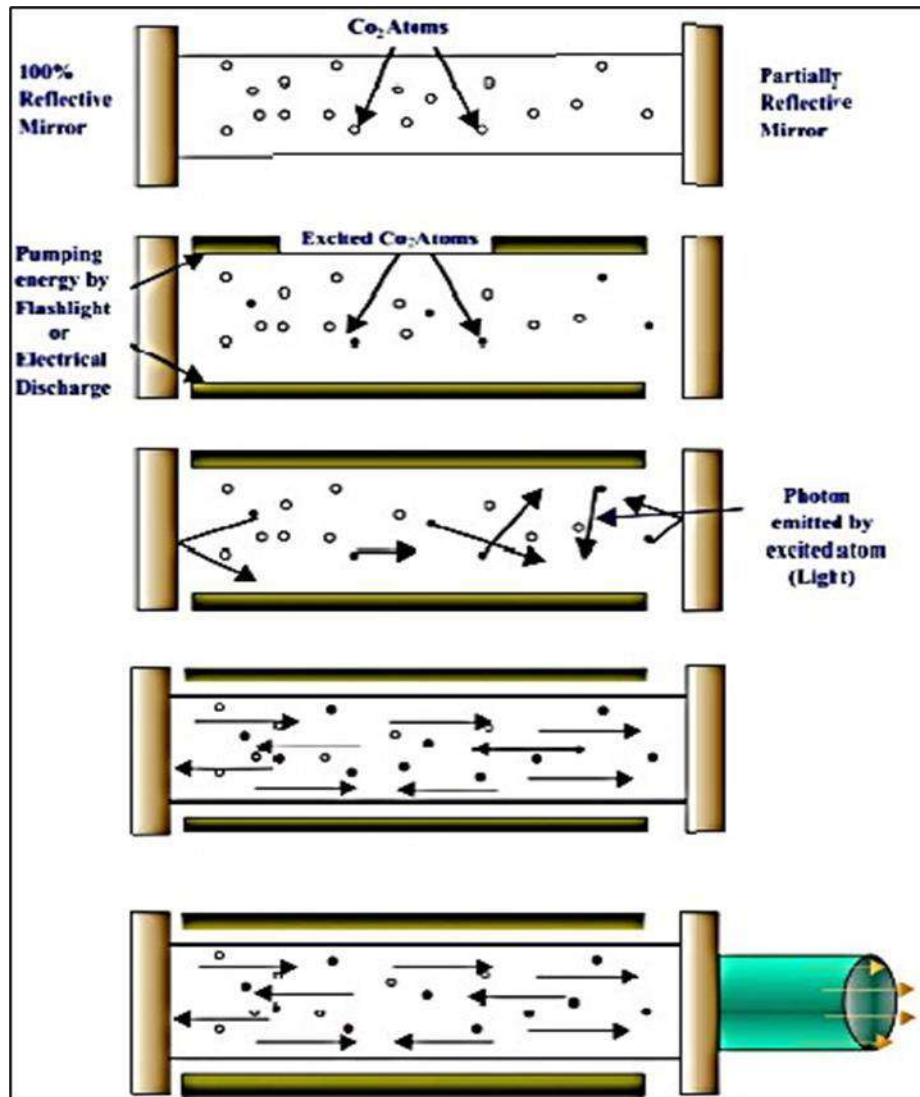


Fig. Laser beam generation principle

Different media used to stimulate the photons generate different wavelengths, but each type of laser has a specific wavelength.

Different types of lasers are available for manufacturing operations which are as follows:

- CO₂ (pulsed or continuous wave): It is a gas laser that emits light in the infrared region. It can provide up to 25 kW in continuous-wave mode.
- Nd:YAG: Neodymium-doped Yttrium-Aluminum-Garnet (Y₃Al₅O₁₂) laser is a solid-state laser which can deliver light through a fibre-optic cable. It can provide up to 50 kW power in pulsed mode and 1 kW in continuous-wave mode.

5.3 Equipment of LBM

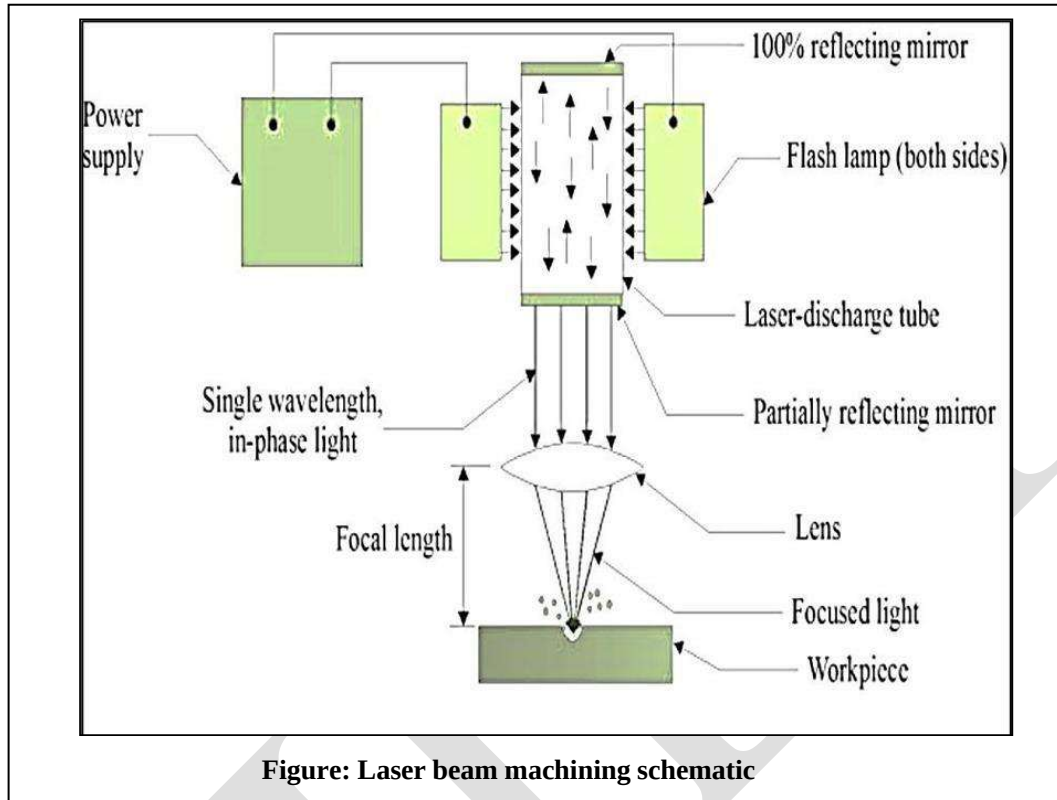


Figure: Laser beam machining schematic

Laser beam machining utilizes the narrow beam of intense monochromatic light which melts and vaporize the material of the workpiece. The setup for this process is shown in the Figure. It mainly consists of:

1. **Laser generation unit:** In this unit ruby rod, flash lamp, power supply, mirrors are used for production of laser beams. The solid-state laser i.e. ruby rod is used in the form of cylindrical crystal with 10 mm diameter and 150 mm long. The ends are finished to close optical tolerances. The flash lamp is wound around the ruby rod and it is connected to electric power supply. The inner surface of the container wall is made highly reflective all the light on the ruby rod. The electrical power supply is designed to give 250 to 1000 watts energy to the flash lamp. The ruby crystal is surrounded by a helical flash tube containing inert gas xenon. The flash tube is connected to a pulsed high voltage source by which the xenon transforms the electrical energy into white light flashes (light energy).
2. **Cooling arrangement:** The ruby rod becomes less efficient at higher temperatures and gives maximum efficiency at when kept at a very low temperature. Hence cooling system is provided

in which liquid nitrogen is used, sometimes air- or water-cooled provision is also made, but it has less effectiveness compared to liquid nitrogen.

3. **Collimating lens:** The highly amplified beam of laser light is focused on workpiece through a lens, it gives high energy density which melts and vaporizes the metal.
4. **Workpiece tables:** The workpiece to be cut is placed on the aluminium work table which is resistant to the laser beam. The laser head is transverse over the workpiece and table can be moved as per requirement.

5.4 Mechanism of Material Removal / Cutting using Laser Beam:

The mechanism of material removal by laser beam is given in the following steps:

- Place the work piece on the table. As there is absence of cutting forces, fewer work holding devices are needed.
- The focal point of the laser is intentionally focused onto the surface of the workpiece for providing the heat in a concentric manner.
- Due to the striking of laser beam, heat is generated at the work-piece surface and as a result, the material vaporizes instantly, producing kerf in the material.
- The movement of machine-axis is through the computer control which helps to achieve the required profiles on the work piece. Heat Affected Zone (HAZ) is minimal in laser as compared to flame cutting.
- To clear the molten metal that has yet not vaporized or clogged on the surface of the workpiece, the assist gas, (inert gas or exothermic gas is used for this propose) under pressure is passed on-to the workpiece. The use of different assist gases with different work materials is given in table below.

➤ Mild steel ➤ Stainless steel ➤ Aluminum ➤ Titanium ➤ Nonmetals	➤ Oxygen ➤ Oxygen or nitrogen (nitrogen leaves an oxide free edge that can improve weldability) ➤ Nitrogen ➤ Argon (an inert gas because of its reactivity) ➤ Air or inert gas
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5.5 Parameters of LBM

1. **Focal length:** The workpiece should be placed close to the lens for machining. If focal length is less, straight holes will be produced. If the focal length is more, taper holes will be produced.
2. **Power density:** The power density of the beam determines whether the beam will perform the function of cutting or welding. For machining operation, the power density should be higher i.e. around $1.5 \times 10^7 \text{ W/cm}^2$.
3. **Focal length:** It is the distance between laser beam and the workpiece. If the laser is very close to workpiece, the divergence of the beam occurs for small focal length in the metal cutting.

5.6 Characteristics of Laser

1. A laser beam is highly monochromatic.
2. Laser ray is highly pure beam of light.
3. It is an intense beam of light.
4. Highly directional.
5. Highly collimated.
6. The light produced by laser is coherent.

These properties allow laser light to be focused, using optical lens, onto a very small spot with resulting high power densities.

5.7 Applications of LBM

1. **Laser Welding:** It is useful for joining sheet metal or stock pieces of about 2.5 mm thick or less.

Many metals and alloys can be welded by laser like low carbon steel, stainless steel, titanium and its alloys, silicon bronze, etc. another advantage of laser welding is no need of grinding after welding.

2. **Laser Cutting:** A laser beam can be used in cutting metals, plastics, ceramics, textiles, cloth and even glass. It can also use for cutting complex shapes with sharp concern and slots. It is useful for cutting of steel, titanium, nickel, certain refractory materials and plastics but cutting of aluminium and copper has been especially problematic.
3. **Laser Engraving:** Laser beam can be utilized for making or engraving so as to produce controlled surface pattern on a workpiece.
4. **Laser Drilling:** It has ability to make small and very small holes of shallow depth. Most of the laser beam machining are useful for drilling of small holes in fuel filters, carburettor nozzles, hypodermic needles, jet engine blades cooling holes, etc. in air craft turbine industry, laser drilling used for making holes for air bleeds etc

5.8 Advantages of laser cutting

- There's no direct contact between tool and workpiece as no physical tool is required
- Tool's absence offers no tool wear problems
- Micro holes with large depth to diameter ratio can be drilled by using LBM
- The properties of machined materials or magnetic materials does not get affected by laser beam machining
- LBM is capable to cut through air, gas, vacuum and even through liquid
- In this process no burrs or chips are produced
- Angular drilling or cutting can be obtained by tilting the workpiece
- Mechanical force is not exerted on the workpiece results in smooth machining of fragile workpiece
- Any metals or non-metals can be machined like tungsten, ceramics, zincromium, etc
- Laser beam is capable to travel longer distances without diffraction and can be focused at longer distances where weld, drill and cut is not easily possible
- It is used for various operations like drilling, engraving, cutting, welding, trimming, etc

5.9 Limitations of laser cutting

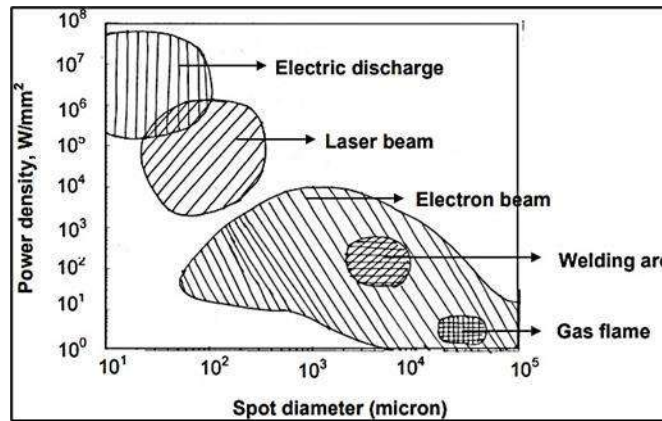
- The capital and operating cost is very high

- The material removing rate is very low
- It cannot cut the reflective and highly conductive material like aluminium, copper and its alloys
- Skilled operator is required for channeling and operating the process
- The output energy of the laser is difficult to control precisely
- Safety procedures are to be followed strictly for safe and trouble-free performance
- Machined holes may have taper from entry to exit
- Flash lamp has limited life

Electron Beam Machining (EBM)

5.10 Introduction to EBM

Electron Beam Machining (EBM) and Laser Beam Machining (LBM) are thermal processes considering the mechanisms of material removal. However electrical energy is used to generate high-energy electrons in case of Electron Beam Machining (EBM) and high-energy coherent photons in case of Laser Beam Machining (LBM). Thus these two processes are often classified as electro-optical-thermal processes. There are different jet or beam processes, namely Abrasive Jet, Water Jet etc. These two are mechanical jet processes. There are also thermal jet or beams. A few are oxyacetylene flame, welding arc, plasma flame etc. EBM as well as LBM are such thermal beam processes. Figure shows the variation in power density vs. the characteristic dimensions of different thermal beam processes. Characteristic length is the diameter over which the beam or flame is active. In case of oxyacetylene flame or welding arc, the characteristic length is in mm to tens of mm and the power density is typically low. Electron Beam may have a characteristic length of tens of microns to mm depending on degree of focusing of the beam. In case of defocused electron beam, power density would be as low as 1 W/mm^2 . But in case of focused beam the same can be increased to tens of kW/mm^2 . Similarly, as can be seen in Fig., laser beams can be focused over a spot size of $10 - 100 \text{ }\mu\text{m}$ with a power density as high as 1 MW/mm^2 . Electrical discharge typically provides even higher power density with smaller spot size.



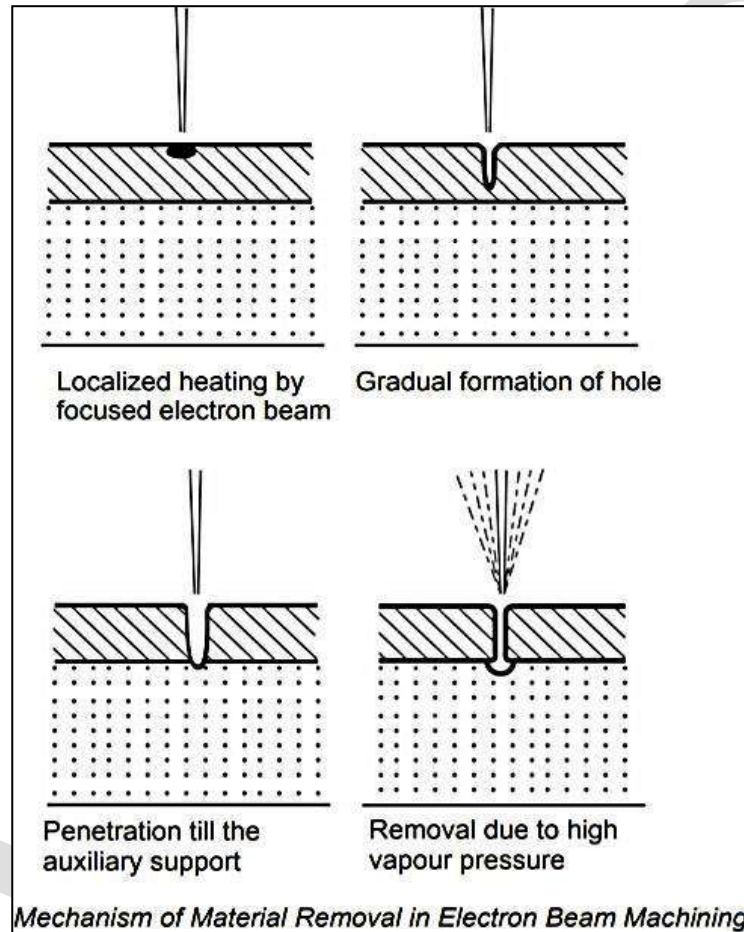
Variation in energy density with spot diameter of thermal beam processes

EBM and LBM are typically used with higher power density to machine materials. The mechanism of material removal is primarily by melting and rapid vaporisation due to intense heating by the electrons and laser beam respectively.

5.11 Principle of EBM

- Electron beam is generated in an electron beam gun. The construction and working principle of the electron beam gun would be discussed in the next section.
- Electron beam gun provides high velocity electrons over a very small spot size. Electron Beam Machining is required to be carried out in vacuum.
- Otherwise the electrons would interact with the air molecules, thus they would lose their energy and cutting ability.
- Thus the workpiece to be machined is located under the electron beam and is kept under vacuum.
- The high-energy focused electron beam is made to impinge on the workpiece with a spot size of 10 – 100 μm .
- The kinetic energy of the high velocity electrons is converted to heat energy as the electrons strike the work material.
- Due to high power density instant melting and vaporization starts and “melt – vaporisation” front gradually progresses, as shown in Figure.
- Finally the molten material, if any at the top of the front, is expelled from the cutting zone by the high vapour pressure at the lower part.

- Unlike in Electron Beam Welding, the gun in EBM is used in pulsed mode. Holes can be drilled in thin sheets using a single pulse. For thicker plates, multiple pulses would be required. Electron beam can also be maneuvered using the electromagnetic deflection coils for drilling holes of any shape.



5.12 Working

- A tungsten filament is heated to temperature above 2000°C and thus cathode is heated. Thus a high beam of electrons is emitted from the cathode.
- These electrons move towards the anode when a high potential difference is applied between the heated cathode and anode.
- Focusing coils are used to focus and direct the beam on the workpiece.
- Magnetic lens focus the electron beam on the work surface.
- The kinetic energy of electron beam is transferred to the workpiece in the form of heat energy as

the electrons strikes the surface.

- The speed of electron is around 75% of speed of light.
- As the high speed electrons strike the work surface the base metal is melted continuously and vapourised at the point where the beam is directed on workpiece. Thus a hole of small diameter is created on the workpiece.
- On the exit side of hole backing material is used. Electron beam after complete penetration into the workpiece partly penetrates into the backing material. Backing materials, melts vapourises and comes out of hole at high pressure.
- The molten material comes out along with vapourised backing material. Thus the machining is performed in a vacuum chamber.

5.13 Equipment and mechanism of metal removal

- Figure shows the schematic representation of an electron beam gun, which is the heart of any electron beam machining facility.
- There are 3 important elements of electron beam machining system i.e., 1) Electron beam gun. 2) Power supply. 3) Vacuum system.

1. Electron beam gun:

- It is used to produce electron beam of desired shape and focus at the predetermined location. A super heated cathode made of tungsten generates the electrons. Due to force of repulsion from cathode electrons move at a very high speed towards the anode. The speed of the electrons passing towards the anode is 70% of the speed of light.
- A magnetic lens is used to shape the electron beam into converging beam. This beam is focused and directed on the work surface. Magnetic lens are used to point the location of beam and make it circular beam falling on the workpiece.

2. Power supply:

- The power supply generates a voltage as high as 150 Kw to accelerate electrons.
- Electron beam gun is usually operates at about 12 Kw. This power density developed at work. Surface is too high and thus it is capable to melt and vaporize the workpiece. Thus metal removal in EBM is basically due to vaporization.

3. Vacuum system:

- The electron beam generation and machining operation takes place in a vacuum chamber. The

vacuum does not allow rapid oxidation of filament.

- There is no loss of energy of electrons as a result of collision with air molecules.

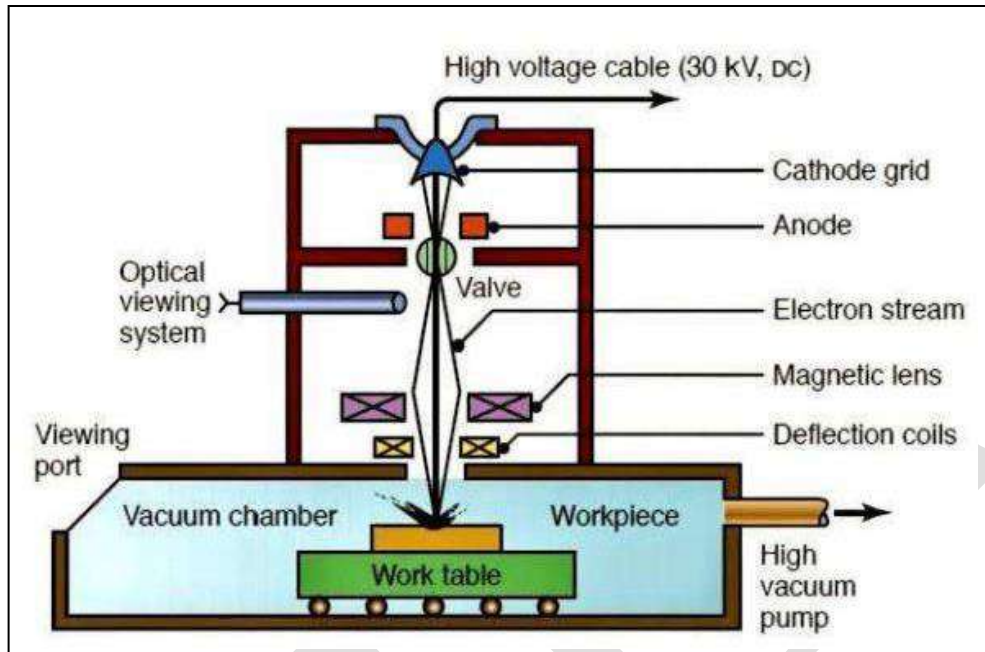


Fig. Schematic presentation of Electron Beam Machining process

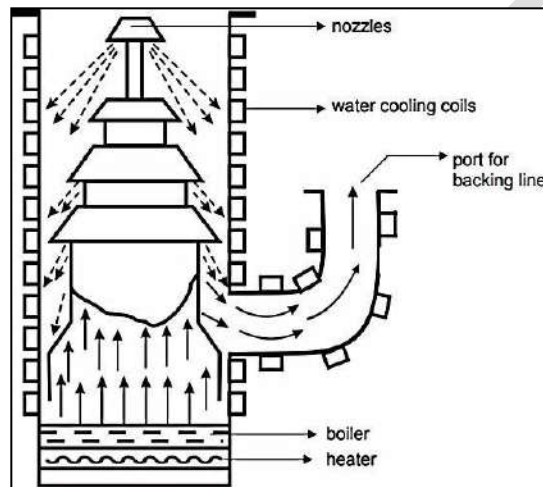
Electron Beam Gun

- The basic functions of any electron beam gun are to generate free electrons at the cathode, accelerate them to a sufficiently high velocity and to focus them over a small spot size. Further, the beam needs to be manoeuvred if required by the gun.
- The cathode as can be seen in Fig. is generally made of tungsten or tantalum. Such cathode filaments are heated, often inductively, to a temperature of around 25000°C.
- Such heating leads to thermo-ionic emission of electrons, which is further enhanced by maintaining very low vacuum within the chamber of the electron beam gun.
- Moreover, this cathode cartridge is highly negatively biased so that the thermo-ionic electrons are strongly repelled away from the cathode.
- This cathode is often in the form of a cartridge so that it can be changed very quickly to reduce down time in case of failure.
- Just after the cathode, there is an annular bias grid. A high negative bias is applied to this grid so that the electrons generated by this cathode do not diverge and approach the next element, the annular anode, in the form of a beam.
- The annular anode now attracts the electron beam and gradually gets accelerated. As they leave

the anode section, the electrons may achieve a velocity as high as half the velocity of light.

- The nature of biasing just after the cathode controls the flow of electrons and the biased grid is used as a switch to operate the electron beam gun in pulsed mode.
- After the anode, the electron beam passes through a series of magnetic lenses and apertures. The magnetic lenses shape the beam and try to reduce the divergence.
- Apertures on the other hand allow only the convergent electrons to pass and capture the divergent low energy electrons from the fringes.
- This way, the aperture and the magnetic lenses improve the quality of the electron beam.
- Then the electron beam passes through the final section of the electromagnetic lens and deflection coil.
- The electromagnetic lens focuses the electron beam to a desired spot.
- The deflection coil can manoeuvre the electron beam, though by small amount, to improve shape of the machined holes.
- Generally in between the electron beam gun and the workpiece, which is also under vacuum, there would be a series of slotted rotating discs. Such discs allow the electron beam to pass and machine materials but helpfully prevent metal fumes and vapour generated during machining to reach the gun.
- Thus it is essential to synchronize the motion of the rotating disc and pulsing of the electron beam gun.
- Electron beam guns are also provided with illumination facility and a telescope for alignment of the beam with the workpiece.
- Workpiece is mounted on a CNC table so that holes of any shape can be machined using the CNC control and beam deflection in-built in the gun.
- One of the major requirements of EBM operation of electron beam gun is maintenance of desired vacuum.
- Level of vacuum within the gun is in the order of 10^{-4} to 10^{-6} Torr. {1 Torr = 1mm of Hg} Maintenance of suitable vacuum is essential so that electrons do not loose their energy and a significant life of the cathode cartridge is obtained.
- Such vacuum is achieved and maintained using a combination of rotary pump and diffusion pump.
- Diffusion pump, as shown in Figure is attached to the diffusion pump port of the electron beam

gun. Diffusion pump is essentially an oil heater. As the oil is heated the oil vapour rushes upward where gradually converging structure as shown in Fig. 9.6.4 is present. The nozzles change the direction of motion of the oil vapour and the oil vapour starts moving downward at a high velocity as jet. Such high velocity jets of oil vapour entrain any air molecules present within the gun. This oil is evacuated by a rotary pump via the backing line. The oil vapour condenses due to presence of cooling water jacket around the diffusion pump.



Working of a Diffusion Pump

5.14 Process Parameters of EBM

The process parameters, which directly affect the machining characteristics in Electron Beam Machining, are:

- The accelerating voltage
- The beam current
- Pulse duration
- Energy per pulse
- Power per pulse
- Lens current
- Spot size
- Power density

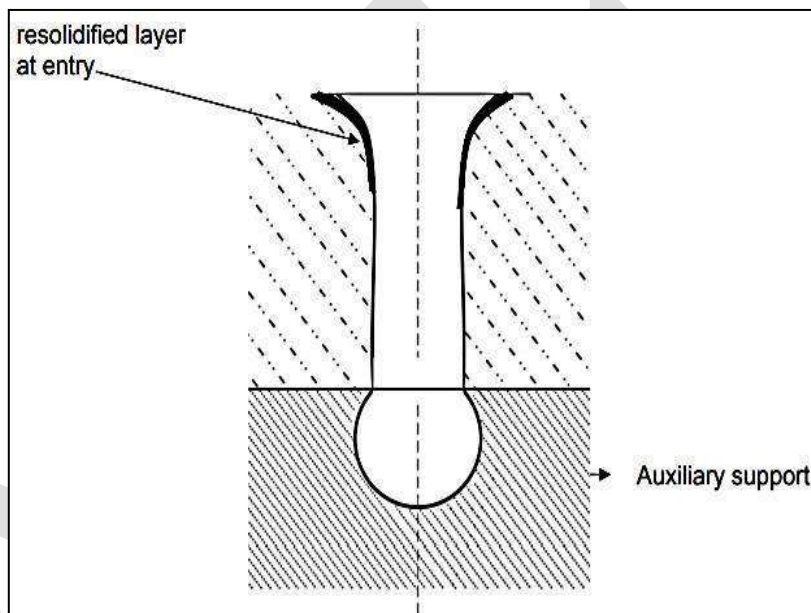
As already been mentioned in EBM the gun is operated in pulse mode. This is achieved by appropriately biasing the biased grid located just after the cathode. Switching pulses are given to the bias grid so as to achieve pulse duration of as low as 50 μ s to as long as 15 ms. Beam current is directly related to the number of electrons emitted by the cathode or

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available in the beam.

Beam current once again can be as low as 200 μA to 1 A.

Increasing the beam current directly increases the energy per pulse. Similarly increase in pulse duration also enhances energy per pulse. High-energy pulses (in excess of 100 J/pulse) can machine larger holes on thicker plates. The energy density and power density is governed by energy per pulse duration and spot size. Spot size, on the other hand is controlled by the degree of focusing achieved by the electromagnetic lenses. A higher energy density, i.e., for a lower spot size, the material removal would be faster though the size of the hole would be smaller. The plane of focusing would be on the surface of the workpiece or just below the surface of the workpiece. This controls the kerf shape or the shape of the hole as schematically shown in Figure.



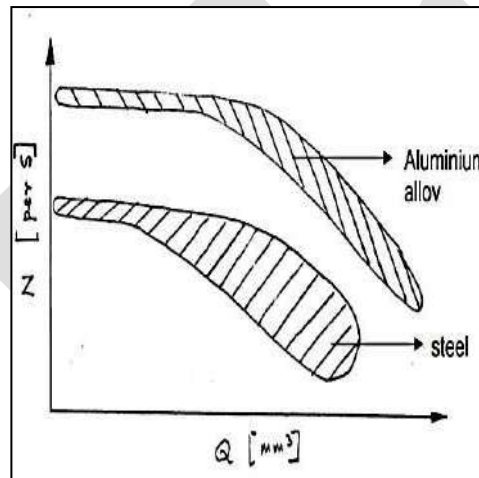
As has been indicated earlier, the final deflection coil can maneuver the electron beam providing holes of non-circular cross-section as required.

5.15 EBM Process Capability

- EBM can provide holes of diameter in the range of 100 μm to 2 mm with a depth up to 15 mm, i.e., with a l/d ratio of around 10.
- A wide range of materials such as steel, stainless steel, Ti and Ni super-alloys, aluminum as well as plastics, ceramics, leathers can be machined successfully using electron beam.

- As the mechanism of material removal is thermal in nature as for example in electro-discharge machining, there would be thermal damages associated with EBM.
- However, the heat-affected zone is rather narrow due to shorter pulse duration in EBM. Typically the heat-affected zone is around 20 to 30 μm .
- Some of the materials like Al and Ti alloys are more readily machined compared to steel.
- Number of holes drilled per second depends on the hole diameter, power density and depth of the hole as well as material type.

Fig. depicts the variation in drilling speed against volume of material removed for steel and Aluminium alloy. EBM does not apply any cutting force on the workpieces. Thus very simple work holding is required. This enables machining of fragile and brittle materials by EBM. Holes can also be drilled at a very shallow angle of as less as 20 to 300.



Variation in drilling speed with volume of material removal for steels and aluminium

5.16 Applications of EBM

- Used for producing very small size holes like holes in diesel injection nozzles, air brakes etc.
- Used only for circular holes.
- Electron beam machining is more popular in aerospace, food processing, chemical clothing etc.
- It is suitable for drilling of holes on metals such as tungsten, molybdenum.
- It is also used for drilling 1000's of holes (diameter less than 1 mm) in very thin plates.
- Micro machining operations on workpieces of thin sections.
- Micro drilling operations (upto 0.002 mm) for thin orifices, dies for wire drawing parts of electron microscopes, fibre spinners, injector holes for diesel engines etc.

- Very effective for machining of metals of low heat conductivity and high melting point.
- To drill gas orifice in nuclear reactors.
- To produce wire drawing dies.
- To make cooling holes in turbine plates.

5.17 Advantages of EBM

- Very small size holes can be produced.
- Surface finish produced is good.
- Highly reactive metals like Al and Mg can be machined very easily.
- The operation is chemically clean.
- Hole location is very accurate.
- Depth of penetration and heat can be easily controlled.
- Excellent method for micro finishing.
- There is no tool pressure or wear.
- There is no contact between tool and workpiece.
- It can cut any known material, metal or non metal that would exist in vacuum.
- No physical or metallurgical damage.
- Heat can be concentrated on a particular spot.
- Close dimensional tolerances can be achieved because problem of tool wear is non existent.

5.18 Limitations of EBM

- High investment cost.
- Low metal removal rate.
- Skilled operator is required for machining.
- High power consumption.
- Only small cuts are possible.

QUESTION BANK

Laser Beam Machining (LBM)

1. Expand LASER.
2. Write a short note on Laser beam machining.
3. What are the advantages and disadvantages of Laser beam machining ?
4. With a neat sketch explain laser beam machining, generation of laser, equipment and mechanism of metal removal.
5. What are the parameters and characteristics of Laser beam machining.
6. What are the applications of Laser beam machining?
7. Discuss the advantages and limitations of LBM. (4 Marks) **(VTU Dec 2011)**
8. Explain with a neat sketch, the principle of operation and mechanism of metal removal in laser beam machining. (8 Marks) **(VTU Dec 2011)**
9. Describe the apparatus used to generate laser. (5 Marks) **(VTU Dec 2012)**
10. Write the limitation of LBM. (5 Marks) **(VTU Dec 2012)**
11. With a neat sketch, explain the mechanism of metal removal in LBM process. (6 Marks) **(VTU June/July 2013)**
12. Explain with sketch the mechanism of metal removal in laser beam machining. (7 Marks) **(VTU June/July 2014)**
13. Enumerate the advantages, limitations and applications of laser beam machining. (6 Marks) **(VTU June/July 2014)**
14. List out the commonly used gases in Laser beam machining. (4 Marks) **(VTU Dec 2014 /Jan 2015)**
15. What are the advantages and applications of Laser Beam Machining (LBM) ? (6 - 8 Marks) **(VTU June/July 2015) (VTU June 2012)**
16. Draw a neat sketch of a typical setup for Laser Beam Machining (LBM) and explain briefly. (8 Marks) **(VTU June/July 2016)**
17. What are the advantages and limitations of LBM ? (4 Marks) **(VTU June/July 2016)**

Electron Beam Machining (EBM)

1. Briefly explain Electron beam machining with neat sketch.
2. Explain the principle of Electron beam machining.
3. What are the applications of Electron beam machining.

4. What are the advantages and disadvantages of Electron beam machining.
5. Briefly discuss the merits and demerits of EBM. (4 Marks) **(VTU June/July 2011)**
6. With a sketch, explain the principle of EBM. (8 Marks) **(VTU June/July 2011)**
7. Comment on the parameters influencing MRR in EBM. (8 Marks) **(VTU June/July 2011)**
8. With a neat diagram, explain the generation and control of an electron beam. (8 Marks) **(VTU Dec 2011)**
9. Sketch and explain electron beam machining process. (8 Marks) **(VTU June 2012)**
10. What are the process parameters of affect on the machining process in EBM ? (4 Marks) **(VTU June 2012)**
11. Explain the generation of electron beam used in EBM with the help of the diagram of the equipment used. (10 Marks) **(VTU Dec 2012)**
12. State the advantages, disadvantages and application of EBM. (6 Marks) **(VTU June/July 2013)**
13. Explain how the electron beam is generated in electron beam machining (EBM) process. (6 Marks) **(VTU June/July 2013)**
14. Explain with sketch the principle of working of electron beam machining process. (7 Marks)
15. What are the important characteristics required for electron beam in Electron beam machining. (4 Marks) **(VTU Dec 2014 /Jan 2015)**
16. Explain with a neat sketch, Electron Beam Machining and list out its advantages. (12 Marks) **(VTU Dec 2014 /Jan 2015)**
17. Explain with sketch, the working of electron beam machining (EBM). (10 Marks) **(VTU June/July 2015)**
18. What are the applications of EBM ? (4 Marks) **(VTU June/July 2015)**
19. With neat sketch, explain working principle of Electron Beam Machining (EBM). (8 Marks) **(VTU June/July 2016)**